

Monitoring industrial ecosystems

ELECTRONICS

Analytical report – 2024 edition

This report was prepared by Liliya Pullmann, Fraunhofer ISI and Carmen Moreno, Technopolis Group for the European Commission Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. However, it does not necessarily reflect the views of the European Commission.

EUROPEAN COMMISSION

Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs
Directorate B – Simplification and networks
Unit GROW.B.4 – Industrial Forum, Alliances and Clusters

European Innovation Council and SMEs Executive Agency (EISMEA)
Unit I.02 - SMP/SME Pillar, Internal Market and Support to Standardisation
E-Mail: EISMEA-SMP-COSME-ECOSYSTEMS@ec.europa.eu

European Commission
B-1049 Brussels

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Executive Summary

This report has been prepared within the 'European Monitor of Industrial Ecosystems' (EMI) project with the objective to analysing the green and digital transformation of industrial ecosystems and progress made over time, in this specific case electronics.

The second edition of the Industrial Ecosystems series has divided the analysis into two sections meaning, 'green transition' and 'digital transition' followed by three subsections each showing: a) industry efforts, b) framework conditions and c) the impact on the environment and on productivity.

The key findings of this year's analysis are being presented below:

Green Transition

What progress has the industry made in taking action for the environment?

- The electronics industrial ecosystem is at the early stages of its green transition, gradually incorporating sustainable practices and technologies to minimise environmental impact. While this shift is driven by regulatory requirements and the need to reduce resource consumption and associated costs, it is also fueled by growing societal pressure for greater environmental responsibility. Challenges persist, highlighting the need for targeted support, investment, and innovation.

- According to Eurobarometer 2024, **20% of electronics firms had carbon-reduction strategies (up from 7% in 2021) in 2024, and 79% were implementing energy-saving measures**. However, attention to material savings and waste reduction has declined since 2021, highlighting the need for renewed focus on these areas.

- Innovations in manufacturing are supporting sustainability efforts. **Advances such as low-temperature processing, optimised EUV lithography, and alternative materials are reducing environmental footprints**. Digital technologies, including AI and IoT can also help improve resource efficiency and minimising waste. Strengthening local production powered by renewable energy further enhances supply chain resilience and reduces emissions.

- Significant challenges remain for **SMEs, which face additional hurdles**, including resource constraints, lack of awareness, and bureaucratic barriers, limiting their ability to adopt sustainable practices.

- **Startups are critical enablers of sustainability within the industrial ecosystem.** Collaborations between startups and established firms have shown promising impacts on market adoption of green technologies and circular business models. Many startups in the EU specialise in developing sustainable materials and processes as well as designing energy-efficient devices and smart technologies that optimise power consumption throughout the product lifecycle.

- **Private investment in environmental technologies remains limited**, with most companies allocating less than 5% of their annual turnover to green initiatives. Venture capital funding has stabilised, with a focus on advanced sustainable materials, but significant regional disparities in investment across the EU persist.

- **Electronics play a pivotal role in enabling the green transition in other industrial ecosystems.** By driving advancements in renewable energy, smart grids, and energy-efficient technologies, the industry contributes significantly to emission reductions and cost savings. For instance, semiconductors facilitate grid optimisation and electrification, supporting sustainable energy systems. However, the EU's ambition to expand semiconductor production under the Chips Act could lead to considerable increase in emissions, underscoring the urgency of climate-conscious strategies.

To what extent do framework conditions such as public financing and skills support the green transition?

- The EU's ambition to expand semiconductor production under the Chips Act risks further increasing emissions unless climate-conscious strategies are implemented. Electricity consumption, a major contributor to the industry's environmental impact, has driven companies to focus on energy-efficient manufacturing, renewable energy adoption, and the development of low-energy-consuming devices.

- Public investments, such as the **European Regional Development Fund (ERDF), have played a vital role in supporting green initiatives in the EU, with 6% of electronics-related projects between 2014 and 2020 supporting the green transition**. However, integration of green technologies into mainstream manufacturing remains limited underscoring the urgent need for scaling.

- The green transition hinges on a skilled workforce equipped to handle renewable energy, circular economy principles, and sustainable manufacturing. **Despite increasing demand, the supply of green skills remains insufficient, with vocational training and higher education programmes lagging in sustainability focus**. Bridging this gap will require enhanced training programmes, industry collaboration, and prioritization of green expertise.

How is the industrial ecosystem's impact on the environment changing?

- Using Exiobase data, **the electronics ecosystem accounted for 1.5% of global greenhouse gas (GHG) emissions in 2022**, with some estimates suggesting up to 4% when broader impacts are included. Key environmental indicators, such as material extraction and water consumption, have remained constant or increased, reflecting limited progress in resource efficiency. While land use peaked in 2021, a slight decrease was observed in 2022.

- **The electronics industry, particularly semiconductor manufacturing, remains energy- and resource-intensive, contributing significantly to environmental challenges**. Scope 1–3 emissions largely stem from process gases and electricity use. Manufacturing emissions are rising due to increasing chip complexity, while the use phase of many devices, such as smartphones and PCs, accounts for the largest share of lifecycle emissions. Moreover, global e-waste generation continues to grow, with low recycling rates and high plastic content in end electronic devices posing environmental and economic challenges. Despite the EU's leadership in regulatory frameworks, recycling rates remain insufficient.

Digital Transition

What is the progress of industrial efforts towards digitalisation?

- In 2024, **33% of companies in the EU electronics ecosystem reported having a digital transformation strategy,** according to the results of the EMI Enterprise survey.

- **Cloud computing and IoT are the most widely adopted technologies, streamlining operations, enhancing connectivity**, and improving data storage. Adoption of AI, big data, and robotics has risen significantly, reflecting a shift toward automation, intelligent process management, and data-driven decision-making. These technologies have enabled productivity gains of 10–15%, primarily through enhanced monitoring, precision, and supply chain resilience.

- However, the **EU lags behind the US and China in many critical areas like cloud computing, AI, and quantum technologies** linked to electronics, underscoring the need for accelerated investment and innovation.

- **Startups are critical drivers of digital innovation, particularly in IoT, AI, and smart manufacturing.** They are advancing specialised chip design, high-performance computing, and advanced packaging techniques. However, their high failure rates - around 90% - highlight challenges such as funding shortages and difficulties in achieving market

fit. Partnerships with established firms and increased R&D investment are helping mitigate these risks.

- **Venture capital investments in young electronics firms have steadily grown, targeting AI, big data, IoT, robotics, and AR/VR.** Among these, AI and AR/VR have attracted the highest funding. Public investment is also significant, with Horizon Europe allocating 70.4% of its EUR 163.57 million funding to Digital Transition projects. EU initiatives like the Chips Act further aim to bolster the sector, supporting infrastructure, innovation, and talent development.

To what extent do framework conditions such as public financing and skills support the digital transition?

- **Horizon 2020 allocated 42.2% of the EU funding to electronic projects that contributed to the digital transition within electronics**. In Horizon Europe 70.4% of the EU funding to date contributes to electronics projects related to the digital transition.

- **Digital transformation is driving demand for advanced skills in AI, big data, IoT, and robotics.** Currently, 32% of job postings in the electronics sector require such expertise. However, the EU faces a significant talent gap, exacerbated by declining STEM graduate numbers in many member states. Key shortages are evident in roles like project management and machine learning. To address the skill gap, companies are investing in on-the-job training, research partnerships, and initiatives such as the Saxony's Semiconductor Talent Incubation Programme.

- While automation reshapes jobs by enhancing productivity and upskilling workers, **smaller firms often struggle with limited infrastructure**.

What is the impact of digital technologies on competitiveness?

- **The digital transition presents vast opportunities for the EU electronics ecosystem, from increased efficiency to global leadership** in cutting-edge technologies. However, achieving its full potential requires addressing persistent challenges such as funding gaps, and implementing innovative technologies.

- **Advanced electronics and semiconductors are foundational to AI, providing the computational power, scalability, and energy efficiency needed for modern applications.** Generative AI, for instance, is revolutionising design, material selection, and quality inspection, while IoT and AI enable energy efficiency and waste reduction, aligning with both digital and green transitions.

- However, the **growing computational demands of AI** have significantly increased energy consumption. Innovations such as AI-specific architectures, low-power designs, photonic technologies, and advanced semiconductor fabrication are critical for mitigating environmental impacts of AI. These developments highlight the indispensable role of advanced electronics in enabling scalable, sustainable AI advancements while supporting broader industrial competitiveness.

1. Introduction

1.1. Objectives

This report has been prepared within the '**European Monitor of Industrial Ecosystems' (EMI)** project, initiated by the European Commission, Directorate General for Internal Market, Industry, Entrepreneurship and SMEs and the European Innovation Council and SMEs Executive Agency (EISMEA). The overall objective of the project is to **contribute to the analysis of the green and digital transformation of industrial ecosystems** and progress made over time.

The EU's updated industrial strategy[1] has identified 14 industrial ecosystems[2] – one of them being **'Electronics'** - that is in the focus of this report. The industrial strategy defined industrial ecosystems as encompassing all players operating in a value chain: from the smallest startups to the largest companies, from academia to research, service providers to suppliers. The notion of ecosystems captures the complex set of interlinkages and interdependencies among sectors and firms across the EU. Industrial transition is driven by technological, economic, and social changes, and in particular by green and digital technologies and the shift to the circular economy. The process is however characterised by complex, multi-level, and dynamic development. To make transition sustainable, technological change needs to be coupled with new business models, the necessary investments, skills, regulatory framework conditions and behavioural change across the ecosystem.

Measuring performance and change is vital to allow policymakers and industry stakeholders to track progress over time and get feedback whether the system is moving in the desired direction. To measure performance, a dedicated **monitoring and indicator framework** has been set up for the purposes of this project with an aim to capture them in regular intervals (see the overview of the monitoring framework in Figure 1).

The indicator framework includes a **set of traditional and novel data sources that allow shedding new light on ongoing transformation patterns.** The novelty of the analysis lies in the exploratory and innovative data sources used across the different chapters. Due to its effort to analyse industrial ecosystems using a more or less standardised set of indicators, the study cannot address all aspects of the green and digital transition. Therefore, additional analysis and industry-specific data sources should be used to supplement a full assessment.

The **methodological report** that sets the conceptual basis and explains the technical details of each indicator is found in a separate document uploaded on the EMI website. Moreover, some of the specific industry codes used throughout this analysis have been also included in Appendix B. The green and digital technologies considered in this study are include the following:

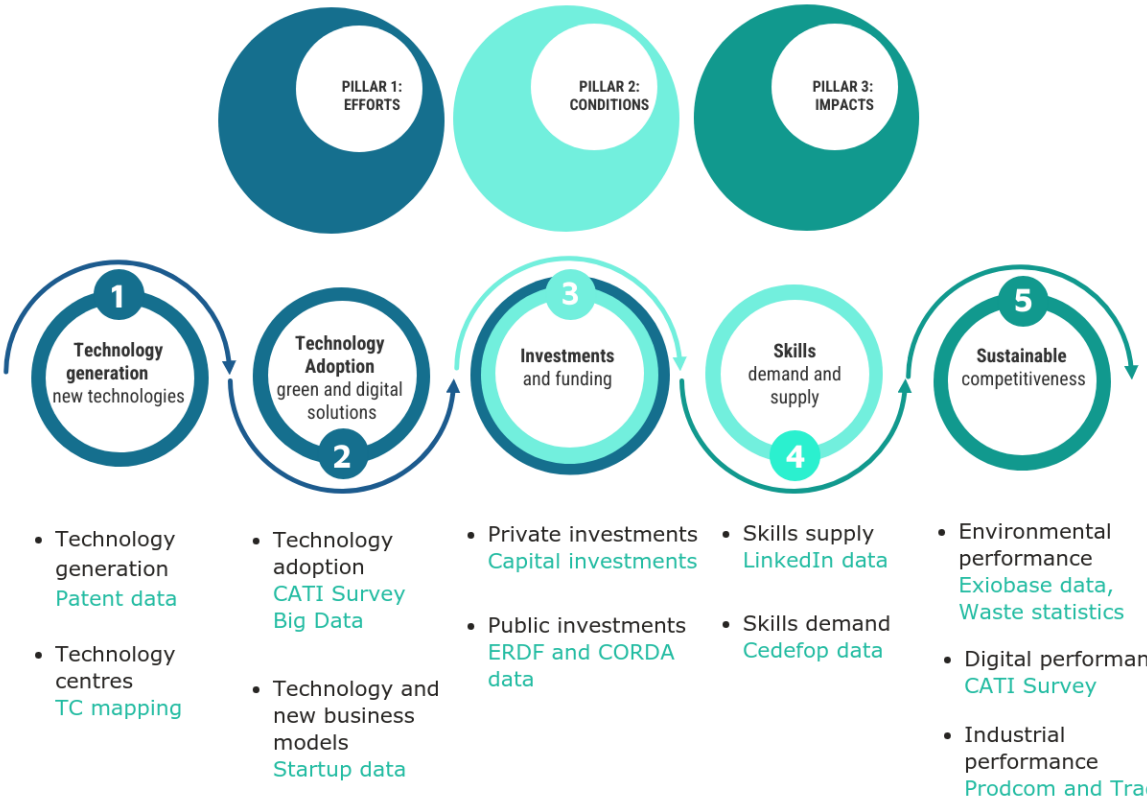
***Green transition technologies*: advanced materials, biotechnology, clean production technologies, recycling technologies, energy saving technologies**

***Advanced digital technologies*: artificial intelligence, cloud computing, augmented and virtual reality, blockchain, robotics, Internet of Things, digital security**

[1] European Commission (2020) A New Industrial Strategy for Europe, COM/2020/102 final and European Commission (2021). Updating the 2020 New Industrial Strategy: Building a stronger Single Market for Europe's recovery, COM(2021) 350 final

[2] The 14 industrial ecosystems include: construction, digital industries, health, agri-food, renewables, energy intensive industries, transport and automotive, electronics, textile, aerospace and defence, cultural and creative culture industries, tourism, proximity and social economy, and retail

Figure 1: Overview of monitoring industrial ecosystems and relevant data sources



Source: Technopolis Group, IDEA Consult and Fraunhofer ISI

This report contributes to the analysis of the **key pillars put forward in the 'Blueprint for the development of transition pathways'**[3] of the Industrial Forum developed in 2022.

1.2. Scoping the electronics industrial ecosystem

According to the definition in the Annual Single Market Report[4], the electronics ecosystem encompasses activities described in the statistical classification of economic activities NACE rev.2[5] as Manufacture of computer, electronic and electrical products (C26), Manufacture of machinery and equipment (C28) and related services. However, the established statistical classification is no longer accurate, as companies in electronics operate across different industrial sectors and can be used only as proxies. All data calculations performed in this study adhered to the definition outlined in the Annual Single Market Report. The electronics ecosystem encompasses a broad area of activities relevant for the development, manufacturing and using all kinds of electronic devices. These span resources and materials, electronic components, electronic boards, stand alone and embedded electronics to the broad range of services that support the development and deployment of electronics in different application fields.[6] For the purpose of this study and

[3] European Commission (2020) Task Force 2 – Support to the development of transition pathways. Blueprint for the development of transition pathways for industrial ecosystems. Retrieved from: https://ec.europa.eu/docsroom/documents/49407/attachments/1/translations/en/renditions/native
[4] https://commission.europa.eu/system/files/2021-05/swd-annual-single-market-report-2021_en.pdf
[5] https://ec.europa.eu/eurostat/documents/3859598/5902521/KS-RA-07-015-EN.PDF
[6] Coulon 2020. Study on the Electronics Ecosystem. European Commission.

report, we focus on the 'electronics industrial ecosystem' using this and the definition outlined in the Annual Single Market Report.

While this study examines the broader electronics ecosystem, we focus particularly on semiconductors given their key role in environmental impact, digital transition and the EU's industrial strategy. This targeted approach helps reduce analytical complexity while addressing the sector's critical importance: semiconductors not only largely determine the environmental footprint and functionality of all downstream electronics, but also represent a key strategic priority for EU technological sovereignty and supply chain resilience.

This strategic focus aligns with the green and digital transition's demands of the electronics industrial ecosystem. As the electronics industry evolves, it presents both challenges and opportunities to reconcile environmental and industrial goals. By fostering a supportive regulatory environment, encouraging investment in sustainable technologies, and addressing skill gaps through education and collaboration, policymakers can help ensure that the sector not only meets current demands but also leads the way in innovation and sustainability. The emphasis on advanced digital technologies will play a crucial role in this transition, making it imperative for companies to adapt and evolve continuously.

The interplay between green and digital transitions offers a pathway for the electronics by leveraging technologies that enhance both digital capabilities and environmental performance, companies can improve efficiency, reduce waste, and lower emissions. The necessity for ongoing research and case studies will further illuminate the potential benefits of these technologies, encouraging more widespread adoption and investment in sustainable practices.

2. Green transition

2.1. Industry efforts to green the industrial value chain

What progress has the industry made in taking action for the environment?

- The electronics industry is gradually shifting towards more sustainable manufacturing to unlock financial and environmental benefits, supported by government mandates and public demand. According to the Eurobarometer 2024, 20% of companies have strategies to reduce their carbon footprint, up from 7% in 2021. 79% of firms are implementing energy-saving measures, while 60% focus on recycling initiatives. However, attention to material savings and waste reduction has decreased compared to 2021. Electricity consumption is a major factor in the industry's environmental impact. According to the EMI Enterprise survey, 45% of electronics firms have invested in energy-saving technologies.

- Overall, the industry is focusing on three key strategies to reduce emissions: 1) energy-efficient facilities and equipment to minimise energy use; 2) supplier collaboration to reduce environmental impact in equipment and materials; 3) design of technologies/devices that consume less energy and contribute to overall emission reductions. Within the industrial ecosystem, progress is particularly evident in areas such as energy efficiency and renewable energy adoption. Innovations in manufacturing processes, such as low-temperature processing, and alternative chemicals are essential for reducing the industry's environmental impact. However, SMEs face challenges in embracing sustainability due to resource constraints, lack of awareness, bureaucratic challenges, and economic uncertainties in adopting sustainable practices. Accurate emissions reporting and effective e-waste recycling also remain critical issues. Addressing e-waste requires improved recycling rates and safer disposal practices.

- Startups role is essential in driving sustainability within the electronics ecosystem. European startups focus on developing eco-friendly and advanced semiconductor materials that are energy-efficient in production and operation. Many startups in the EU adopt circular economy models, encouraging recycling and reducing waste. Notable impact is achieved through collaboration with established firms to implement circularity models or to bring sustainable innovation to the market.

- Eurobarometer findings indicate that most companies invest less than 1% (31%) or between 1% and 5% (39%) of their annual turnover in environmental technologies, with few planning to increase this significantly. While some firms consider larger projects, overall future plans suggest a limited outlook for increased investment in environmental measures. Investments in renewable energy are growing across Europe, supported by EU directives to reduce carbon emissions and increase renewable energy share. Companies prefer locations with strong renewable energy infrastructure for semiconductor production due to lower energy costs and stable supply.

- Venture capital investments in the electronics sector related to the green transition have remained stable, peaking in 2023. These investments highlight a focus on advanced sustainable materials, attracting the most venture capital. Despite these developments, overall investment in environmental technologies remains relatively limited, with significant regional variations across the EU.

This section reports first on the progress of firms within the industrial ecosystem towards the green transition, focusing on the adoption of environmental technologies and circular business models. It also analyses how startups and young companies that provide environmental solutions exclusively in the electronics industry contribute to the transformation of the industrial value chain. Moreover, it examines the level of investment by electronics companies in the green transition including green technologies, renewable energy, and circular economy solutions.

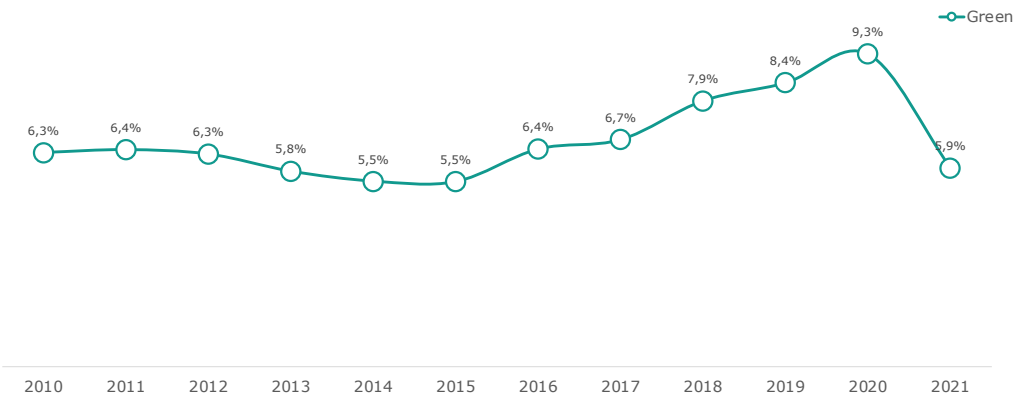
2.1.1. Technology generation

Technology generation has been analysed based on patent data. In electronics, patent applications are essential for protecting new inventions and technological advancements, offering inventors exclusive rights to their innovations. Utility patents are the most common patents granted for electronics inventions as they protect the functional aspects of an invention[7] such as a novel method of improving battery efficiency in smartphones.

Technology developments through patents are captured for the electronics industrial ecosystem based on the patenting activities related to the specific sectoral activities as outlined in the NACE classification of the Annual Single Market Report 2021 and based on patent-based classifications[8].

Looking at the technologies related to the green transition as presented in the methodological framework, the Figure below presents the share of all patent applications relevant for the green transition of the electronics ecosystem in all patents filed from the industry. A steady increase can be observed in the share of patent applications until 2020 with a decrease from 2020 to 2021[9].

Figure 2: Share of patent applications in green transition technologies in all patent applications filed for the electronics industrial ecosystem

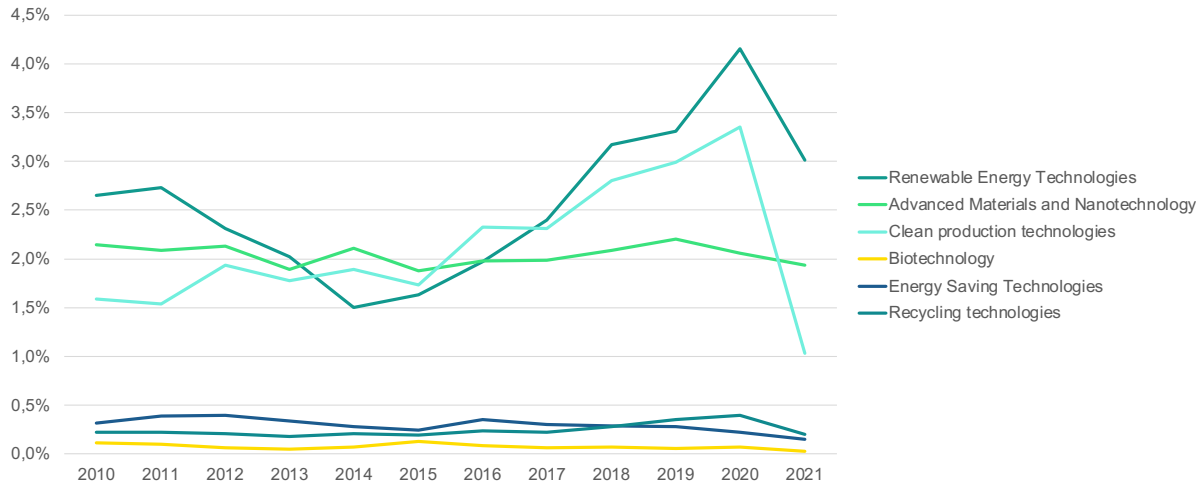


Source: Fraunhofer based on PATSTAT

At the level of green technologies, renewable energy technologies, advanced materials, and clean production technologies are the three leading technologies for the electronics industrial ecosystem in terms of patenting activities (see Figure 3). Trends over time indicate that the share has decreased since 2020.

[7] https://patentpc.com/blog/electronics-inventions-patent-basics-understanding
[8] The analysis is based on 'transnational patents' (Frietsch/Schmoch, 2010) (i.e. PCT/WIPO filings or direct applications at the EPO, excluding double counts) and was conducted on an extended version of the EPO's Worldwide Patent Statistical Database that Fraunhofer ISI implemented locally. Technologies relevant to ecosystems, in this case the construction industrial ecosystem, are defined based on a search that refers to patent classifications (IPC) and/or use keywords to identify relevant applications across classes. The detailed methodology is presented in the Methodological Report.
[9] World Construction Network (2024). Q2 2024 update: patent activity in the construction industry. Retrieved from: https://www.worldconstructionnetwork.com/data-insights/patent-activity-construction-industry/?cf-view

Figure 3: Share of respective green transition technologies in overall patent application filed in the electronics industrial ecosystem



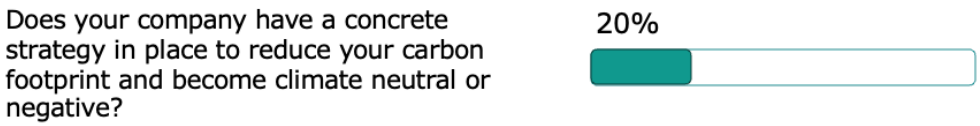
Source: Fraunhofer based on PATSTAT

2.1.2. Uptake of environmental technologies and circular business models

The electronics industry is to transition to sustainable manufacturing practices, highlighting that early adoption is crucial to unlocking both financial gains and environmental benefits. This shift involves rethinking traditional production methods to minimise waste, reduce energy consumption, and integrate eco-friendly materials. The adoption of technologies and circular business models in the electronics industrial ecosystem has been explored in detail by the Eurobarometer 2024 and has been complemented by a CATI survey conducted as part of the EMI project over the period July-September 2024.

According to the Eurobarometer 2024 survey results, 20% of companies in the electronics industrial ecosystem have a concrete strategy to reduce carbon footprint or become climate neutral or negative. This has been a move forward from 7% reported in 2021.

Figure 4: Share of companies in electronics industrial ecosystem that has adopted strategy to reduce carbon footprint



Source: Eurobarometer survey 2024.

In terms of adoption of green measures, as depicted in Figure 5, 79% of companies aimed at measures aiming to save energy, and 60% aiming to recycle in 2024. In addition, progress has been made to switch to greener suppliers. Nonetheless, the electronics ecosystem paid less attention to save materials and minimise waste in 2024 than in 2021.

These findings are in line with many electronics companies in the EU are adopting sustainable practices, such as improving energy efficiency, and addressing e-waste. In addition to measures aimed at improving resource efficiency and supporting climate neutrality, electronics enterprises have also directly implemented specific green technologies.

Figure 5: Share of firms in electronics in the EU27 indicating the adoption of environmental measures

Environmental measures	***Share of adoption (2021)***	***Share of adoption (2024)***
Saving energy	63%	79%
Recycling, by reusing material or waste within the company	62%	60%
Saving materials	69%	54%
Minimising waste	69%	49%
Switching to greener suppliers of materials	15%	43%
Saving water	40%	40%
Designing products that are easier to maintain, repair or reuse	40%	39%
Selling your residues and waste to another company	38%	20%
Using predominantly renewable energy	10%	17%

Source: Eurobarometer survey 2024

Figure 6: Adoption of green technologies by companies in electronics

Green technologies	***Share of adoption***
Energy-saving technologies	45.18%
Waste management technologies	20.67%
Advanced materials	18.99%
Clean production technologies	16.76%
Biotechnology	3.21%
Carbon capture technologies	2.70%

Source: EMI Enterprise survey 2024

The EMI survey indicates that **45% of electronics companies invested in energy saving technologies**. There is widespread agreement based on research that electricity consumption is the most significant factor in reducing the environmental footprint of the electronics and semiconductor industry (see Semi et al. (2023)[66]; McKinsey (2022); Greenpeace (2023)). It currently accounts for around 83% of semiconductor device carbon dioxide emissions along its entire lifecycle. The study of Gupta et al. (2020) indicates a radical decrease of carbon emissions due to the use of renewable energy. For example, under the US grid, about 60% of Intel's and 45% of AMD's emissions come from hardware use and energy consumption. However, the use of renewable energy reduces operational emissions — solar and wind emit only 41g and 11g of CO_2 per kilowatt-hour, compared to 301g for the US grid.

To reduce the major carbon emissions along the value chain, Semi, Semiconductor Climate Consortium, and Boston Consulting Group emphasise three key actions Semi et al. (2023)[66]:

- Establishing energy-efficient manufacturing facilities and offices;
- Collaborating with suppliers to reduce electricity usage and enhance energy efficiency in equipment and materials, and
- Designing energy-efficient devices.

The growing focus on sustainability within the industry, as shown by the increasing number of related research papers and corporate sustainability reports, is a positive trend that aligns with global efforts to combat climate change.

Companies in the microelectronics sector exhibit an increasing motivation to embrace sustainable practices. There is a growing recognition of the need to develop and promote green technologies, driven by regulatory requirements, market demand, and a commitment to environmental responsibility. Rising consumer and societal preferences for eco-friendly products are placing increased pressure on companies to innovate and adopt sustainable practices. Businesses are beginning to see sustainability as a competitive advantage, with economic benefits like energy efficiency and waste reduction leading to cost savings and improved operational efficiency. Moreover, investments in sustainability initiatives can contribute to industry resilience by stabilising access to resources like energy and raw materials. Technological advancements are another key driver, enabling the development of more energy-efficient, eco-friendly materials, products, and manufacturing processes. As demonstrated across various sectors, the electronics industry plays a pivotal role in advancing broader sustainability goals by developing and supplying eco-friendly products that promote efficiency and reduce environmental impact. By designing energy-efficient devices, utilising recyclable materials, and incorporating space-saving techniques, the electronics sector minimises its environmental footprint while improving efficiency across industries and social spheres. This commitment to sustainability not only meets growing market expectations but also helps create a more resilient and resource-conscious industry.

Accordingly, electronics companies are increasingly setting ambitious emissions targets to align with decarbonisation goals. This particularly applies to large global market players, as their operations have a significant environmental impact. Many of these companies are motivated by reputation and image reasons, recognising that consumers and investors are becoming more environmentally conscious. Therefore, the drive toward ambitious emissions targets is not just about compliance; it's also a strategic move to strengthen the market position and foster the long-term growth in an increasingly eco-aware landscape.

Many EU-based microelectronics and semiconductor companies have implemented various initiatives focused on energy efficiency, renewable energy, waste management, and water conservation. The progress reported in recent sustainability reports highlights significant achievements and a strong commitment to environmental stewardship. Key targeted areas are:

- **Energy:** Companies are actively investing in energy-efficient technologies and optimising manufacturing processes to reduce energy consumption. There is a strong shift towards renewable energy sources, with many companies increasing the share of renewables in their energy mix. On-site renewable energy installations and power purchase agreements are common strategies.
- **Carbon Neutrality and Emissions Reduction:** Several companies have set ambitious carbon neutrality targets and are making progress towards achieving them. Many of them leverage advanced emission abatement technologies and energy efficiency measures that contribute to notable reductions in CO_2 emissions.
- **Sustainable Manufacturing Processes**: Companies are increasingly adopting greener production techniques, focusing on materials and chemicals that are less harmful to the environment. Some harmful chemicals and materials are being replaced with greener alternatives.
- **Water and Waste Management:** Efficient use and recycling of water are critical components of sustainability initiatives, with high water recycling rates achieved. Some companies have waste management programmes focusing on reducing hazardous waste and increasing recycling rates.

Table 1 summarises various mitigation strategies and innovations that electronics and semiconductor companies can implement to significantly reduce their environmental

impact. Many companies are actively embracing these strategies, recognising their dual benefit.

Table 1: Mitigation Strategies for Reducing Environmental Impact in Electronics and Semiconductor Industries

Mitigation Strategies and Innovations	Details
Energy Efficiency Improvements	
Upgrading Tools	Replace old equipment with energy-efficient alternatives
Optimising Cleanrooms	Adjust process parameters, reduce air pressure, and enhance humidity control to cut energy use
Heat Recapture	Implement technologies to recover and reuse heat
Energy Efficiency in Factories	Use LEDs for lighting and implement efficient energy management systems
Renewable Energy	
Green Energy Procurement	Source renewable electricity (solar, wind, hydropower) to reduce carbon footprint
On-Site Renewable Energy	Install photovoltaics to produce parts of required electricity in-house
Process Gas Emissions Reduction	
Alternative Gases	Research and adopt chemicals with lower environmental impact
Gas Abatement	Capture and refine unused process gases for reuse
Process Optimisation	Regulate process parameters to minimise emissions
Sustainable Manufacturing Practices	
Circular Economy	Recycle and reuse materials, and proper dispose of waste
Sustainable Packaging	Reduce waste through eco-friendly packaging practices
New Materials	Explore renewable source materials and green technologies

Source: own compilation

This growing trend indicates a positive shift within the industry towards more responsible and eco-friendly practices. However, challenges persist, especially for **SMEs that might face significant hurdles in adopting sustainable practices, including lack of awareness and intrinsic motivation to pursue sustainability actively, resource constraints, bureaucratic hurdles, and economic uncertainty** (for details see Table Sustainability Challenges and Needs of Electronics SMEs in Annex). A critical need also exists for demonstrating the real value of integrating sustainable technologies, as SMEs

are often motivated by clear, quantifiable benefits.[10] Addressing these issues requires targeted strategies that focus on education, resource availability, bureaucratic support, and technological integration.

While sustainability is increasingly recognised as important, **many companies in the EU (and beyond) tend to prioritise immediate financial benefits over long-term sustainability goals**, which can hinder progress. Furthermore, significant disparities persist between different countries and regions in the EU.

Experts point to (see: Hess 2024) existing drawbacks hindering reliable estimations of the real environmental impact of the industry. One of key issues is an inaccurate reporting of market-based scope 2 emissions, which encompass indirect emissions from the generation of purchased electricity, steam, heating, and cooling. The market-based approach allows firms to claim emissions reductions by purchasing Renewable Energy Certificates (RECs) or engaging in power purchase agreements (PPAs). However, this can lead to an overestimation of actual renewable energy usage, as companies might buy RECs without sourcing their electricity from renewable sources. In contrast, the location-based method calculates emissions based on the average intensity of the electricity grid, providing a more accurate reflection of actual emissions. This discrepancy can mislead stakeholders about a company's true environmental impact, making it appear that companies are progressing more toward carbon neutrality than they actually are. Such inaccuracies undermine environmental initiatives and hinder collective efforts to reduce overall GHG emissions. Therefore, standardised reporting practices with consistent guidelines are needed to assess genuine progress toward sustainability goals.

Addressing the e-waste challenge is a critical aspect of the electronics ecosystem, as the rapid advancement of technology as well as the broad Digital and Green Transitions lead to increased electronic waste generation. Overcoming the e-waste crisis requires increased recycling rates, safer and more efficient recycling practices and processes, robust legislation, and a shift towards repairable and durable products[76]. While there have been some advancements in e-waste recycling among several electronics companies in the EU, the overall progress can be seen as slow and inefficient in many respects. Many challenges remain. A more coordinated and consistent approach is needed to enhance the effectiveness of e-waste recycling efforts and drive meaningful change across the industry.

Overall, while there has been progress in meeting the requirements of the WEEE Directive, compliance is not uniform across the EU. High-performing countries have established effective systems, whereas others face challenges that hinder their ability to meet the collection and recycling targets. Despite some progress in collection, largely driven by EU regulations and initiatives, a **number of EU-countries faces challenges, such as limited infrastructure and public engagement, or inefficient recycling processes**. The lack of proper recycling infrastructure and technology hampers the ability to recycle e-waste effectively, resulting in a significant amount of e-waste ending up in landfills.

Furthermore, a limited availability and lack of access to reliable data to conduct effective life cycle assessments (LCAs) presents a significant barrier to electronics companies, especially SMEs, to facilitate better decision-making processes and implement sustainable practices.

Innovations optimising manufacturing for sustainability

Advancements and optimisations in electronics and their manufacturing processes are crucial for reducing the industry's carbon footprint. There is a significant potential for emission reductions through innovation and process optimisation. The development and uptake of new technologies and processes is crucial to achieve major improvements.

[10] Experts emphasised that the challenge frequently lies in establishing a clear connection between sustainable practices and economic success. SMEs need convincing arguments, such as operational cost savings or access to new customer segments, to engage actively with sustainability.

As was indicated before, dry etching and EUV lithography are critical steps in high-end semiconductor manufacturing that consume considerable energy. Moreover, dry etch processes often rely on fluorinated compounds that have high global warming impact. To address these challenges, innovations in alternative materials and energy-efficient technologies are essential. Developing greener etching processes and optimising EUV performance can significantly reduce the environmental footprint of semiconductor manufacturing. IMEC´s research[11] indicates a significant potential for lowering of the environmental impact of these processes through the implementation of innovative technologies and techniques summarised in Table 2.

Table 2: Mitigation Strategies for Reducing Environmental Impact in Semiconductor Lithography and Etch Processes

Process Component	Mitigation Strategy
Dry Etch	**Ultra-Thin Resists and Underlayers and Minimal Passivation** to minimise fluorinated compounds and their environmental impact **Low Process Temperatures**: Decreases energy consumption. **High-NA-Compatible Metal Line Etch**: Demonstrated to have ~6% of the process gas emissions relative to the reference process before abatement.
Lithography (EUV Scanners)	**Greener Energy**: Shifting to renewable energy sources. **Reducing Multi-Patterning Steps**: Cuts down on energy-intensive process steps. **Lower Photoresist Dose**: Reduces material usage. **Enhanced Scanner Throughput**: Increases efficiency. **High-NA EUV Single Exposure**: Replacing the conventional low-NA litho-etch-litho-etch (LELE) process with High-NA EUV single exposure can reduce emissions by about 30%. Achieved by eliminating process steps despite the increased power consumption of the High-NA scanner.

Source: own compilation based on [32]

The IDTechEx report 'Sustainable Electronics Manufacturing 2023-2033'[12] emphasises the growing incentives for sustainable innovation in printed circuit boards (PCBs) and integrated circuits. Driven by government mandates and increasing public demand for sustainability, the industry is transitioning, although some companies remain hesitant due to initial costs and regulatory hurdles. Key opportunities for reducing environmental footprint in manufacturing include[12]:

1. **Low-Temperature and Rapid Processing**: Additive manufacturing methods can drastically lower material waste and water consumption, with PCB production using up to 95% less water.
2. **Flexible PCBs**: These require design efficiency that aims at minimising material waste (e.g., through compact layouts and optimised layer stacks), as well as deploying innovative processing techniques and materials that benefit from low-temperature processing.
3. **Digitisation**: Implementing AI and IoT technologies can enhance efficiency and minimise waste through process automation and early detection of material misuse.

[11] https://www.imec-int.com/en/press/imec-presents-levers-reduce-co2-equivalent-footprint-lithography-and-etch-advanced-technology

[12] IDTechEx Research (2024): Sustainable Electronics Manufacturing 2023 - 2033. Online available at https://www.idtechex.com/en/research-report/sustainable-electronics-manufacturing-2023-2033/902.

4. **Supply Chain Resilience**: Recent disruptions have urged manufacturers to diversify their supply chains, investing in local production supported by access to renewable energy.

2.1.3. Environmental startups in electronics

Sustainability is one of the topics actively taken up by innovative startups in Europe and other regions. As the industry seeks to reduce its environmental impact, many startups specialise in developing eco-friendly materials and processes. Startups in the EU are at the forefront of creating eco-friendly materials or focusing on advanced semiconductor materials, which are more energy-efficient in both production and operation. Energy efficiency is another key area where startups make a significant contribution. They design energy-efficient devices and smart technologies that optimise power consumption, leading to lower energy usage throughout the product lifecycle. Furthermore, many startups are adopting circular economy practices, such as product-as-a-service models, thereby encouraging recycling and reducing waste. Collaboration of startups developing sustainable technologies with established firms further amplifies their impact, as they bring innovative sustainable solutions to market.

Innovations in printed and organic electronics is further key topic for startup activities in the EU with established hubs in Saxony (Germany), Berlin, Netherlands, and France. One example is a German startup Leaftronics focusing on development of compostable PCBs from tree leaves. This approach has already garnered interest from some major industry stakeholders, highlighting its potential to significantly address existing challenges. If successful, the innovation could revolutionise the way materials are sourced and recycled in the electronics industry.

The analysis of startup generation has been compiled through a database encompassing Crunchbase and Net Zero Insights[13] data, and it has been focused on microelectronics and semiconductors. A total of **836 electronics startups were identified** between **2015 and 2023** in the context of this monitoring exercise. When zooming into the Electronics startups which develop a green technology or solution[14], we find 97 companies created for the considered period. As reflected in the Figure below, there is a number of startups which activity focuses on advanced sustainable materials, for instance developing silicon alternatives and other semiconductor materials or composites, such as graphene, nanomaterials, GaN, and SiC for higher performance and efficiency. The company EpinovaTech AB is a company that specialises in manufacturing high-quality gallium nitride (GaN) materials and chips[15].

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[13] www.crunchbase.com and https://netzeroinsights.com/
[14] As defined in the context of this project, please see more information in the EMI methodological report
[15] http://www.epinovatech.com
[16] https://www.founderio.com/de/startup/3370121

existing challenges. If successful, the innovation could revolutionise the way materials are sourced and recycled in the electronics industry.

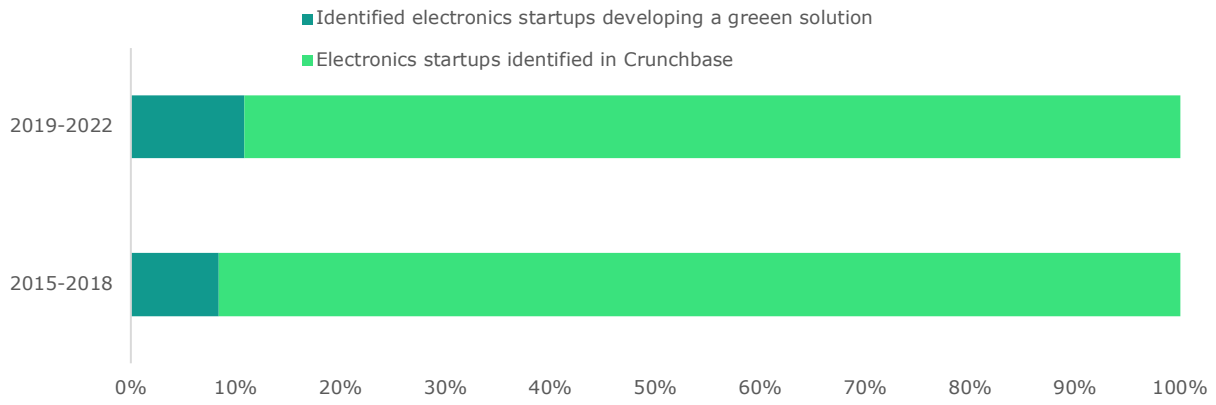
A number of startups also bring novel solutions and technologies to the market that enable cleaner production, mostly related to the manufacturing of printed electronics.

- For instance, **Safe-PCB**[17] is a manufacturing company that design and produces small and medium series of printed circuit board.

The share of green startups focused on energy-saving technologies has been steadily increasing.

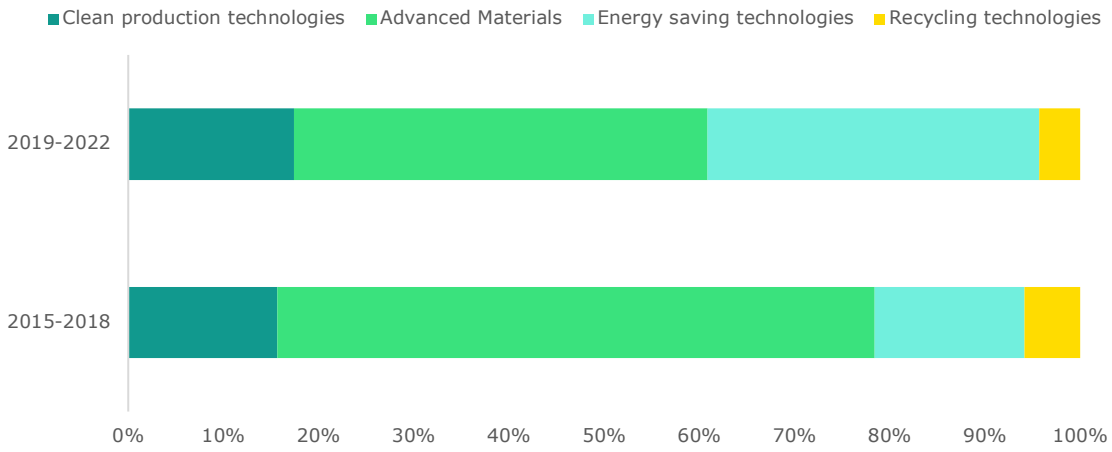
- An example is **Hoursec**[18], which has developed AI-On-Chip technology inspired by brain computing, particularly synaptic functions. This technology acts as a sampling device that maximises the ratio of information transmitted to energy consumed, saving both time and energy while ensuring privacy for the end user.

Figure 7: Share of environmental startups in electronics created between 2015 and 2022



Source: Technopolis Group, Crunchbase and Net Zero Insights

Figure 8: Share of startups created, by advanced digital technology between 2015 and 2023.



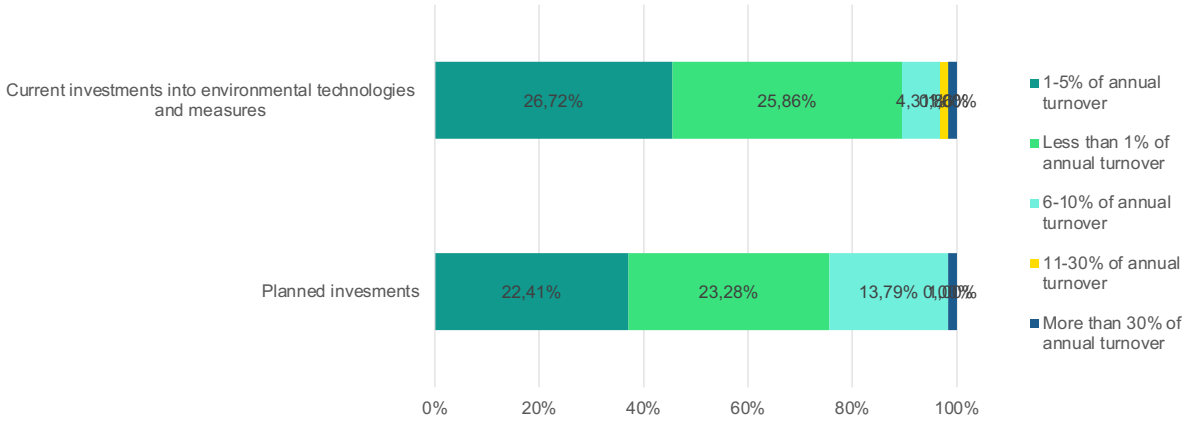
Source: Technopolis Group, Crunchbase and Net Zero Insights

[17] https://www.safe-pcb.com/safe_pcb/
[18] https://hoursec.ch

2.1.4. Private investments

Findings from the Eurobarometer indicate that most companies invest less than 1% (31%) or between 1% and 5% (39%) of their annual turnover in environmental technologies and measures. While certain categories of investments are expected to increase—such as the share of companies planning to invest 6-10% of annual turnover, indicating some larger projects in the pipeline, overall, future plans do not suggest a positive outlook for increased environmental measures.

Figure 9: Current and planned investments in environmental technologies and measures



Source: EMI Enterprise survey 2024

This transition is already in progress but varies significantly across regions and countries. Investments in renewable energy continue to grow across Europe, driven by EU directives aimed at reducing carbon emissions and increasing the share of renewables in the energy mix. The ongoing support from governments enhances the feasibility of transitioning to cleaner energy sources. Access to renewable energy plays a crucial role, as companies in regions with strong renewable energy infrastructure can benefit from lower energy costs and a more stable supply. Thus, when selecting locations for modern semiconductor production facilities, companies prefer locations that are well endowed with renewable energy sources.

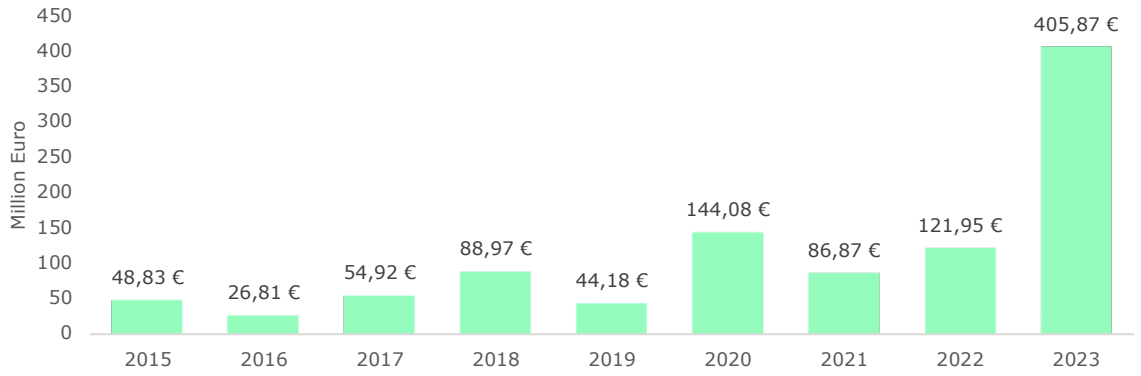
Venture capital investments have been tracked by the Crunchbase and Net Zero Insights dataset. The data looks at venture capital investments in young companies (age 10 years or less) and gives insights into the amount of investment related to the green transition, as well as investments related to technologies underpinning the transition. Venture capital data remain sufficiently well reported that general trends can be observed however some data lags for the most recent year (2023) may be present.

Venture capital investments in the electronics industrial ecosystem related to the green transition have remained stable, with a peak in 2023. This peak is largely attributed to a select series of specific investments, such as:

- **Aledia**[19], a French company founded in 2011, develops and manufactures innovative light-emitting diodes based on a unique 3D architecture using gallium-nitride on-silicon microwires (Advanced sustainable materials). It underwent a round of EUR 116 m in 2023.
- **SMART-Photonics**[20] (Advanced sustainable materials), a pure-play foundry for Indium Phosphide photonics semiconductors, raised EUR 61.7 m in 2023.

[19] http://www.aledia.com
[20] http://smartphotonics.nl/

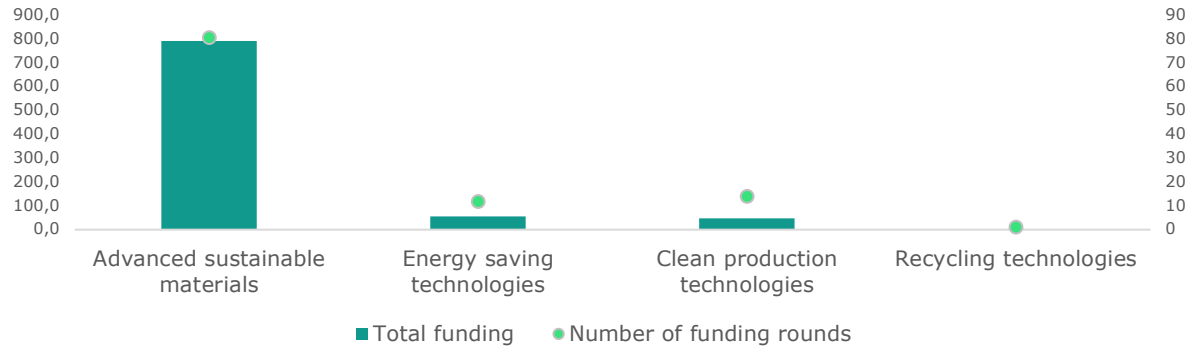
Figure 10: Venture capital investment into electronics young companies – green transition



Source: Technopolis Group, Crunchbase and Net Zero Insights

Innovative electronics startups related to advanced sustainable materials have attracted most venture capital investment (see Figure below).

Figure 11: Venture capital investments into young electronics companies developing different green technologies



Source: Technopolis Group based on Crunchbase and Net Zero Insights

2.2. Framework conditions – assessment of the broader ecosystem supporting the green transition

To what extent do framework conditions such as public financing and skills support the green transition?

Public Investments

Public investments play a major role in advancing the green transition within the EU's electronics industrial ecosystem. The European Regional Development Fund (ERDF) and Framework Programmes provide significant funding, demonstrating high-level commitment to sustainable development. 4,401 ERDF projects related to electronics were identified from 2014 to 2020, totalling EUR 5.1 bn in funding. About 6% of this funding supported the green transition. In terms of funding, 10% of the identified electronics projects contribute to the green transition. Of these, 44% support advanced green technologies, such as materials and energy efficiency initiatives.

Supply and Demand of Green Skills

The electronics sector's green transition relies on a workforce skilled in renewable energy integration, circular economy principles, and data analytics. Despite progress in HEIs, Eurostat data shows a low share of graduates in environmental protection technology across the EU. Vocational training often lacks necessary updates and sustainability focus, widening the skills gap. Demand for green skills is on the rise, with job ads increasingly seeking expertise in energy-efficient design and sustainable manufacturing. Addressing the skills gap requires enhanced education and training programmes, industry collaboration, and prioritising green skills development to support the electronics industry's Green Transition.

Demand for Sustainable Business Processes

The electronics industry is increasingly focusing on sustainable business practices as part of its approach to reduce environmental impact. Optimising resource and energy consumption is a priority for many electronics companies due to substantial cost savings and regulatory requirements. Consequently, companies are increasingly adopting energy-efficient manufacturing processes, energy management systems, renewable energy sources, and smart technologies to improve their sustainability performance. In addition, Green Transition acts as a catalyst for innovation in electronics, especially in microelectronics. Innovations include advanced materials, biodegradable substances, and sustainable design processes. Circular economy strategies, such as product-as-a-service and remanufacturing, are gaining traction. However, adoption varies across EU countries, with Western and Northern Europe leading due to strong regulatory frameworks. Notable examples include Fairphone's modular design, Apple's Take-Back Programme, and Philips' Circular Lighting initiative.

Electronics as enabler of Green Transition

Electronics are a key enabling technology for the Green Transition, as demonstrated by its role in the widespread adoption of renewable energy and smart grids. They provide the technological foundation necessary for efficient energy generation, conversion, storage, and smart grid management. By enhancing grid efficiency, electronics contribute to substantial cost savings and emissions reductions. As the demand for renewable energy continues to grow, advancements in electronic technologies remain crucial for improving efficiency, reliability, and sustainability within energy systems.

Experts recommend strengthening the EU’s green electronics transition through long-term sustainability strategies for competitive advantage. Key recommendations include developing mixed strategies with financial incentives and clear regulations to support green technologies. Strengthening EU production capacities and simplifying access to finance for SMEs and startups is essential. Establishing EU-wide training programmes for sustainable practices and fostering cross-border collaborations will help address the skill challenge. Simplifying administrative processes and creating a centralised EU database for sustainability compliance will further support the Green Transition. Moreover, implementing comprehensive CO2 pricing and taxation systems is key to incentivise sustainable practices and minimise greenwashing. Experts emphasise the need for a unified regulatory framework that encompasses all aspects of the ecosystem — from raw material extraction to end-of-life disposal, strong enforcement mechanisms, and standardised reporting to ensure compliance and foster fair competition across the industry.

Framework conditions that support the green transition refer to various structural and institutional elements that create an enabling environment for businesses to transition towards more sustainable and environmentally friendly practices. These conditions are crucial for driving the adoption of green technologies and fostering circular economies. Key components of these framework conditions include the generation of underlying technologies, public policy, skills demand and supply and demand-side factors among others that are analysed in the sections below.

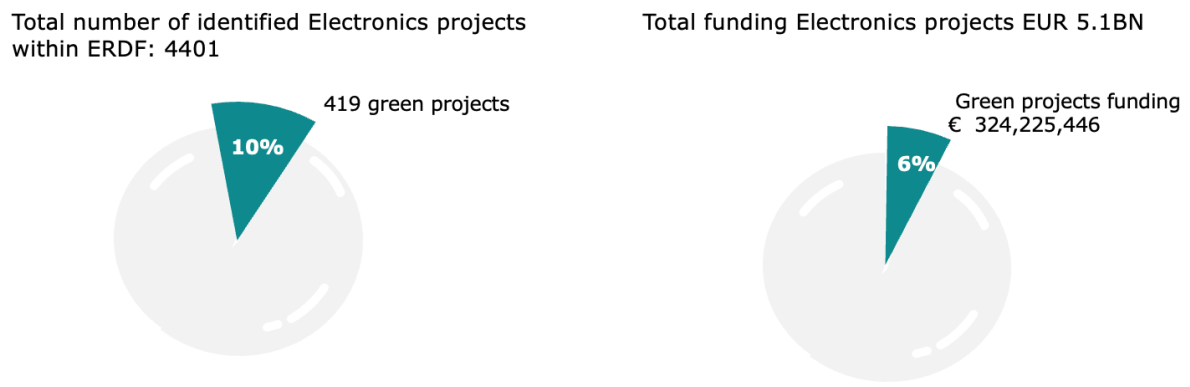
2.2.1. Public investments

Public investments affect the attractiveness of the industrial ecosystem and advance the greening of the industrial ecosystem while also signalling a high-level commitment to sustainable development, making the ecosystem and the green transition therein more attractive to businesses, investors, and other stakeholders. In the context of this study, the **investments made in the green transition in the electronics industrial ecosystem through the public policy programmes** – i.e., the European Regional Development Funds, Cohesion funds, and the Framework Programmes – are examined through the combination of the databases Cohesion Open Data and Kohesio, as well as the database Community Research and Development Information Service (Cordis).

A key source of public funding that enables the green transition is the **EU's regional development funds**. For this analysis, two data sources have been used and merged, namely the Cohesion Open Data maintained by the European Commission and the Kohesio data. The latter contains more than 1.7 million projects and approximately 500 000 beneficiaries financed throughout the funding period 2014–2020[21]. For more information on the data sources and the approach, see the methodological report.

ERDF projects[22] that are related to the electronics industrial ecosystem could be identified in the data based on the codes of economic activity and additional keyword searches in the project descriptions. The total number of **ERDF projects in the electronics ecosystem** that could be identified in 2014-2020 was 4 401, good for a total funding of EUR 5.1 bn. The analysis shows that **about 6%** (or EUR 5.19 bn) **of the funding to electronics ERDF projects supported the green transition[23]**.

Figure 12: Number of electronics industry related projects co-financed by the ERDF



Source: Technopolis Group based on Kohesio

Some examples of electronics projects supporting the green transition are described in the Box below.

[21] European Union (2022) Linking data: Kohesio platform. Retrieved from: https://data.europa.eu/en/publications/datastories/linking-data-kohesio-platform
[22] The European Regional Development Fund (ERDF) supports projects aimed to strengthen economic, social and territorial cohesion in the European Union. It aims to correct imbalances between regions enabling investments in a smarter, greener, more connected and more social Europe that is closer to its citizens. The ERDF finances programmes in shared responsibility between the European Commission and national and regional authorities in Member States.
[23] The projects related to the green transition were filtered using a combination of keyword text mining and AI powered LLMs

Box 1: ERDF electronics projects supporting the green transition

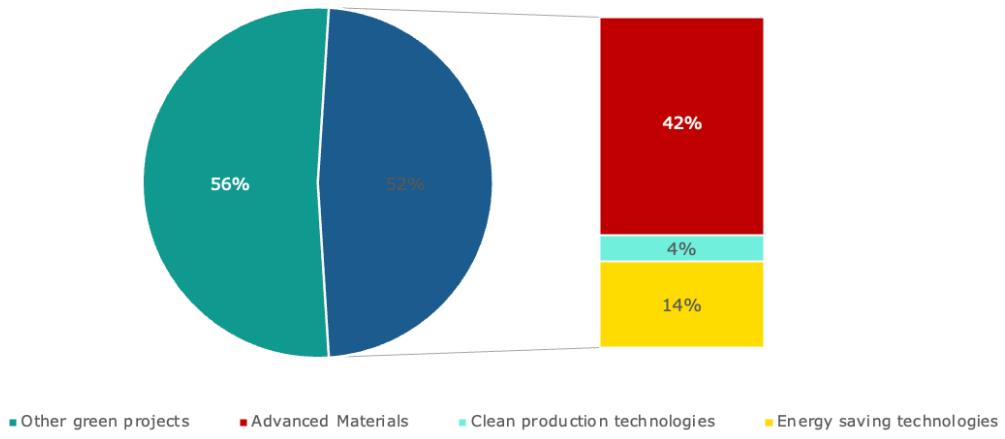
The French ERDF project 'OPGAN' concerns the introduction of new materials such as Gallium Nitrure (GaN). Falling under the advanced sustainable materials area, the project received an EU budget of EUR 247 266.

A Lithuanian ERDF project on the Development of an advanced optoelectronic materials through smart molecular engineering ran from 2018 to 2022. The project received a total EU funding of EUR 698 184.

Source: The authors based on Kohesio

In terms of funding, 10% of the identified electronics projects contribute to the green transition (or 419 projects). Of this set of green projects, **44% of the funding goes to projects active in advanced green technologies**, such as advanced materials, energy savings and clean production technologies. In particular, Figure below shows that 42% support the development of advanced materials, and 14% invest in energy efficiency for instance through the acquirement of new lab machinery.

Figure 13: Share of projects in 'advanced green technologies' in the total of ERDF electronics projects that contribute to the green transition



Source: Technopolis Group based on Kohesio

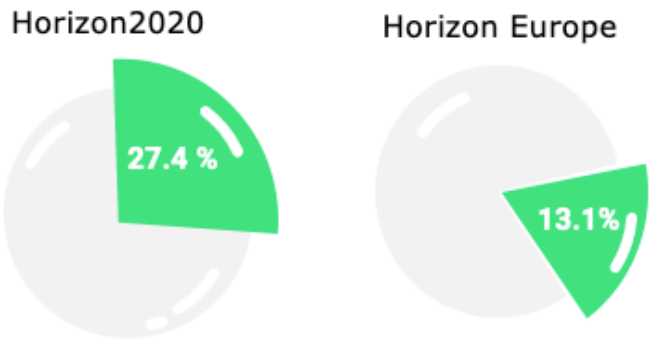
Framework Programmes

The European Commission Framework Programmes for Research and Innovation cover both Horizon 2020 (2014-2020) and Horizon Europe (2021-2027) in the analysis of the electronics industrial ecosystem. While the Horizon 2020 programme funded EUR 300 m to projects classified under the electronics industrial ecosystem[24], Horizon Europe has thus far[25] provided EUR 165.57 m in funding.

As shown in the Figure below, under Horizon 2020, nearly one-third (27.4%) of EU funding for electronics projects contributed to the green transition. For Horizon Europe, 13.1% of EU funding for electronics projects supports the green transition. These shares highlight the significant emphasis the Framework Programmes place on the green transition.

[24] The methodological report provides detailed insights into the methodology. In particular, the CORDIS data have been merged with the relevant EuroSciVoc data. EuroSciVoc connects projects to relevant scientific concepts, providing a deeper understanding of the research focus.
[25] Cut-off date: March 2024

Figure 14: Share of Horizon 2020 and Horizon Europe electronics projects that contribute the green transition



Source: Technopolis Group analysis

Two examples of an electronic project funded under Horizon 2020 and Horizon Europe supporting the green transition is described in Box 2.

Box 2: Horizon 2020 and Horizon Europe electronics projects supporting the green transition

The project GreenDiamond[26] is funded under **Horizon 2020 under the title** "Green Electronics with Diamond Power Devices". The project started in 2015, and it has raised a total amount of EUR 38.9 m.

The **GreenDiamond project** focuses on improving power electronic devices to enhance the transmission and conversion of low-carbon energy. Diamond, known for its exceptional thermal and electronic properties, is seen as the ultimate semiconductor for high-power electronics. Recent breakthroughs, such as the availability of electronic grade single crystals and advancements in MOS channel technology, have made it possible to develop diamond power transistors, which offer improved efficiency, higher blocking voltages, and reduced thermal requirements. This could increase the efficiency of power converters by four times, reducing losses by 75%. The project's goal is to fabricate a 10kV diamond transistor and an AC/DC converter, with a consortium of experts leading the effort. Europe holds a competitive edge in diamond technology, which is crucial for developing future green electronics.

The project ENRICH is funded under Horizon Europe under the title 'Reinventing Power Electronic Architectures for a Fully Integrated Power Converter-on-a-Chip'. The project started in and it has raised a total of EUR 4.12 m.

The **ENRICH project** addresses the growing need for efficient, compact, and lightweight power electronics in the transition to de-carbonised energy and electric transportation. It proposes a fundamental redesign of the power conversion chain by integrating novel multilevel converter topologies with new wide bandgap semiconductor materials, particularly Gallium Nitride (GaN). This approach eliminates magnetic components and enables the monolithic integration of the entire inverter on a single chip, creating a high-power GaN power processor. By overcoming current integration challenges, this technology aims to achieve record power densities and provide significant technological and socioeconomic benefits, especially in automotive, aerospace, and renewable energy sectors.

Source: Technopolis Group based on Cordis

The **European Regional Development Fund (ERDF)**[27] plays a crucial role in supporting the green transition of electronics by providing financial resources to regions within the European Union to promote sustainable development. The ERDF has a particular focus on

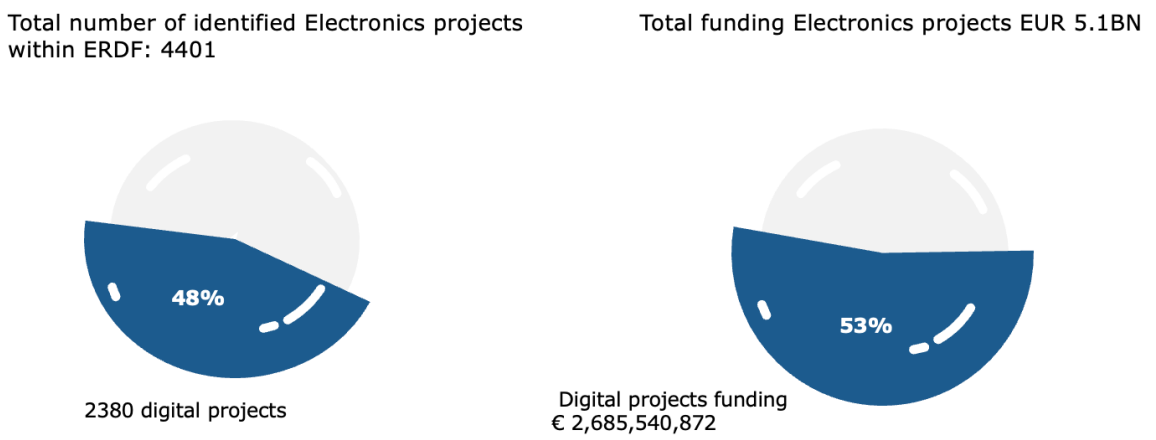
[26] European Commission (2024) FlAgship demonstration of industrial scale production of nutrient Resources from Mealworms to develop a bioeconomY New Generation. Retrieved from: https://cordis.europa.eu/project/id/837750/reporting

[27] https://ec.europa.eu/regional_policy/funding/erdf_en

less-developed regions with the objective to reduce regional disparities. In this section, we investigate to what extent the ERDF has been used to foster the twin transition in electronics. This has been done based on the in-depth data mining of the Kohesio database.

Data on ERDF shows that projects that aim at the digital transition of electronics represent 48% of all electronics projects co-funded by the ERDF and 53% of the total funding going into electronics, underlining the key enabling condition of electronics in the digital transition.

Figure 15: Number of projects and volume of public funding (EUR) allocated to electronics projects co-funded by the ERDF



Source: Technopolis Group based on data from Kohesio including all ERDF co-financed projects

Box 3: ERDF electronic projects supporting the digital transition

Spain developed a project titled '**Design and development of a SoC (system on chip) device with silicon integration of controlled magnetic field technology (CMC) and microcontroller system with advanced embedded algorithms for detection of anomalies by edge computing'** from 2019 to 2021, funded by the EU with a budget of EUR 709 056.

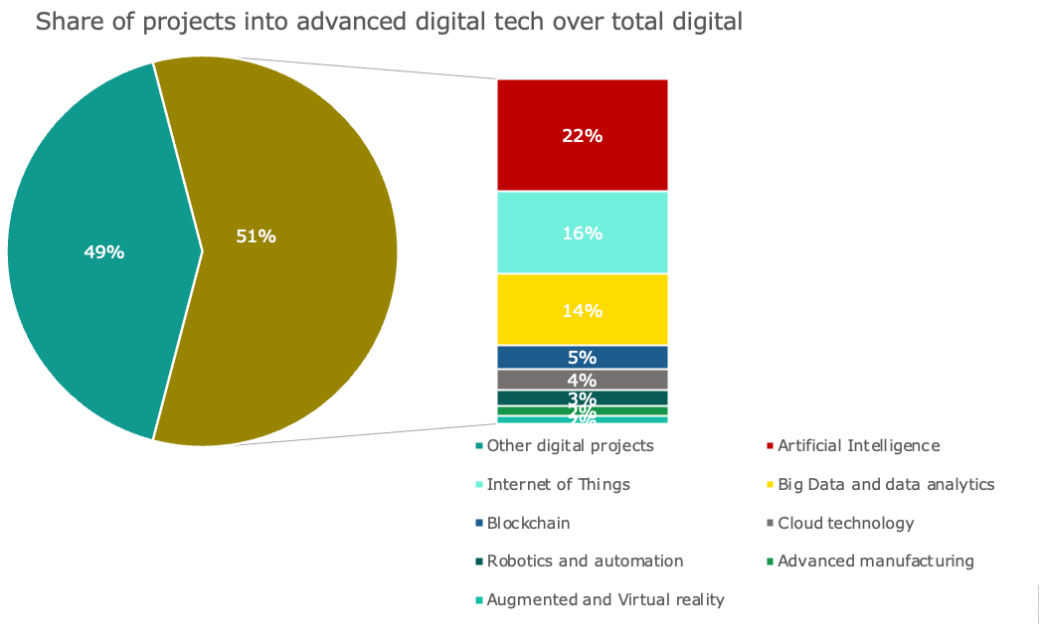
The Netherlands developed a project **focused on the development of technology that enables electronic printing**, from 2016 to 2020, funded by the EU with a budget of EUR 284 651

Source: authors based on Kohesio

Furthermore, when examining the ERDF electronics projects that contribute to the digital transition, 51% of the funding in these projects is going towards projects related to advanced digital technologies[28] (see Figure below). Most of these projects are related to AI, Internet of things, followed by big data and data analytics.

[28] As identified for the purpose of this project, e.g., projects related to IoT, AI, Cloud, Blockchain, advanced manufacturing etc.

Figure 16: Share of ERDF funding allocated to specific green topics.



Source: Technopolis Group based on Kohesio

2.2.2. Skills supply and demand underpinning the green transition

This section analyses the demand and supply side of the labour market in terms of green transition skills (green skills relevant for environmental protection, implementation of green technologies, or the circular economy). Companies in the electronics industry are faced with the challenge of up- and reskilling their employees amid a rapidly changing world of work and the digital and green transformation in electronics[29]. As customers increasingly prioritise sustainability in their purchases, basic sustainability skills are also becoming more important for electronics staff. The availability of skills and resources influences the organisation's decision to adopt green technologies[30] . During the 4th OECD Digital4SMEs roundtable, multiple stakeholders stressed the importance of the internal capabilities of the enterprise, notably the skills of employees, for successful green transformation.

Supply of skills relevant for the green transition

The successful green transition in the electronics sector is fundamentally dependent on the availability of a skilled labour force capable of designing and implementing sustainable practices and technologies. A sustainable mindset in electronics design is central for creating products that minimise environmental impact and promote resource efficiency. It involves considering the entire lifecycle of a product, from material sourcing to disposal, and prioritising renewable resources and sustainable practices.

Skills in renewable energy integration are critical, as workers must learn to incorporate solar and wind energy into manufacturing processes effectively. Understanding circular economy principles is another key skill area highly relevant, allowing companies to adopt sustainable business models. Furthermore, professionals in data analytics and digital technologies are urgently needed to enhance resource optimisation and waste reduction efforts within companies.

[29] EuroCommerce et al (2022). Contribution 29 November 2022- Pact for Skills – Skills Partnership for the CCI Ecosystem.
[30] McKinsey & Company, (2022), Transforming the EU CCI and wholesale sector.
https://www.mckinsey.com/industries/CCI/our-insights/transforming-the-eu-CCI-and-wholesale-sector

Skills in the MINT fields (Mathematics, Informatics, Natural Sciences, and Technology), particularly in electronics, physics, and chemistry are in high demand in context of the green transition of electronics industry, as they are directly linked to the development and implementation of sustainable technologies. Despite the growing importance of sustainability and environmental issues, the Eurostat data on tertiary graduates in environmental protection technology reveals an extremely low share within the total number of graduates across the EU27.

The positioning of EU countries regarding the development of green skills varies significantly. Study on policies and approaches in education for environmental sustainability in the EU, Mulvik et al. (2021)[31] indicates that insufficient government support for education for sustainable development is a significant barrier in many EU countries. Many EU countries face different challenges hindering them to implement effective education programmes for environmental sustainability.

Overall, there has been progress in establishing green electronics programmes at Higher Education Institutions (HEIs) across the EU in the recent decade, which increasingly focus on sustainability, sustainable design of electronics, energy efficiency, and environmentally friendly manufacturing practices. However, variations exist in programme depth, research opportunities, and industry collaboration. Institutions like RWTH Aachen, Technical University of Munich, Technical University of Dresden, Technical University Delft and École Polytechnique de Paris are at the forefront, offering comprehensive programmes that integrate sustainability into engineering curricula. These programmes not only convey theoretical knowledge but also provide practical application expertise through industry collaborations, research initiatives, and project-based learning. However, it is important to note that these programmes are often not part of the core curriculum but are offered as optional courses. At the same time, significant disparities exist among individual EU-countries and regions, with Eastern and Southern European countries displaying a variety of challenges. Many relevant programmes at HEIs in Widening Countries exhibit limited industry engagement, lack of integration of practical projects and resource constraints, affecting both the depth and breadth of their programmes.

While many universities in the EU are adapting their strategies, the vocational training does not reflect the same level of engagement or curriculum updates. Vocational training often lacks the sustainability focus and still needs to be duly incorporated into curricula. The comments from industry experts highlight a widespread concern about insufficient aligning of educational programmes at both – tertiary and vocational levels - with industry needs, ensuring that the workforce is prepared to tackle future sustainability challenges.

Notwithstanding the progress in integrating dedicated courses at HEIs, the gap between skills demand and skills supply in green technologies is constantly increasing. Boston Consulting Group points out to a critical shortage of green energy workers, with an anticipated skills gap of 7 million by 2030.[32] Economist Impact[33] indicates that businesses are insufficiently implementing green skills programmes for their workforce. Experts also indicated to growing needs of the industry and skills gaps in sustainable design and energy management, urging companies to engage in upskilling initiatives more actively. Close collaboration of academia with industry as well as continuous updating and investing in training programmes, internships, and practical experiences will be crucial in preparing the next generation of professionals to address the needs in a rapidly evolving electronics landscape.

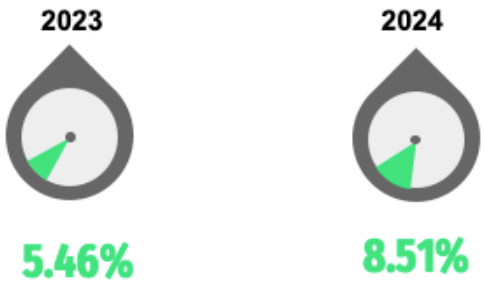
[31] Mulvik, I., et al. (2021): Education for environmental sustainability: policies and approaches in European Union Member States. Hg. v. Luxembourg: Publications Office of the European Union, 2021. European Commission. Online available at https://www.oekolog.at/dokumente/97/Education_for_Environmental_Sustainability_-_EC_Report_-_2022.pdf.
[32] https://www.bcg.com/publications/2023/will-a-green-skills-gap-put-climate-goals-at-risk
[33] https://impact.economist.com/sustainability/green-skills-outlook/infographic

To gain insights into the supply of skilled professionals, the LinkedIn database has been used[34], which has a total of 1.32 million professionals working in the electronics and semiconductor industry and with a profile on LinkedIn as of April 2024. While this means that an estimated 51% of electronics professionals are active on the platform, it is to be noted that some jobs, functions and countries are better represented than others[35], such as Germany and France. The data shows that about 52% of electronics professionals active on LinkedIn has a master's degree, and 30% a bachelor's degree, hence there is a bias towards professionals with higher educational level.

Notably, within the registered professionals on LinkedIn employed in the electronics industry, **8.51% indicated to have one type of green skill in 2024**, a share that increased compared to 2023, where 5.46% were found to have at least one green skill (see Figure 17). **Green skills, which include expertise in areas such as environmental management, recycling, and the use of sustainable materials, are becoming increasingly prevalent in the sector.**

Companies like ASML, STMicroelectronics, Infineon Technologies, GE Vernova, and Intel Corporation are among the top employers of professionals for green skills. Many professionals with green skills in the electronics industry also have backgrounds in engineering and manufacturing, with the most common job positions being owners and computer technicians.

Figure 17: Share of professionals in electronics with skills relevant for the green transition



Source: Technopolis Group based on LinkedIn

Demand for skills relevant for the green transition

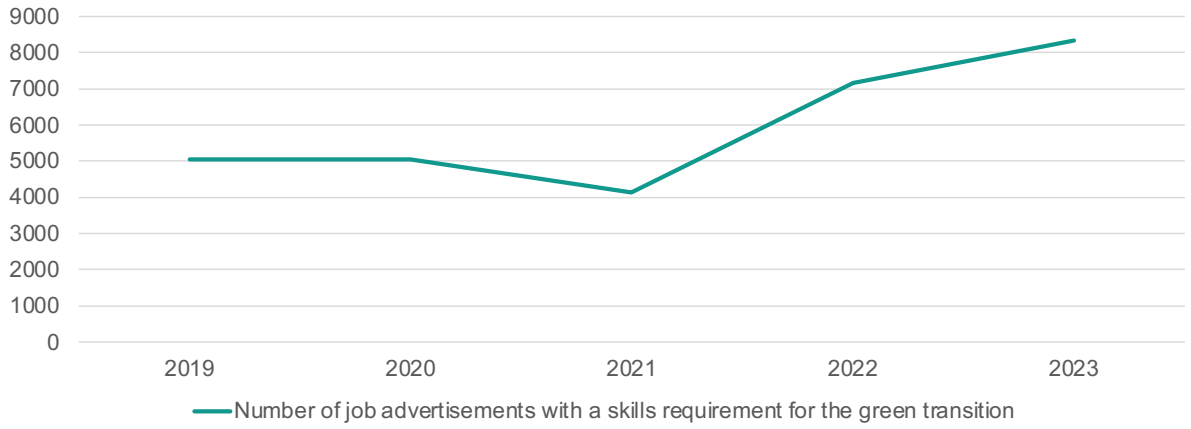
Skills demand in the electronics industrial ecosystem has been analysed following the skills intelligence insights of Cedefop, the European Centre for the Development of Vocational Training[36]. This dataset covers the EU27 Member States (plus UK) and is based on the collection and analysis of more than 530 online job advertisement sources (424 distinct websites) which are open-access sites. The dataset provides information on most requested occupations and skills across European countries based on established international classifications, e.g., ISCO-08 for occupations, ESCO for skills, and NACE rev. 2 for sectors. These websites reflect the characteristics of the local labour markets in terms of geographical distribution, ranging from 50 sites in Belgium and the Netherlands, to three in Luxembourg. For each job advertisement, there is data on the characteristics of the job, the employer, and job requirements[37].

[34] In particular, LinkedIn is the largest professional network platform with rich information like profile summary, job title, job description and field of study, which can be used for the identification of skilled professionals with skills relevant to the green transition.
[35] The methodological report outlines how the report corrects this representativeness bias, i.e., by comparing LinkedIn population to the active population and derive a corrective weighting.
[36] https://www.cedefop.europa.eu/en/tools/skills-online-vacancies
[37] This methodology has its limitations, as described in the methodological report. Some limitations are, for instance, that only online job advertisements are represented and that the volume and quantity of the data depend on the selection of portals and sources.

Specific to the electronics industrial ecosystem[38], there were **379 720 unique job advertisements** from companies in the year 2023 in the EU27, from which **2% included a requirement for green skills.**

The total number of such job ads increased notably from 2021 to 2023, indicating a growing interest in green skills. This upward trend points to a heightened demand for professionals skilled in areas such as energy-efficient design, sustainable manufacturing, waste reduction, and environmental compliance. The increase also underscores the sector's strategic shift toward integrating sustainability into the companies' operations.

Figure 18: Share of online job advertisements with a requirement for environmental skills in construction



Source: Technopolis Group based on analysis of Cedefop data

2.2.3. Demand for sustainable business processes

Optimising resource and energy consumption is a significant priority for many electronics companies, primarily due to the substantial cost savings it can deliver. By improving it, companies can significantly reduce operational expenses. Although the initial investment in energy-efficient technologies may be high, long-term savings often justify these expenses. Additionally, stringent regulatory requirements surrounding energy use and emissions compel organisations to focus on energy optimisation. Moreover, many countries offer financial incentives, such as tax relieves or grants, for those investing in energy-efficient technologies. Therefore, electronics companies are increasingly adopting a range of technologies and practices to optimise energy consumption within their operations.

To reduce energy consumption and lower operational costs while enhancing the overall sustainability performance, electronics companies in the EU increasingly adopt:

- **Energy-efficient manufacturing processes** by leveraging advanced techniques, such as lean manufacturing and automation, companies can minimise energy waste and improve operational efficiency.
- **Energy management systems (EMS)**, which allow companies to monitor and control energy usage across their facilities. These systems enable real-time data analysis and reporting, providing insights into energy consumption patterns and helping identify areas for improvement.
- **Renewable energy sources** such as solar and wind power to power their facilities.
- **Energy-efficient equipment**: Companies are upgrading to machinery and tools that consume less energy, including LED lighting and high-efficiency motors, resulting in substantial savings during production.

[38] In the case of the retail industrial ecosystem the dataset was filtered for the NACE industries as defined in the Annual Single Market Report.

- **Smart grid technology** and **Industrial Internet of Things (IoT):** help optimise energy consumption by improving load management and automating energy usage based on real-time data.
- **Waste heat recovery systems** to capture and reuse heat generated during manufacturing processes.
- **Employee training and engagement:** By educating staff on energy-saving practices and fostering a culture of sustainability, companies can ensure that everyone is aligned with their sustainability goals.

However, while energy optimisation is a major sustainability area for electronics companies, it is not the sole focus for many companies. Other aspects, such as **resource management, waste reduction, and sustainable sourcing**, are also critical. Companies are increasingly adopting circular economy principles, emphasising recycling and product lifecycle management.

Overall, electronics companies across the entire EU are increasingly prioritising resource optimisation as part of their sustainability and economic efficiency efforts. However, the progress made by individual companies located in different EU-countries might differ considerably. As was mentioned before, in many Western European and especially Scandinavian countries sustainability is often supported by robust regulatory frameworks and a strong cultural emphasis on environmental responsibility. Conversely, in Eastern and Southeastern European countries, the emphasis on sustainability is weaker due to limited regulatory support and financial resources. Companies in these regions often face challenges in implementing comprehensive sustainability programmes.

Variability also exists at the company level. While large companies with well-known brands are leading the way with their commitments to green practices, others lag behind due to differing levels of investment and strategic focus on sustainability. This particularly applies to small companies.

Green transition as key innovation driver in electronics

The **Green Transition is rapidly emerging as a catalyst for innovation within the electronics sector**, particularly in microelectronics. As global awareness of environmental issues continues to rise, companies are increasingly prioritising the development of sustainable technologies. This shift is not only a response to regulatory requirements but also a conscious strategic move to help companies regain their competitive edge in a changing market landscape.

In the microelectronics sector, especially low energy consumption and high energy efficiency have become paramount requirements. Companies are responding to these by creating products that minimise energy use. Several key innovations are driving this transition. The development of advanced materials, biodegradable and recyclable substances is transforming the electronics industry. Many companies in the EU are working on materials that can be more easily recycled or reused at the end of their lifecycle.

These innovations are part of a broader trend towards sustainable design, which aims to integrate environmentally friendly practices into product development.

Sustainable Design Processes

The advantages of sustainable design are significant, including reduced environmental impact through minimised waste and pollution, enhanced energy performance and lower greenhouse gas emissions, compliance with regulations, and market differentiation that attracts eco-conscious consumers. Overall, incorporating green technologies in electronic design supports environmental goals while driving substantial economic benefits, making them essential for long-term industry growth.

Sustainable electronic design focuses on environmentally friendly technologies to create efficient products. Key aspects include the use of energy-efficient components, such as low-power microcontrollers and energy-efficient transistors, which help reduce overall energy consumption. Selecting recyclable materials like aluminium simplifies end-of-life

management, while integrating smart power management systems optimises power usage based on device needs. Modular design allows for upgradable components and interchangeable parts, extending product lifespan and facilitating repairs.

Advanced technologies, such as AI, significantly enhances the sustainable design of electronics by optimising processes and improving efficiency. AI-driven algorithms help select eco-friendly materials and explore lightweight designs, decreasing energy use in products. Additionally, AI can support lifecycle assessments, allowing companies to evaluate environmental impacts from production to disposal.

Sustainable product design has become priority to many electronics companies. They are increasingly focusing on sustainable design practices of the electronics components and end devices. However, support of sustainable design in electronics varies widely across EU countries, with some countries showing strong initiatives and others facing significant challenges. Germany and France, for example, have a number of robust regulations and R&D programmes encouraging sustainable practices in different industries, including electronics. Further EU countries with a strong public support include the Netherlands, Belgium, Sweden, and Finland. Overall, Scandinavian countries stand out in their commitment to green innovations, including the electronics ecosystem. They implemented policies that promote eco-design principles, supported by innovation grants for startups and SMEs focused on sustainability. Moreover, these countries have strong R&D capabilities in sustainable development and design of electronics, which significantly impacts their focus of activities.

Conversely, many so-called Widening Countries[39] struggle with insufficient policies and funding for sustainability initiatives that hinder their progress. Many of them exhibit limited support for sustainable design. The reasons for it are manifold including lack of innovation potential, low public and business awareness of sustainability issues, or insufficient collaboration between government, industry, and academia. Cultural and economic contexts also matter, with economic growth frequently prioritised by national governments in these countries over sustainability.

In interviews, **industry experts emphasised the importance of advanced chip design capabilities for the future of the electronics industry**. By integrating sustainability practices into electronics design and continuously enhancing design capabilities, Europe has a potential to regain its competitive edge in the electronics sector. However, while there is a push for greener designs, the implementation can be challenging due to existing processes and the need for new standards. To address these challenges, stakeholder collaboration is crucial. Development of design capabilities requires continuous investing in research and development as well as providing education and training for sustainable electronics design.

Circular business models

A recent PWC study has found that **implementing a circular strategy can lead to 12% cost savings and a 10% reduction in CO_2 emissions of the electronics industry**[40]. They point out that product-as-a-service (PaaS) and remanufacturing have so far proved to be the most effective strategies for consumer and industrial electronics and ICT, while remanufacturing is preferred for automotive electronics. Although, each model requires initial investment and transformation, gradually implementing circular input strategies can yield valuable results without major changes. Companies can combine strategies, such as using circular inputs in manufacturing alongside a PaaS model, to maximise savings. Encouraging factors for circularity include digitalisation, partnerships, and regulatory pressures, while challenges involve initial investments. Despite challenges, there is a strong business case for adopting circular models.

[39] Bulgaria, Croatia, Cyprus, Czechia, Estonia, Greece, Hungary, Latvia, Lithuania, Malta, Poland, Portugal, Romania, Slovakia, and Slovenia.
[40] PWC (2023): Future Proofing the Electronics Industry. The case for circular business models. PWC. Online available at https://circulareconomy.europa.eu/platform/sites/default/files/2023-11/Circular%20electronics.pdf.

Overall, the pace of adoption of circular economy strategies is influenced by factors, such as government commitment, public engagement, and industry readiness to embrace sustainable practices, which vary significantly across EU countries. **Western and Northern European countries, such as Germany, the Netherlands, Finland and Sweden lead in implementing circular practices**, driven by strong regulatory frameworks, public awareness, and industry engagement. In contrast, the majority of widening countries is in the early stages of this transition.

Overall, the **electronics industry is hesitant to invest in circular business models** due to limited B2B customer demand and the misconception that climate progress conflicts with financial performance[40]. Consequently, electronics companies in the EU are slowly progressing towards circular business models. Nevertheless, several firms have made some noteworthy attempts in implementing sustainable practices. Examples presented in Box 4 highlight the positive steps taken by some companies toward achieving circularity, demonstrating the potential for broader adoption in the future.

Box 4: Examples of electronics circular business models adopted by companies in the EU

Fairphone – modular design

Fairphone, a Dutch company, is reshaping smartphone manufacturing with a strong commitment to ethical practices. Aiming to create a fairer and more sustainable electronics industry, Fairphone responsibly sources raw materials and ensures fair labor practices in production. Their modular design promotes repairs and upgrades, reducing electronic waste. Fairphone's approach challenges industry norms and advocates for a fundamental shift in smartphone production and consumption.

Apple's Take-Back Programme

Apple is advancing a circular economy through its Take-Back Programme, available in 99% of the countries where its products are sold. Central to this initiative is Apple Trade In, which allows customers to recycle any Apple device for free in 25 countries. This programme encourages responsible disposal and recycling, minimising electronic waste and facilitating the recovery of valuable materials for reuse in new products.

Philips - Circular Lighting and Health Solutions

Philips has shifted its business model to provide lighting and healthcare solutions as services rather than products. For example, in its "Circular Lighting" initiative, Philips offers lighting as a service (LaaS) to businesses, allowing them to use energy-efficient lighting without owning the fixtures. Philips retains ownership, ensuring the products are maintained, refurbished, and recycled at the end of their life. This model promotes resource efficiency and reduces waste.

Siemens - Digitalisation and Circular Economy

Siemens promotes circularity through its "Digital Industries" division, which helps manufacturers optimise their production processes using digital twins and IoT technologies. By enabling predictive maintenance and efficient resource use, Siemens assists companies in reducing waste and extending the lifecycle of their equipment. Siemens also focuses on recycling and reusing materials within its own manufacturing processes.

IKEA - Sustainable Electronics Integration

IKEA has begun integrating sustainable electronics into its product offerings, especially in smart home solutions. The company encourages customers to return used electronics for recycling and has introduced products designed for easy disassembly and recycling. IKEA's commitment to sustainability includes using renewable materials and ensuring that all of its products are designed for a circular economy by 2030.

BSH Hausgeräte's BlueMovement: Pioneering Circular Economy in Home Appliances

BSH Hausgeräte, encompassing brands like Bosch, Siemens, Neff, and Gaggenau, is adopting a circular economy through its BlueMovement initiative, which utilises a product-as-a-service business model. This model extends appliance lifespans by including repair services, aiming to

reduce resource dependence and waste generation. BlueMovement addresses this by offering services that include delivery, installation, repair, and pickup of appliances at the end of their contracts. Appliances are repaired and reused, with functional parts salvaged for new units. The initiative ultimately saves resources and decreases reliance on virgin materials, while leasing high environmental rating appliances helps customers lower their energy, water, and detergent consumption.

Electrolux - Appliance Repair and Recycling Services

Electrolux has introduced a circular business model that emphasises appliance repair and recycling. The company offers repair services for its home appliances, encouraging customers to fix rather than discard broken items. Electrolux has also established recycling programmes that ensure old products are disposed of responsibly, with materials recovered for reuse. Their goal is to design all appliances with sustainability in mind by 2030. In 2022 Electrolux has launched "Appliance-as-a-Service" programme for landlords, beginning with Sweden's SKB as its first customer.
This subscription-based model allows landlords to pay a monthly fee for installation, maintenance, and repair of appliances, extending product lifespans and reducing environmental impact. Electrolux oversees refurbishment and recycling when replacements are needed, aligning with its sustainability goals under the "For the Better 2030" framework.

Infineon Technologies – Recycling and more

The company emphasises closed-loop recycling, collaborating with specialised recycling partners to recover valuable materials and re-use them effectively. Infineon focuses on the 6R concept— Reduce, Reliability, Repair, Reuse, Refurbish, and Recycle — to guide its operations and product development.

Source: compilation based on own research

2.2.4. Case Study: Electronics as an enabler of renewable energy and smart grids

The role of electronics in the large-scale adoption of renewable energy is multifaceted and essential. From enabling efficient energy generation, conversion and storage to facilitating smart grid management and optimising energy consumption, electronics provide the technological foundation necessary for integrating renewable resources into the energy landscape. The integration of these technologies enhances the grid efficiency and supports renewable energy sources, contributing to substantial cost savings, job creation, and emissions reductions. As the demand for renewable energy grows, advancements in electronic technologies remains crucial for further improvement of efficiency, reliability, and sustainability in energy systems.

Electronic components are vital for the renewable energy transition, enabling efficient power management and integration of renewable sources. **Power electronics, including inverters and converters, play a key role** in this. For example, inverters convert DC from solar panels to AC for feed-in into the grid and use, while converters manage electricity flow, with types like DC-DC for voltage adjustment and AC-DC for electric vehicle charging. Advancements in semiconductor and materials technology, such as using Silicon Carbide (SiC) and Gallium Nitride (GaN) in power electronics, are enhancing the efficiency of these components. Together, they ensure effective conversion, storage, and utilisation of energy from renewable sources like solar and wind, significantly supporting sustainability and the growth of renewable energy.[41]

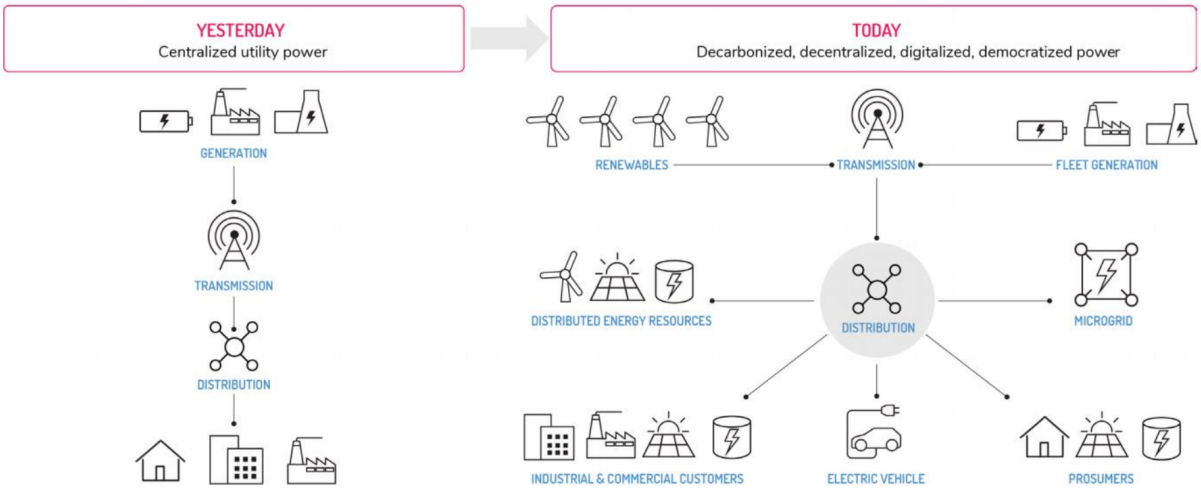
Overall, advanced semiconductors, such as those based on SiC and GaN, play a pivotal role in meeting growing power demands making the integration of renewable energy sources more feasible. These materials also contribute to improved energy storage solutions, optimising battery management systems for better load balancing. With the

[41]https://www.power-and-beyond.com/the-role-of-power-electronics-in-renewable-energy-and-sustainability-a-aa557714d10512a85f04edccc1799e32/

increasing importance of e-mobility, environmentally friendly battery technology has become essential.[42] Overall, advanced electronics are fundamental to integrating renewable energy into efficient power management systems, driving the transition to a sustainable energy future. According to the market intelligence Yole, the power electronic device market is set to grow significantly, with projections reaching EUR 31 bn ($36.5 bn) by 2029, driven by the adoption of wide-bandgap semiconductors like SiC and GaN. Renewable energy applications, including stationary battery energy storage systems and high-voltage transmission, are creating significant demand for advanced converters and higher voltage ratings[43].

When using wind power as a specific example, **advancements in power semiconductor and power electronics technologies have driven a substantial transformation in wind power systems and helped reduce energy costs**, as the growing capacity of wind turbines correlates with a significant rise in the use of power electronics. Moreover, the efficiency of energy conversion is critical for large-scale wind power generation. As indicated by the increasing utilisation of power electronics in individual wind turbine systems, efficient energy conversion contributes directly to lowering the cost of energy. The transformation of wind power systems, evidenced by the shift from 50 kW in the 1980s to those approaching 15 MW nowadays, highlights the economic benefits of power electronics in improving efficiency and facilitating compliance with grid requirements, ultimately making renewable energy competitive in the energy market[44].

Figure 19: Electronics: invisible enabler of Smart Grid



Electronics and semiconductor technologies are pivotal in transforming the energy landscape into a decentralised, decarbonised, digitalised, and democratised power system. **Advanced electronics are essential enablers of smart grids**, facilitating the transition to renewable energy and delivering significant economic benefits. As was described above, they are essential for the integration of distributed energy resources like solar panels and wind turbines, enabling efficient energy management through smart grids and microgrids. Digital technologies, powered by advanced electronics, enable real-time monitoring and data analytics, allowing for informed decision-making in energy consumption. Miniaturisation technologies and increased power density are crucial for applications like data centres and 5G networks, making high-voltage DC deployment essential for managing rising power loads. Compact DC/DC converters are vital for efficient energy management,

[42] https://www.eetimes.eu/abbs-radjassamy-on-the-drivers-and-enablers-of-next-gen-power-systems/
[43] Yole (2024): Power Converters for xEV, Green Energy and Industrial 2024. Yole Intelligence.
[44] Blaabjerg, Frede; Yang, Yongheng; Kim, Katherine A.; Rodriguez, Jose (2023): Power Electronics Technology for Large-Scale Renewable Energy Generation. In: *Proc. IEEE* 111 (4), S. 335–355. DOI: 10.1109/JPROC.2023.3253165.

while reliable electronics combined with IoT analytics enhance grid performance monitoring. Stakeholders - including consumers, industries, and government - increasingly use renewable energy, supported by smart-grid interconnectivity.

Being the major enabling technology, supply chain challenges – as was demonstrated in recent years, particularly regarding the availability of microelectronic components – have a direct impact on the digitalisation of energy grids, halting progress in integrating renewable energy sources into the existing systems. As a result, the production of energy-efficient technologies has also been adversely affected.

Expert recommendations for strengthening EU's green electronics transitions.

Experts stress that firms, which adopt a long-term focus on sustainability are better equipped to enhance their competitive advantage, positioning themselves as market leaders by fulfilling both current and future regulatory and market demands. Continued investment in research, collaboration, and education will be essential to fully realise the potential of green technologies and ensure the industry's viability in an increasingly eco-conscious marketplace. In particular, the following efforts are required:

- Policymakers should develop a **mixed strategy that combines extensive market instruments with targeted technology programmes**. This can include financial incentives, grants, and subsidies tailored to support the adoption of green technologies. Procurement policies that prioritise green technologies, along with regulations that encourage, or mandate sustainability practices can be part of such strategy.
- **Increased funding for R&D should be allocated specifically to green technologies** within the electronics sector. This includes further fostering partnerships between industry and academia to drive innovation and develop sustainable solutions.
- **Strengthen production capacities for green electronics technologies manufactured in the EU**. Europe should aim to implement strategic measures similar to the USA's Inflation Reduction Act, while enforcing new product standards.
- **Commitment to long-term planning investments** supported by establishing clear regulations and schemes that favour long-term investments to guide companies in their investment decisions. To promote this, policymakers should establish clear regulatory frameworks and supportive schemes. Financial incentives, such as tax breaks and accelerated depreciation for green technology investments, can further encourage commitment.
- To enable SMEs and other companies to conduct life cycle assessments (LCAs) and implement sustainable practices based on informed decisions, the establishment of a centralised EU-wide database would be essential. This database should provide reliable, standardised sustainability data from suppliers, addressing the current data accessibility challenges faced by SMEs. Additionally, training programmes should be offered to help SMEs effectively utilise the database.
- Creating **skills development programmes for Green Electronics**: Establish EU-wide training programmes to upskill the workforce in areas like sustainable design, green manufacturing, and circular economy practices. These programmes should be supported by both industry and academic institutions and funded through public-private partnerships.
- **Encourage and financially support collaboration between key European players**: Develop further IPCEI-type cross-border cooperations where electronics companies, universities, and research institutions can collaborate on green technology projects. These hubs can act as centres of excellence for sustainable electronics and foster partnerships to address common challenges.

- **Support actions should build upon existing initiatives like the European Green Digital Coalition**[45] as a source of ecosystem specific information as well as a foundation of future, more substantial networks of collaboration.
- **Encouraging collaboration across the EU and globally**: Develop cross-border cooperations where electronics companies, universities, and research institutions can collaborate on green technology projects. These hubs can act as centres of excellence for sustainable electronics and foster partnerships to address common challenges.
- **Support Startups and SMEs in Green Electronics**: Startups represent a critical driving force behind sustainability innovations in the electronics industrial ecosystem. Further effective support of these businesses through grants, seed funding, and low-interest loans to minimise risks as wells as establishing mentorship programmes is therefore of paramount importance to promote green innovations in electronics. Additional simplifying regulations will ease the process of starting and operating a business, reducing bureaucratic hurdles. To effectively support startups and SMEs in green electronics while strengthening their competitiveness, policymakers should focus on better aligning the objectives of Digital and Green Transitions.
- **Simplify access to finance for companies working on green electronics**, especially for smaller players. In the context of the European Investment Bank (EIB) recent commitment to deepen Europe's venture capital markets grants for SMEs developing sustainable electronics should receive specific attention.
- **Europe should support its electronics companies in international collaborations** conducive to developing more sustainable solutions. In particular, such outreach should be directed at EFTA nations, the UK as well as global value partners.

While experts emphasise the availability of supportive regulations and policies in the EU, there are some related challenges and issues that need to be effectively addressed:

- **Consistency in policies is essentia**l to avoid contradictions that can undermine efforts toward a Green Transition. Currently, the European regulatory framework for the electronics ecosystem is not free of contradictions. To attenuate that situation, clearer communication and coordination between different regulatory bodies should be established.
- **Regulations should be clear and actionable for all companies**, regardless of size. As the progress in Green Transition requires substantive investment, they should remain stable over time. Future regulation should reflect an understanding of the different challenges faced by both large and small companies, emphasising the need for tailored regulations that support sustainability at different levels of the industrial ecosystem.
- **Industry-wide green labels on products** based on coherent standards and requirements can help enhance the marketability of European sustainable products while promoting better environmental practices.
- Many companies face additional administrative efforts as a result of the variety of well-intentioned sustainability requirements. Especially SMEs face considerable challenges, which has the potential to hinder the speedy adoption of green technologies. **Simplifying regulations and reducing bureaucratic obstacles** can create a more conducive environment for the Green Transition. For example, the establishment of a centralised EU-wide database could provide SMEs with essential information to comply with sustainability regulations. This also

[45] https://digital-strategy.ec.europa.eu/en/policies/european-green-digital-coalition

necessitates creating of user-friendly tools and resources that can help SMEs navigate through the complexities of compliance with sustainability regulations.

- The very real danger of diminishing self-sufficiency in critical industries in Europe needs to be weighed against the benefits of environmental regulations. Currently, these regulations restrict European firms' access to certain chemicals and hinder their ability to carry out specific production processes. The reduction of available materials and processes in Europe has led to an in-creasing dependency on foreign suppliers.

In the view of many experts, however, impactful mechanisms to incentivise a more substantial Green Transition of the electronic eco-system are still missing:

- It is essential to establish a **comprehensive CO_2 pricing mechanism**: Develop implement a robust and transparent CO_2 pricing system that applies to all sectors and implement it gradually, yet with determination.
- A **CO_2 taxation system and a CO_2 pricing mechanism for imported goods** could shift market behaviour more effectively towards sustainable products and practices than current practices, which often include loopholes and exceptions that undermine genuine efforts toward sustainability.
- Measures need to be put in place to **avoid greenwashing by mandating companies to provide verifiable proof of their green practices** rather than, for example, allowing them to rely on certificates that may not reflect actual practices and by developing standardised metrics for reporting environmental impact.

Overall, experts advocate for a cohesive strategy to address sustainability challenges within the electronics industry, emphasising the need for a unified regulatory framework that encompasses all aspects of the ecosystem — from raw material extraction to end-of-life disposal. Strong enforcement mechanisms with standardised guidelines for reporting frameworks are essential to ensure compliance and track progress. A singular approach to regulations, free from exceptions and loopholes, would foster a fair competitive environment, ensuring that all companies adhere to the same standards.

2.3. The impact of the industrial ecosystem on the environment

What progress has the industry made in taking action for the environment?

- Semiconductor and electronics manufacturing is energy- and resource-intensive, with Scope 1–3 emissions showing that most emissions stem from process gases and energy use, especially in front-end fabrication processes. Supply chain complexity presents challenges at every stage, from raw material extraction to manufacturing and transportation. Responsible sourcing of raw materials mitigates environmental and supply risks, enhancing supply chain resilience. Leading firms prioritise sustainable sourcing, but smaller companies often face resource constraints. Furthermore, rapid technological advancement leads to increased electronic waste. Depending on different assessment scopes, electronics currently contribute to 1.5-4% of global greenhouse gas emissions. Significant emissions occur during the use phase and disposal of electronic devices. - For many electronic devices, especially PCs **and smartphones, the use phase now accounts for the largest share of lifecycle emissions**, though manufacturing emissions are rising with increased chip complexity. Global **e-waste generation is rising rapidly, with low recycling rates and high plastic content posing environmental and economic challenges**, despite EU leadership in regulatory frameworks and collection targets.

- At the same time, **advanced electronics play a critical enabling role in energy efficiency, grid optimisation, and electrification**, supporting the broader green transition across sectors. Innovative electronics contribute significantly to reducing emissions and improving device efficiency.

- The EU's ambition to boost semiconductor production could result in a noticeable increase in emissions, potentially outpacing other polluting industries, underscoring the urgent need for climate-conscious expansion strategies. For the industry to achieve sustainability, there must be a long-term commitment to proactive sustainability planning, investment in R&D, and transitioning to eco-friendly materials and processes.

As awareness of climate change grows, the electronics industry faces increasing pressure to reduce its carbon footprint and enhance transparency regarding its environmental impact. The EU's intention to increase the production capacity under the Chips Act and to strengthen the semiconductor supply chain in Europe will inevitably increase the industry's environmental footprint. Addressing this issue will require immediate and effective strategies to mitigate the environmental impact of the industry[46]. Consequently, it is essential that the expansion of manufacturing capacity goes hand-in-hand with prioritising climate and environmental considerations.

As the electronics industry strives for sustainability, it faces several key challenges:

- **Supply Chain Complexity**: The global electronics supply chain is complex, with sustainability challenges at every stage — from raw material extraction to manufacturing and transportation. Ensuring transparency on environmental and ethical practices is difficult, requiring comprehensive data across the supply chain.
- **Energy Consumption**: The energy-intensive nature of electronics production and usage poses ongoing challenges. Optimising carbon footprints by reducing greenhouse gas emissions throughout the lifecycle demands innovative solutions and a commitment to renewable energy.
- **E-Waste Dilemma**: Rapid technological advancements lead to increased electronic waste (e-waste) as devices become obsolete. Proper disposal, recycling, and responsible end-of-life management are crucial to mitigate the environmental impact and conserve valuable resources.

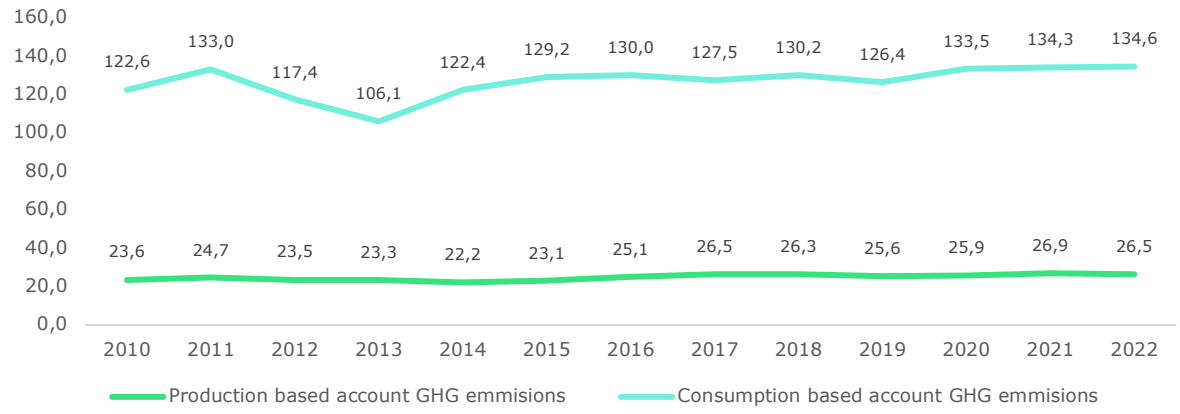
This section presents the indicators that capture the environmental impact of the electronics industrial ecosystem based on Exiobase data. It also reflects on the progress made over time and to what extent targets are being reached.

The global electronics industrial ecosystem due to its resource intensity has a large impact on the environment. In 2022, the industrial ecosystems accounted for 1.5% of greenhouse gas (GHG) emissions among all 14 industrial ecosystems, based on a production-based assessment using Exiobase data. However, other research[47] using a different methodology estimates that this sector could be responsible for up to 4% of global GHG emissions. The variation between these figures stems from differences in the scope of analysis, with the higher estimate potentially considering broader indirect emissions, such as those from digital infrastructure and content consumption.

The Figure below reveals that from 2010 to 2022, the CO2 emissions of the electronics industrial ecosystem slightly increased (from 23.6.6M to 26.5M tones following production-based accounting). Following the consumption-based accounting, Exiobase data shows that the CO2 emissions also slightly increased (from 122.6M to 134.6M).

[46] Hess, Julia (2024): Chip Production's Ecological Footprint: Mapping Climate and Environmental Impact. Stiftung Neue Verantwortung. Online available at https://www.interface-eu.org/publications/chip-productions-ecological-footprint
[47] https://www.idtechex.com/de/research-article/decarbonizing-the-electronics-industry-with-sustainable-manufacturing/28329

Figure 20: CO2 emissions of the electronics industrial ecosystem in EU27 (consumption and production accounts)



Source: Technopolis Group based on Exiobase data, calculations in 2024

(Micro)electronics and semiconductor manufacturing being the backbone of electronics industry – belong to the most energy-intensive and resource demanding branches. High energy consumption is particularly linked to the complex processes, cleanroom requirements, and sophisticated equipment involved during various processing steps. According to Chen et al. 2013[48], large semiconductor fabs consume around 100 megawatt-hours of power each hour, which is significantly more than other sectors like automotive or food processing require. Advanced fabs[49] need even more energy, so that the energy cost share in a semiconductor manufacturing company can very between 10% - 30% of total operating costs, depending on the region and operational efficiency. Therefore, along with the need of policy compliance to reduce the environmental footprint, companies have a strong economic incentive to reduce their energy costs and are increasingly focusing on energy efficiency. This includes investments in renewable energy sources and energy-efficient manufacturing technologies.

Scope 1 – 3 GHG emissions (without use phase)

The emissions profile of electronics companies is categorised according to the Greenhouse Gas (GHG) Protocol Corporate Standard 2004[50] – a widely adopted corporate accounting and reporting standard. According to this standard, energy consumption and emissions are broken down into:

- **Scope 1 emissions**: direct emissions from organisations' owned or controlled assets and processes.
- **Scope 2 emissions**: indirect emissions from the purchased energy.
- **Scope 3 emissions**: all indirect emissions, upstream[51] or downstream[52], (not included in Scope 2) across the company's value chain.

[48] Chen, Steve; Gautam, Apoorv; Weig, Florian (2013): Bringing energy efficiency to the fab. McKinsey. Online available at https://www.mckinsey.com/~/media/mckinsey/dotcom/client_service/operations/pdfs/bringing_fabenergyefficiency.ashx

[49] Other costs of advanced fabs are also significantly higher, primarily driven by high equipment and operating costs.

[50] WBCSD and WRI (2004): The Greenhouse Gas Protocol. A Corporate Accounting and Reporting Standard. Revised version. World Business Council for Sustainable Development. Online available at https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf.

[51] Upstream emissions include emissions from mining and refining of metals (e.g., copper, gold), other sourced materials and inputs, emissions from supplier operations and energy use for manufacturing components, transportation, and waste disposal.

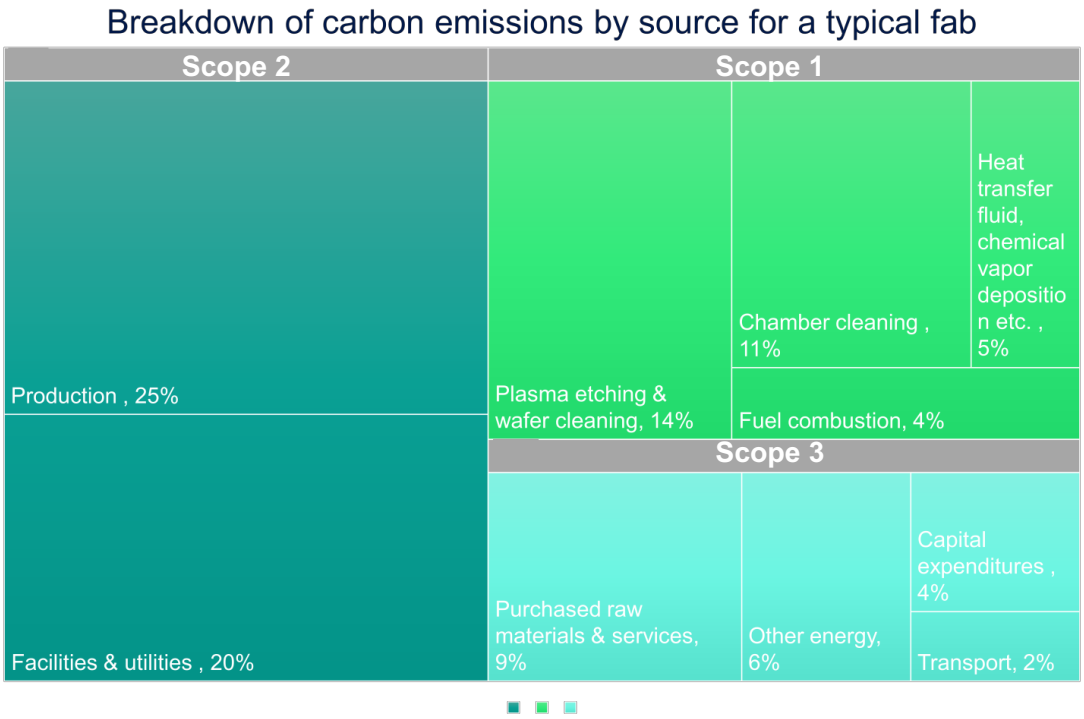
[52] These consist of emissions from the use of electronic devices (e.g., energy consumption during use phase), product end-of-life disposal or recycling, and downstream transportation and distribution, including emissions from packaging materials.

Based on the estimations provided by Greenpeace (2023)[53] and McKinsey (2022)[54], GHG emissions in a typical semiconductor fab are distributed as follows:

- **34% as Scope 1 emissions**, which are direct emissions from on-site sources. GHGRP 2022[55] and Greenpeace 2023 have found that approx. 12.2% from these emissions are attributable to stationary combustion, while the largest share - 87.8% stems from process gases[56] and heat transfer fluids. Nearly half of them arise from plasma etching and wafer cleaning, while 37.7% come from chamber cleaning.
- **Scope 2 emissions account for 45%** largely representing indirect emissions related to the energy consumption of production processes. McKinsey (2022) estimated that 56% of these emissions originate from production, and 44% from facilities and utilities. Cleanroom operations and specific processes like lithography, deposition, and etching are major energy consumers.
- **Approx. 21% of GHG emissions fall under Scope 3**, which includes indirect emissions from the broader supply chain and product lifecycle. They consist of emissions from materials during the extraction and processing, other energy-related emissions consumed throughout the supply chain, transportation emissions stemming from the logistics, as well as emissions related to equipment and machinery used.

[53] Greenpeace (2023): Invisible Emissions. A forecast of tech supply chain emissions and electricity consumption by 2030. Greenpeace. Online verfügbar unter https://www.greenpeace.org/static/planet4-eastasia-stateless/2023/04/620390b7-greenpeace_energy_consumption_report.pdf.
[54] McKinsey (2022): Keeping the semiconductor industry on the path to net zero. Online available at https://www.mckinsey.com/industries/semiconductors/our-insights/keeping-the-semiconductor-industry-on-the-path-to-net-zero.
[55] GHGRP (2022): GHGRP Overview Report.
[56] Process gases, used in semiconductor manufacturing processes, such as perfluorocarbons, hydrofluorocarbons, nitrogen trifluoride, and nitrous oxide, have high global warming potentials and contribute significantly to GHG emissions.

Figure 21: Overview of GHG emissions of a typical fab



Source: Greenpeace (2023); McKinsey (2022). Note: Emissions do not include emissions associated with the use phase of final devices.

The overall ecological footprint of semiconductor manufacturing can be divided into three major phases: **design, front-end manufacturing, and back-end manufacturing**. While chip design itself has a negligible environmental impact, it significantly influences resource use and energy consumption of end devices. **Front-End Manufacturing (wafer fabrication) has the most significant environmental impact**, characterised by high energy consumption, extensive use of hazardous chemicals, and substantial water usage. The front-end manufacturing phase is closely linked to the upstream environmental footprint, which includes the production and provision of inputs (raw materials, chemicals, and energy sources). Back-End Manufacturing (assembly, test and packaging) contributes to waste generation and also involves hazardous materials, though at a lower intensity. Back-end manufacturing relates to the downstream environmental footprint, which encompasses the impacts of the final product throughout its lifecycle. However, distinguishing between these footprints is challenging due to diverse business models in the semiconductor industry, with most companies reporting only aggregated data in their corporate social responsibility reports.

As feature sizes of chips shrink and complexity increases, energy needs for their fabrication increase. The dilemma for the industry is that, while it has an essential role in delivering technologies that reduce emissions across various sectors and promote the digital and green transition, the industry's own carbon footprint continues to expand. This is exemplified by IMEC's assessment of the environmental footprint of CMOS logic technologies (see: Boakes et al. (2023)[57], which reveals that **transitioning from less advanced nodes (28 nm) to more advanced nodes (14 nm) results in a 33% increase in energy and water consumption, with the carbon footprint is rising**

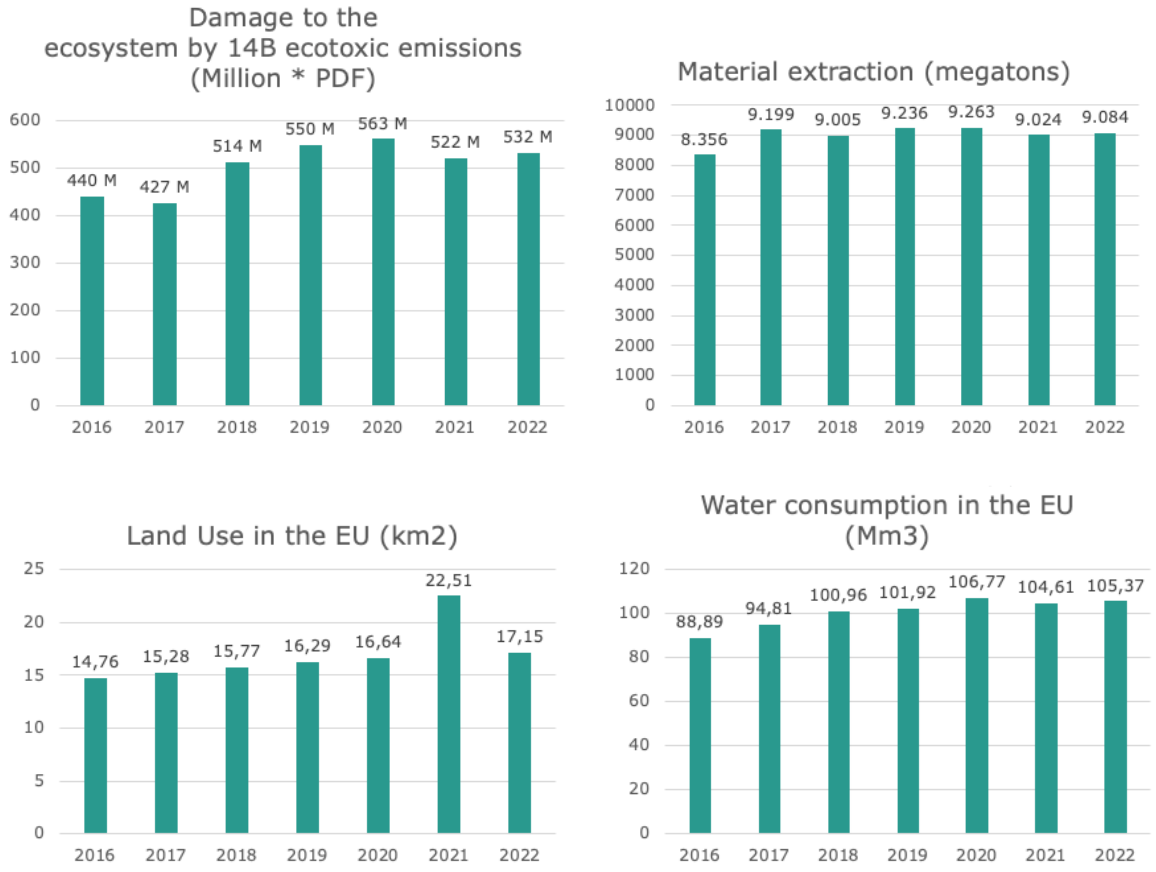
[57] Boakes, L.; Garcia Bardon, M.; Schellekens, V.; Liu, I-Y.; Vanhouche, B.; Mirabelli, G. et al. (2023): Cradle-to-gate Life Cycle Assessment of CMOS Logic Technologies. In: 2023 International Electron Devices Meeting (IEDM). 9-13 Dec. 2023. 2023 International Electron Devices Meeting (IEDM). San Francisco, CA, USA, 12/9/2023 - 12/13/2023. Institute of Electrical and Electronics Engineers. Piscataway, NJ: IEEE, S. 1–4.

from approx. 75 kg CO_2 equivalent per wafer to around 100 kg CO_2.[58] Furthermore, the study highlights higher emissions from fluorinated gases used in etching and deposition processes, underscoring the substantial environmental impact of advanced semiconductor manufacturing. This underlines the need for improved sustainability practices and technological innovations to mitigate environmental impacts[57].

According to data from Exiobase (see Figures below), the following observations can be made for the EU27:

- Damage to the ecosystem decreased from 2020 to 2021, however, it has risen again in 2022.
- Material extraction has remained relatively constant over the years, although a slight increase was recorded from 9024 megatons in 2021 to 9084 megatons in 2022.
- Land use in the EU saw a notable peak in 2021, reaching 22.51 thousand km², but decreased again in 2022 to 17.15 thousand km².
- Water consumption has increased by nearly 20 million cubic meters (Mm³) since 2016. However, since 2020, water usage has remained relatively steady, with a slight increase from 104.61 Mm³ in 2021 to 105.37 Mm³ in 2022.

Figure 22: Damage to the environment, material extraction, land use and water consumption by the electronics industrial ecosystem (production account)



Source: Technopolis Group based on Exiobase data, calculations in 2024, Notes: In the case of damage to the ecosystem, pdf means the proportion of species in an ecosystem that are potentially lost (or negatively affected) due to a specific environmental pressure or impact — such as pollution, land use, or climate change.

[58] In annex you can find a short overview explaining the progression and characteristics of semiconductor technology nodes N28, N2, and A14, highlighting advancements and interplay in transistor size, energy efficiency, applications, manufacturing complexity, and environmental impact.

Challenges and opportunities to reduce Scope 3 emissions

Especially the reduction of the Scope 3 emissions is a complex challenge. It appears that many smaller suppliers are not aware of their carbon footprint or understand emissions within different scopes. Additionally, small and medium-sized enterprises (SMEs) often have limited resources and expertise in sustainability and decarbonisation. Consequently, many suppliers are primarily concerned about the high costs associated with implementing climate protection measures, complicating major market players, such as, Schneider Electric's efforts, to drive emissions reductions throughout its supply chain.[59]

Identifying and comparing scope 3 emissions pose challenges due to a lack of standardised reporting categories and calculation methods. Key issues include[60]:

- **Data Comparison**: Companies may use internal calculations or external data, complicating accuracy assessments.
- **Inconsistent Reporting**: Changes in the number of reported categories each year hinder long-term comparisons for the same company.
- **Higher Reported Emissions**: Companies that report extensively on scope 3 likely show higher total GHG emissions.

The semiconductor industry needs entirely new compounds to reduce its ecological footprint. Current reliance on fluorinated gases and wet chemicals contributes significantly to greenhouse gas (GHG) emissions and environmental contamination. Finding alternatives is complicated by process dependencies and regulatory uncertainties, necessitating proactive sustainability planning and R&D investment. The industry has committed itself to phase out harmful substances, such as PFAS (per- and polyfluoroalkyl substances), driven by regulatory pressures and the growing awareness of environmental and health impacts. There are efforts to develop new and more sustainable substances. For example, Merck – a major provider of chemicals to semiconductor companies – is developing green solvent formulations that are safer and more efficient.[61] As semiconductor technology advances, the need for innovative and eco-friendly chemicals continues to grow, driving research towards less toxic alternatives. However, according to experts' estimations, transitioning from harmful compounds will take time to over 15 years, highlighting the need for a long-term commitment to sustainability[62].

The scope 3 emissions include water consumption. A large semiconductor fab can consume up to 38 million liters of water per day. Most of this water is used for producing ultrapure water (UPW) through multi-stage treatments, with on-site water reuse and recycling being common practices. However, water management strategies are influenced by external factors. For instance, fabs in high water-scarcity regions like Taiwan prioritise water reuse and recycling, achieving rates of around 80%. In contrast, fabs in Europe typically have much lower recycling rates.[63]

Adopting circular economy strategies can reduce Scope 3 emissions by extending product lifecycles and minimising waste. Strategies like product redesign for longevity, modularity, and upgradability decrease resource demand. Recycling and reuse initiatives recover materials, reducing the need for virgin resources and related emissions.

Responsible and sustainable sourcing of raw materials

[59] https://theclimatechoice.com/en/schneider-electrics-path-to-scope-3-decarbonization/
[60] Hess, Julia (2024): Chip Production's Ecological Footprint: Mapping Climate and Environmental Impact. Stiftung Neue Verantwortung. Online available at https://www.interface-eu.org/publications/chip-productions-ecological-footprint.
[61] https://www.merckgroup.com/en/research/science-space/envisioning-tomorrow/scarcity-of-resources/green-solvents.html
[62] Hess (2024).
[63] Hess (2024).

Responsible and sustainable sourcing of raw materials in the EU electronics sector encompasses environmental, legal, social, and ethical dimensions. Manufacturers in the EU increasingly prioritise materials with a lower ecological footprint and collaborate with suppliers who employ environmentally friendly extraction methods. Overall, the movement toward responsible and sustainable sourcing is driven by a combination of regulatory pressures[64], consumer expectations, and the and the desire to enhance brand reputation through ethical practices. Additionally, companies recognise the importance of supply chain resilience, as sustainable sourcing mitigates risks related to material availability and supply disruptions. Investor pressure is also significant, with a growing emphasis on environmental, social, and governance (ESG) criteria for public investments or subsidies.

The extraction of critical raw materials for semiconductor manufacturing, such as palladium, copper, cobalt, and rare earth elements, significantly impacts the environment through habitat destruction, water and soil pollution, and high energy consumption. Silicon, the primary material for wafers, has a lower ecological footprint compared to compound semiconductors like gallium arsenide or germanium. The latter have higher melting temperatures, leading to increased energy consumption and greenhouse gas (GHG) emissions during production. As a result, the ecological footprint of compound semiconductors is generally higher than that of silicon wafers[65].

While leading electronics firms in the EU prioritise sustainable sourcing and actively invest in responsible supply chains, such as Philips. Nokia, Bosch, Infineon, Ericsson, Siemens, Schneider Electric and others, the level of commitment can vary significantly among smaller companies, which may face resource constraints. Many companies adopt sustainability initiatives and set targets to reduce their environmental impact, often collaborating with R&D organisation, industrial partners and governmental organisations to address common challenges. The adoption of certifications, such as ISO 14001 for environmental management, further demonstrates their commitment to sustainable practices.

Use phase of electronics

A significant share of emissions from electronic devices – which all depend on semiconductor components and their building blocks - is produced during the use phase and subsequent disposal. According to Semi et al. (2023)[66], semiconductor devices produced in 2021 have a lifetime CO_2 equivalent footprint of 500 MT, with 16% coming from the supply chain, 21% – from manufacturing, and 63% – from device use. However, the total emissions differ significantly by technology platform, with higher-performance hardware leading to a greater carbon footprint during manufacturing while reducing the emissions during the use phase due to higher energy efficiency of electronic devices. This trade-off is particularly evident in personal computing devices, as demonstrated by the study Gupta et al. (2020)[67]. The study analyses the environmental impact of personal computing devices and reveals that hardware manufacturing, particularly for companies like Apple, accounts for over 74% of their total carbon emissions, totalling 25 million metric tons of CO_2 in 2019. Integrated circuits contribute approx. 33% of Apple's emissions, surpassing the 19% from product use. As energy efficiency of end devices improves, the share of emissions from manufacturing has risen — from 40% to 75% for newer iPhones.

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[64] Key role play EU legal regulations, such as **RoHS Directive** (Restriction of Hazardous Substances), **WEEE Directive** (Waste Electrical and Electronic Equipment), **REACH Regulation** (Registration, Evaluation, Authorisation, and Restriction of Chemicals)**, EU Conflict Minerals Regulation**: (Aims to prevent the trade of minerals that finance armed conflicts, mandating companies to ensure that their supply chains are free from conflict minerals), **Corporate Sustainability Reporting Directive (CSRD).**

[65] Hess, Julia (2024): Chip Production's Ecological Footprint: Mapping Climate and Environmental Impact. Stiftung Neue Verantwortung. Online available at https://www.interface-eu.org/publications/chip-productions-ecological-footprint.

[66] Semi; Semiconductor Climate Consortium; Boston Consulting Group (2023): Transparency, Ambition, and Collaboration: Advancing the Climate Agenda of the Semiconductor Value Chain. Online available at https://discover.semi.org/rs/320-QBB-055/images/Transparency-Ambition-and-Collaboration-BCG-SEMI-SCC-20230919.pdf.

[67] Gupta, Udit; Kim, Young Geun; Lee, Sylvia; Tse, Jordan; Lee, Hsien-Hsin S.; Wei, Gu-Yeon et al. (2020): Chasing Carbon: The Elusive Environmental Footprint of Computing. Online available at http://arxiv.org/pdf/2011.02839v1.

Emissions during the use phase also largely depend on the type of device and additionally rely on consumers' behaviour, which underscores the critical need for a holistic approach to addressing the environmental impact of electronics. As one expert rightly pointed out, an effective transition of the electronics industrial ecosystem should be conceptualised as 'Triple Transition', which includes Green, Digital, and Social Transitions. For the Green Transition to be successful, society must shift its mindset towards more sustainable consumer practices, emphasising energy efficiency, waste reduction, and the adoption of renewable resources. This involves enhancing skills and educating the workforce and consumers on sustainability.

Electronics as major enabler of the Green Transition

The enabling role of electronics for the Green Transition is widely recognised. Innovative electronics play a fundamental role in driving the Green Transition across various sectors. By enabling efficient energy management and conversion, advanced electronic technologies support the integration of renewable energy sources, enhance the grid stability, and facilitate the electrification of transportation. This transformative impact is essential for creating sustainable energy systems and reducing dependence on fossil fuels, ultimately contributing to a greener and more sustainable future.

Advanced electronics technology is pivotal in enhancing the energy efficiency of end devices, leading to substantial savings in energy consumption, costs, and emissions. Just to name a few examples of how innovations in electronics contribute to higher efficiency while saving energy and emissions:

- **Dynamic Voltage and Frequency Scaling (DVFS)**, which allows processors to adjust their power usage based on current workload demands enabling reduction of power consumption during operation, resulting in energy savings of up to 30%.[68]
- **Low-power electronics components** are crucial for improving energy efficiency and reducing energy consumption across a wide range of devices and applications. Components, such as low-power transistors, microcontrollers, and sensors, are designed to operate with low energy while maintaining high performance.
- **Evolution towards chiplet technologies and 3D integration** offers a path to more sustainable manufacturing by addressing cost and performance challenges[69], particularly in automotive applications.
- **Advanced electronics technology facilitates better integration and management of renewable energy sources**. Key components, such as power converters, inverters, and smart sensors enable the seamless conversion of energy generated from sources like solar and wind into forms that can be effectively utilised within the existing electrical infrastructure.
- **Micro-energy harvesting systems**, generating microwatts to milliwatts, are increasingly viable due to advancements in microelectronics. Micro-energy harvesting uses sources like photovoltaic cells, piezoelectric converters, and thermoelectric generators, combined with energy-harvesting power management ICs (EH PMICs) and capacitors. This technology can extend battery life or eliminate the need for batteries in IoT applications, thereby reducing environmental impact and enabling functionality in remote locations. Micro-energy harvesting is therefore **Organic light-emitting diodes (OLEDs)**, increasingly used in smartphones and TVs. They contribute significantly to environmental sustainability through several key factors. Firstly, they are energy-efficient, consuming less power compared to traditional lighting technologies. Additionally, OLEDs generally use fewer toxic

[68] https://www.sciencedirect.com/topics/computer-science/dynamic-voltage-and-frequency-scaling

[69] 3D integration offers a path to more sustainable manufacturing by stacking multiple layers of electronic components vertically, which reduces the overall footprint and material usage. This method enhances performance by shortening interconnect distances between components, leading to faster data transfer and lower power consumption. Additionally, it addresses cost challenges by allowing for more efficient use of materials and reducing the need for larger, more complex single-die solutions.

materials than other technologies. Many of the materials used in OLEDs can be recycled, promoting a circular economy and reducing landfill waste crucial for reducing the environmental footprint of billions of IoT devices.[70]

Through these and other innovations, end devices can significantly reduce energy consumption and contribute to the green transition, which not only lowers operating costs, but also minimises greenhouse gas emissions. Their collective contribution leads to a more sustainable future, demonstrating the vital role of advanced electronics in addressing environmental challenges. One significant advancement enabled by innovative electronic technologies is the development of more energy efficient processors.

Box 5: Estimating the Impact of Enhanced Server Processor Efficiency on Global Data Center Energy Consumption and CO_2 Emissions

According to the International Energy Agency, in 2022 data centres accounted for 1 – 1.5% of global electricity consumption worldwide. In 2022, data centres in the EU consumed an estimated 45–65 TWh of electricity, representing 1.8 – 2.6% of the region's total electricity consumption. Germany, France, the Netherlands, and Ireland accounted for nearly two-thirds of this energy use. [71] In the future, data centres could consume far more of global electricity, accelerated due to the AI demand for electricity.

Server processor power consumption significantly impacts overall data center efficiency. According to Vertiv, saving 1 W at the server processor results in a total data center savings of 2.84 W due to reductions in other components.[72] We can estimate the general savings in power consumption of data centers worldwide based on the given data.

Assumptions

1) According to Vertiv[73], saving 1 W at the server processor results in a total data center savings of 2.84 W due to reductions in other components.
2) Global data center electricity consumption was between 240 TWh and 340 TWh in 2022 (IEA)[74]
3) The global average carbon intensity of electricity generation is approx. 450 grams of CO2 per kWh. This figure is widely used in energy and environmental assessments.
4) Processor Power Consumption Proportion assumed to be 30% of total data center power consumption (e.g. see (Lin et al. 2021; Dayarathna et al. 2016)).[75]
5) Assumed 10% reduction in processor power consumption (as an example of improvement).

Estimations

- Processor power consumption (30 % of total consumption):
  - Lower bound: 240 TWh x 0.3 = 72 TWh
  - Upper bound: 340 TWh x 0.3 = 102 TWh
- Potential energy savings
  - Lower bound: 72 TWh x 0.1 x 2.84 = 20.45 TWh
  - Upper bound: 102 TWh x 0.1 x 2.84 = 30 TWh
- Equivalent CO2 reduction (450 g per saved kWh):
  - Lower bound: 20.45 x 109 kWh x 0.45 kg CO2/kWh = 9.2 x 109 kg CO2
  - Upper bound: 30 x 109 kWh x 0.45 kg CO2/kWh = 13.5 x 109 kg CO2

[70] https://www.eetimes.eu/whats-happening-in-the-world-of-micro-energy-harvesting/
[71] Kamiya, G. and Bertoldi, P. (2024): Energy Consumption in Data Centres and Broadband Communication Networks in the EU. European Commission, Joint Research Centre. Online available at: https://data.europa.eu/doi/10.2760/706491, JRC135926.
[72] Vertiv (2023): Prioritizing your data center IT efficiency actions. Onlive available at: https://www.vertiv.com/4a162d/globalassets/documents/white-papers/energy-logic-2-wp-en-na-sl-24632_51018_2.pdf
[73] Vertiv (2023).
[74] https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks
[75] E.g. see Lin et al. (2021): A hardware-aware CPU power measurement based on the power-exponent function model for cloud servers. In Information Sciences 547, DOI: 10.1016/j.ins.2020.09.033; Dayarathna et al. (2016): Data Center Energy Consumption Modeling: A Survey. IEEE Commun. Surv. Tutorials (IEEE Communications Surveys & Tutorials) DOI: 10.1109/COMST.2015.2481183.

- Potential CO2 savings
  - Lower bound: 9.2 million tons CO2
  - Upper bound: 13.5 million tons CO2

Conclusion:

Improving server processor efficiency by 10% in data centers results in a substantial reduction in energy consumption and, subsequently, in a significant decrease in greenhouse gas emissions, ranging from 9.20 to 13.5 million metric tons in CO2 emissions savings annually.

E-Waste

E-waste increasingly contributes to the environmental crisis, exacerbated by the growing number of electrical and electronic devices. According to the Global E-Waste Monitor, approx. 62 million tons of e-waste were generated globally in 2022, with less than a quarter being recycled. This waste contains valuable materials, such as gold, silver, and copper, worth about $62 billion annually, but also hazardous substances that pose health risks. Small equipment, such as toys and appliances, represents a third of global e-waste (20 million tons), yet has a low recycling rate of 12%. Small IT and telecommunication devices, totalling 5 million tons, have a recycling rate of 22%[76].

EU countries are global leaders in the e-waste management, guided by the WEEE and RoHS Directives.[77] The WEEE Directive sets collection and recovery criteria. The collection rates have improved in many EU countries in recent years. Despite this progress, the majority of EU member states have not met their legally binding collection targets. Challenges remain, including low consumer engagement in returning e-waste and the need for better data on collection rates. The EU has addressed part of this challenge through proactive measures like the Single Charger Initiative, which mandates USB-C standardisation for small electronics. This policy alone helps to eliminate significant amount of e-waste from redundant chargers while extending product lifespans through interoperability and simplifying recycling processes through reduced connector variety.

However, the ongoing Green and Digital transitions are expected to further boost e-waste. For example, e-waste from photovoltaic panels is expected to rise from 0.6 million tons in 2022 to 2.4 million tons by 2030[76]

Figure 23: E-waste collection rates (in percent) in the EU-countries



Source: the global e-waste statistics: https://globalewaste.org/country-sheets/

One further challenge is a large plastic content in the electronic end products. E-waste contains various types of plastics that complicate recycling processes and contribute to the environmental pollution. Addressing this issue requires improved recycling technologies

[76] Baldé, Cornelis P. et al. (2024): Global E-waste Monitor. International Telecommunication Union (ITU) and United Nations Institute for Training and Research (UNITAR).
[77] WEEE (Waste Electrical and Electronic Equipment) and RoHS (Restriction of Hazardous Substances) are two important directives in the European Union aimed at promoting environmental sustainability and public health.

and recycling infrastructure, better product design, and effective policies to manage e-waste sustainably.

Overall, recycling e-waste still presents significant challenges. The lack of standardised recycling policies across regions leads to inefficiencies and increased environmental risks. Moreover, the presence of toxic substances such as lead and mercury necessitates specialised handling. Economic factors further complicate the situation, as the initial high costs represent a serious obstacle from the perspective of many companies.

3. Digital transition

What is the progress of industrial efforts towards digitalisation?

The electronics industrial ecosystem is undergoing rapid digital transformation, driven by the adoption of advanced technologies, the innovation of startups, and increasing venture capital investments.

Technology development

The share of digital patents in the electronics ecosystem showed growth until 2020, followed by a decline. Patent activity is highest in advanced manufacturing, robotics, and micro/nanoelectronics. However, Europe lags behind the U.S. and China in key digital areas like cloud computing, blockchain, cybersecurity, and quantum technologies. Competitive pressure is growing as the U.S. and China invest heavily in cutting-edge technologies.

Uptake of Digital Technologies

In 2024, 33% of companies in the electronics industrial ecosystem reported having a digital transformation strategy, according to the EMI survey. Cloud computing and IoT emerged as the most widely adopted technologies, highlighting their importance in connectivity, data storage, and streamlining operations. Adoption rates for AI, big data, and robotics also grew significantly compared to 2023, indicating a broader shift toward automation, intelligent process management, and data-driven decision-making. These technologies are being leveraged not only to enhance efficiency and productivity but also to foster innovation and improve adaptability in response to changing market demands.

Electronics Digital Startups

Startups play a critical role in driving digital innovation in the electronics sector, with a focus on IoT, AI, and smart manufacturing. Startups are particularly active in areas such as high-performance computing, specialised chip design, and advanced packaging techniques. Despite their high innovation potential, startups face challenges such as high failure rates - around 90% - due to difficulties in securing funding or achieving product-market fit. However, increased R&D investment and partnerships with established companies are helping mitigate these risks.

Venture Capital Investments

Venture capital investment in young electronics companies has steadily grown. It primarily targets AI, big data, IoT, advanced manufacturing, robotics, and AR/VR. Among these, AI, big data, and AR/VR have attracted the highest levels of funding, underscoring their transformative potential in the industry.

3.1. Industrial efforts in transitioning towards digital electronics

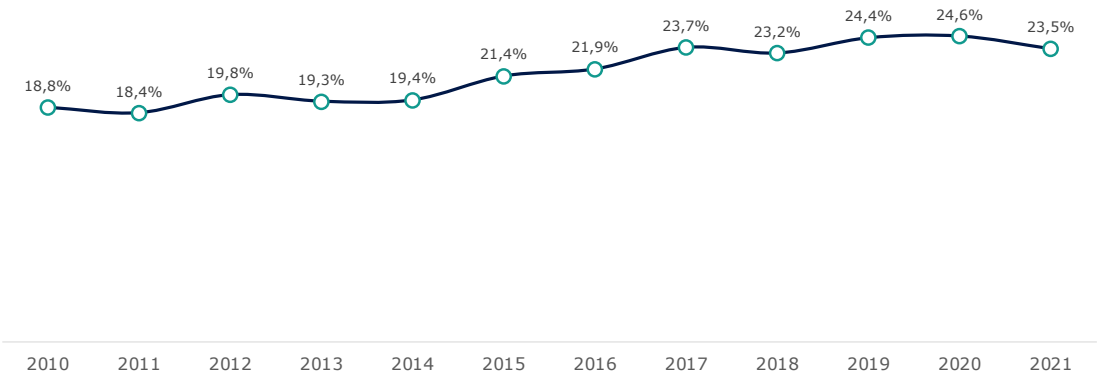
This section analyses the uptake of digital technologies, the evolution of electronics tech startups and private investments in the electronics industrial ecosystem. It explores the progress electronics have made in integrating digital technologies to enhance operations, streamline processes, and improve customer experiences over the past years.

3.1.1. Technology generation

Zooming into the technologies related to the green transition as presented in the methodological framework, Figure below presents the share of all digital patents in all patents filed for the electronics industrial ecosystem among the EU27 industrial ecosystems

overall. The Figure shows a rising trend until 2020, followed by a decline in the most recent year available.

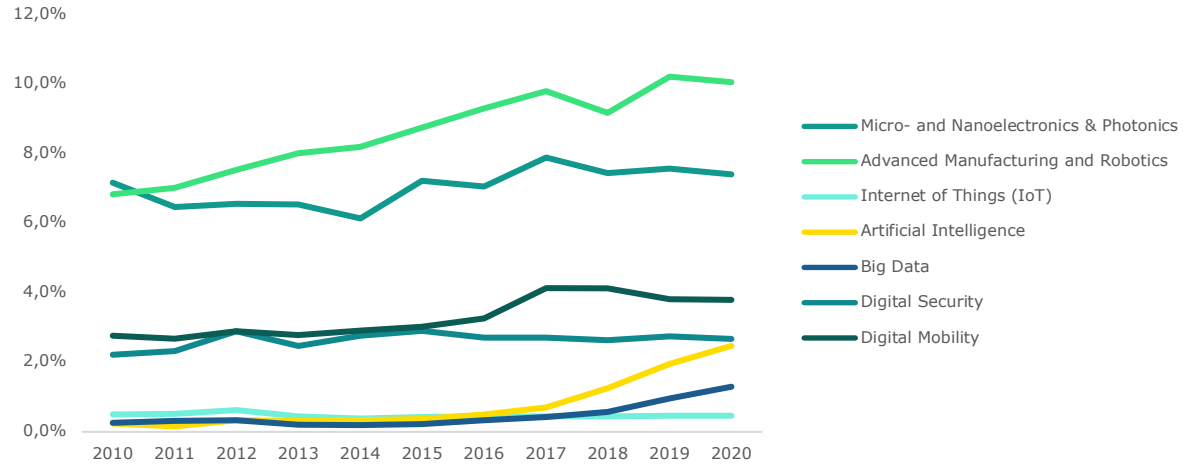
Figure 24: Share of patent applications in digital transition technologies in all patent applications filed for the electronics industrial ecosystem



Source: Fraunhofer based on PATSTAT

At the level of digital technologies, advanced manufacturing and robotics, micro and nanoelectronics & photonics are the two leading technologies for the electronics industrial ecosystem in terms of patenting activities (see Figure 25 below).

Figure 25: Share of respective digital transition technologies in overall patent application filed in the electronics industrial ecosystem



Source: Fraunhofer based on PATSTAT

Despite the existing R&D potential, Europe lags behind the leading regions, first and foremost USA, in many advanced and future relevant digital technology areas, highly relevant for the electronics industrial ecosystem. Combining data in publications, patents, and venture capital funding, McKinsey & Company (2023)[78] has found that Europe particularly falls behind in Cloud and Edge Computing, Block Chain, Cybersecurity, and future programming (Software 2.0; no-code and low-code programming). This also

[78] McKinsey & Company (2023): Reinventing our economy from within. How Europe's start-up ecosystems can learn from each other to ignite and scale up entrepreneurship. Online available at https://www.mckinsey.de/~/media/mckinsey/locations/europe%20and%20middle%20east/deutschland/publikationen/2023-09-22%20reinventing%20our%20economy%20from%20within/reinventing%20our%20economy%20from%20within.pdf.

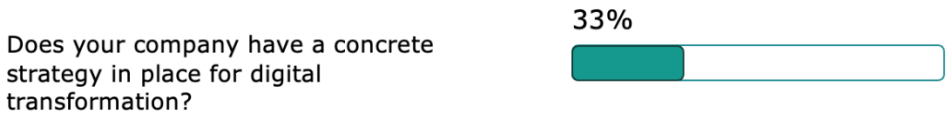
applies, albeit to a lesser extent, to the applied AI - especially in the field of robotic process automation, optimised decision making, and Next Generation Computing (e.g. Quantum computing, neuromorphic computing).

Competitive pressure on leading-edge technologies in electronics is mounting with China and USA having particularly ambitious goals and allocating considerable resources.

3.1.2. Uptake of digital technologies

According to the results of the EMI survey 2024, **33% of companies have a concrete strategy in place for digital transformation in the electronics industrial ecosystem**.

Figure 26: Share of companies in the electronics industrial ecosystem that has adopted a strategy for the digital transformation



Source: EMI Survey 2024

In 2024, cloud computing and the Internet of Things (IoT) emerged as the most widely adopted digital technologies within the industrial ecosystem, reflecting their essential role in enabling connectivity, data storage, and streamlined operations. However, adoption rates for artificial intelligence, big data, and robotics also saw significant growth compared to 2023, indicating a broader shift towards automation, data-driven decision-making, and intelligent process management across industries. As shown in the Figure below, this trend suggests an accelerating commitment to leveraging advanced technologies not only for efficiency and productivity gains but also for fostering innovation and improving adaptability in response to evolving market demands.

Figure 27: Share of companies in the electronics industrial ecosystem that has adopted digital technologies

Digital Technologies	***Share of adoption (2023)***	***Share of adoption (2024)***
Cloud	**20.6%**	**47.49%**
Internet of Things	6.62%	**41.34%**
Artificial Intelligence	6%	**37.43%**
Big Data	11.47%	**22.91%**
Robotics	10.66%	**15.64%**
Edge Computing		**14.53%**
Augmented and Virtual Reality	9%	9.5%
Blockchain	5%	5.59%

Source: EMI Enterprise survey 2024

3.1.3. Electronics digital startups

The digital transition of the electronics industrial ecosystem is significantly supported by innovative tech startups, which contribute to the development of new technologies and

solutions. In Europe as well as in other major regions, there has been a notable surge in startups focused on areas such as IoT, AI, and smart manufacturing[79].

There are several subdomains within the electronics ecosystem offering significant business opportunities for startups. Particularly the rapidly evolving AI sector is driving demand for innovative electronics that supports high-performance computing and advanced AI applications. This opens up huge opportunities for startups to capitalise on the need for new architectures, specialised chips, and advanced packaging techniques. The demand for advanced chips is expected to remain strong in the future, providing a stable environment for businesses focused on innovative solutions. There is an increasing flow of venture capital into AI hardware technologies, creating opportunities for European startups to attract funding and scale their innovations, supported by the EU and government initiatives. Moreover, with a focus on sustainability and advanced manufacturing techniques, European startups can differentiate themselves by developing eco-friendly solutions[80].

Further important focus for startup activities represents the development of AI solutions and tools that support chip design and fault prediction, as well as to improve electronics manufacturing processes, optimising for efficiency and cost.

However, the startup landscape is also characterised by high mortality rates. Statistics indicate that around 90% of startups fail[81], often due to challenges like finding a product-market fit or securing adequate funding. Nonetheless, the high innovation dynamics of the industrial ecosystem is evident through increased investment in research and development, as well as growing collaboration between startups and established companies. These partnerships facilitate knowledge transfer and accelerate the adoption of new technologies.

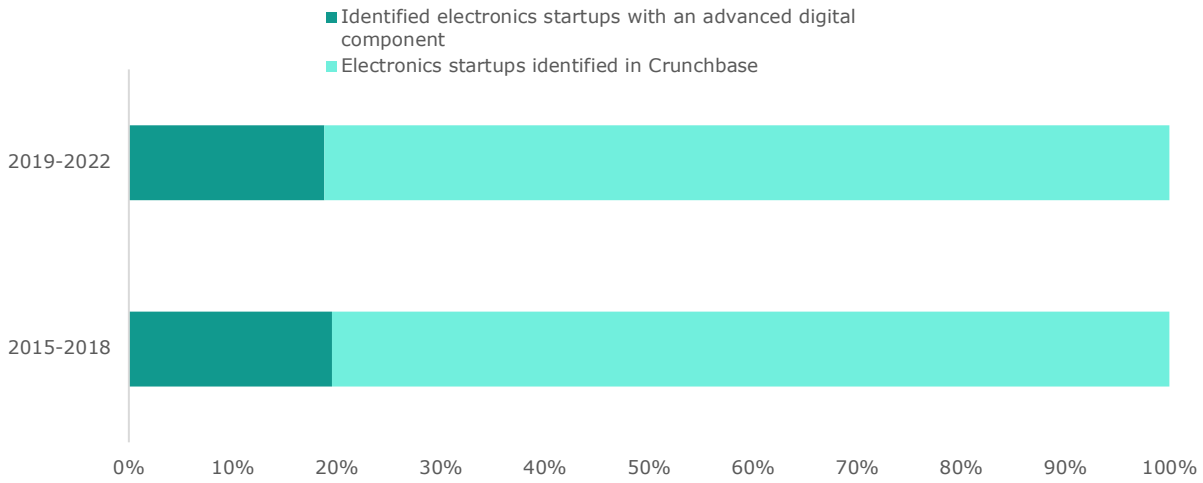
The presence of numerous startups reflects a vibrant ecosystem eager to innovate.

Industry experts underscored the significant potential of startups in driving green innovations. Startups often have inherent motivations to create products that align with sustainability goals and play a vital role in pushing the boundaries of sustainable practices in the electronics sector. Many EU countries have initiated startup programmes that support new ventures focusing on green innovation, including in the field of electronics. Experts mentioned a strong interest from startups in participating in funding opportunities aimed at promoting sustainable practices. This enthusiasm suggests a vibrant ecosystem for innovation in sustainability.

Capturing information on startup creation allows to give insight into the novel technologies being generated related to the electronics industrial ecosystem. The analysis of startup generation has been compiled through a database encompassing Crunchbase and Net Zero Insights[82] data. A total of **842 electronic startups were identified** between **2015 and 2023**. Among these startups we find companies focusing on different sub areas such as embedded systems, semiconductors, microelectronics or chip technology.

[79] EE Times (2023a): 100 Silicon Startups worth watching in 2023. 23 edition. Hg. v. EE Times. Online available at https://www.eetimes.com/product/silicon-100-startups-worth-watching-in-2023-final/.
[80] Pelé, Anne-Françoise (2024): VC Discernment Is Critical Amid AI Surge, Celesta's Top Exec Says. In: *EE Times Europe*. Online available at https://www.eetimes.eu/wp-content/uploads/2024/08/24_09_EETE_all-FINALx3.pdf.
[81] https://www.investopedia.com/articles/personal-finance/040915/how-many-startups-fail-and-why.asp
[82] www.crunchbase.com and https://netzeroinsights.com/

Figure 28: Share of startups created, by advanced digital technology between 2015 and 2023.



Source: Technopolis Group, Crunchbase and Net Zero Insights

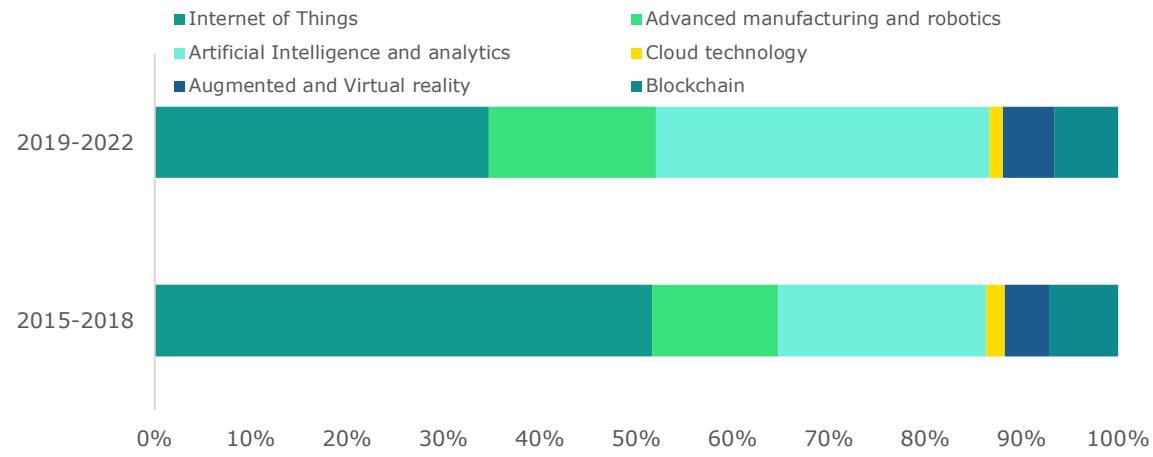
Zooming into those startups which develop or use advanced digital technologies[83], the Figure below reveals which technologies are leading in startup creation.

Notably, startups developing innovative solutions that either incorporate or enable the spread of internet of things devices stand out with a total of 105 startups identified over the indicated period. An example of is seen on the startup Qualinx[84] is a pioneer in Digital-RF technology developing an ultra-low power single chip solution for GNSS and IoT radios

Artificial Intelligence is the second most important technology for startup creation, with total of 59 startups founded related to this technology in the period of 2015 to 2022. The applications of AI are wide, for instance

There are also startups focused on advanced manufacturing and robotics, which mostly entail IC Packaging solutions and 3D printing for printed electronics. For instance, Danish startup ATLANT 3D Nanosystems[85] is a develops atomic layer 3D printing technology.

Figure 29: Technologies underpinning digital tech startups in the electronics industrial ecosystem



Source: Technopolis Group, Crunchbase and Net Zero Insights

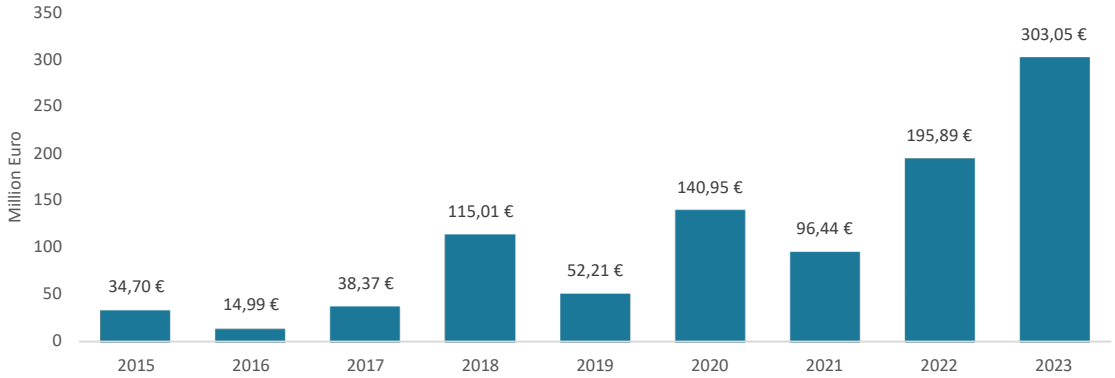
[83] As defined in the scope of this project.
[84] http://www.qualinx.io
[85] https://www.atlant3d.com/

3.1.4. Industrial investments into digital technologies

A Crunchbase and Net Zero Insights data set gives insight into venture capital investments in young companies (age 10 years or less) and gives insights into the amount of investment related to the broader digital transition as well as the investments related to the technologies underpinning the transition, notably AI and big data, internet of things, advanced manufacturing technology and robotics as well as Augmented reality. The data set contains some data lags for the most recent year (2023) however venture capital data remain sufficiently well reported so general trends can be observed.

As depicted in the Figure below, venture capital investments in the electronics industrial ecosystem related to the digital transition has been steadily increasing and peaked in 2023 at EUR 303.05 million.

Figure 30: Venture capital investment into electronics young companies – digital transition



Source: Technopolis Group based on Crunchbase and NO

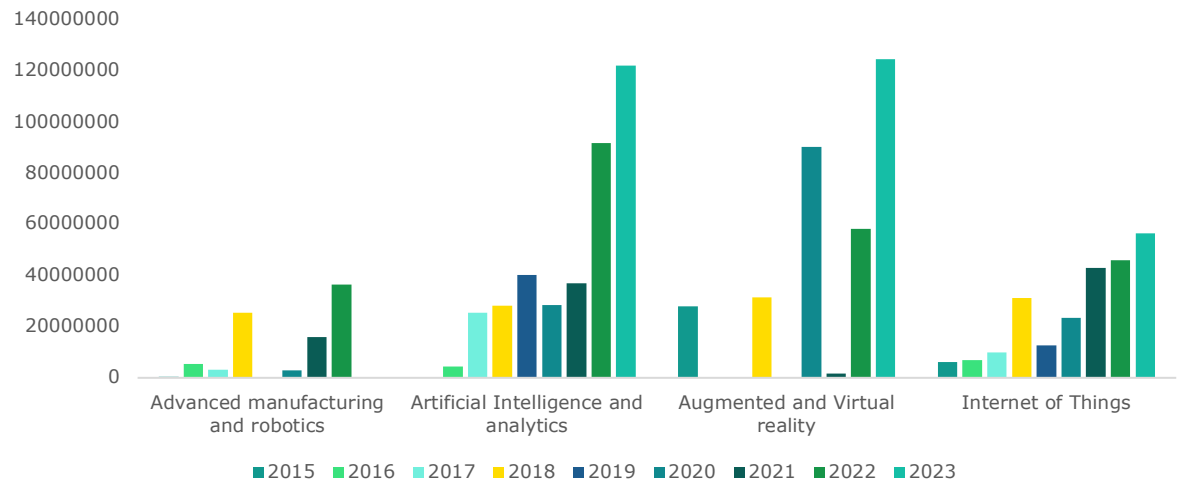
Investments in technologies such as AI and big data, the Internet of Things, advanced manufacturing and robotics, and augmented and virtual reality (AR/VR) show varying trends. Notably, AI, big data, and augmented reality are seeing the highest levels of investment. In the case the companies identified under AR/VR, are focused on developing solutions that enable AR/VR such as the company Swave Photonics[86] is a fabless semiconductor firm that designs and manufactures holographic chips based on diffractive photonics technology.

Details on the funding rounds of select young companies based on the Crunchbase and Net Zero Insights dataset include:

- **Axelera AI,** Axelera AI is an artificial intelligence chip manufacturer founded in 2021 that creates AI hardware and software platforms for edge computing. It has raised EUR 63 M in four funding rounds, with the last one being a Series A in 2023.
- **Swave Photonics,** Swave Photonics is a fabless semiconductor firm founded in 2022 that designs and manufactures holographic chips based on diffractive photonics technology. Since 2022, it has raised a total of EUR 10M during two seed funding rounds.
- **Blickfeld**, Blickfeld offers LiDAR sensors and detection software for autonomous mobility and IoT applications. It has raised EUR 56 M in four funding rounds, with the last one being a Series A in 2021 of EUR 26M.

[86] https://swave.io

Figure 31: Venture capital investments into digital transition of electronics startups



Source: Technopolis Group based on Crunchbase and NO

3.2. Framework conditions – assessment of the broader ecosystem supporting the digital transition

To what extent do framework conditions such as public financing and skills support the digital transition?

Public Investments

The EU has supported the electronics sector through Horizon funding programmes. Horizon 2020 allocated EUR 300 m, with 42.2% tied to digital transition projects. Horizon Europe has so far allocated EUR 163.57 m, with 70.4% supporting digital transition projects.

Skills for Digital Transition

Under the EU Chips Act, the European Union will require additional 350,000 professionals to meet its goals. 32% of recent job ads in the electronics sector require advanced digital skills, such as AI, big data, and IoT expertise. However, the electronics industry faces significant challenges in attracting talent. While STEM graduates are critical for driving Digital Transition, most EU countries are experiencing shortages, with many reporting low or declining numbers of graduates in electronics, automation, and ICT-related fields. A significant talent gap persists, particularly in project management, and machine learning roles. To tackle the skill gap challenge, companies are investing in on-the-job training, forging partnerships with universities, and supporting initiatives such as the Semiconductor Talent Incubation Programme in Saxony.

Otherwise, the adoption of advanced technologies, particularly AI, is reshaping jobs in the electronics industry by driving automation, upskilling, and productivity while risking role displacement. Large firms lead AI adoption due to greater resources, while smaller ones face challenges like limited infrastructure. Generative AI enhances design, material selection, and quality inspection. Technologies like IoT and AI also enable energy efficiency, waste reduction, and sustainable innovation, aligning Digital and Green transitions.

Advanced Electronics as an Enabler for AI

Advanced electronics and semiconductors play a critical role in driving AI advancements. They provide the computational power, scalability, and energy efficiency essential for modern AI applications. However, escalating performance requirements have

significantly increased energy consumption. While Energy-Efficient Performance (EEP) improvements have reduced energy consumption per workload, greater computational demands often offset these gains. Innovations including energy-efficient hardware, AI-centric architectures, photonic technologies, low-power design, and advances in semiconductor fabrication are key in mitigating the environmental impact of AI advancement, emphasising the indispensable role of advanced electronics in enabling scalable, sustainable AI development.

Impact of Digital Technologies on Industrial Competitiveness

The electronics industry in the EU is experiencing high demand for advanced digital technologies like AI, robotics, and IoT, which are enhancing efficiency, precision, and competitiveness. Digital tools also enhance supply chain resilience by improving monitoring, early risk detection, and disruption management. According to the EMI survey, the adoption of advanced digital technologies has led to productivity gains of 10-15%, with AI and robotics delivering the most significant improvements.

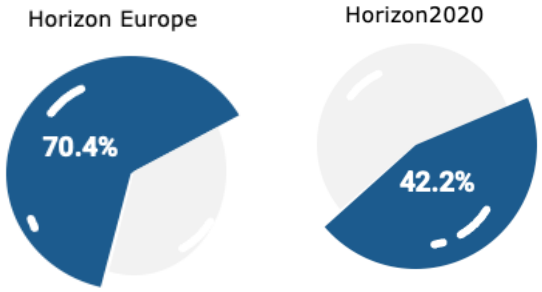
3.2.1. Public investments

Framework Programmes

In this study, funding data from the EU Framework Programmes for Research and Innovation, in particular from Horizon 2020 (2014-2020) and Horizon Europe (2021-2027) was further analysed. As outlined in the previous Chapter, the analysis of the CORDIS data demonstrates that the Horizon 2020 programme provided EUR 300 m to projects classified under the electronics industrial ecosystem[87]. Horizon Europe thus far[88] provided EUR 163.57 m to those projects.

When zooming into the share of the electronic projects that contributed to digital transition, the results indicate that in Horizon 2020 **42.2% of the EU funding** contributed. In Horizon Europe **70.4% of the EU funding** to date contributes to electronics projects related to the digital transition.

Figure 32: Share of Horizon 2020 and Horizon Europe electronics projects that contribute to the digital transition



Source: Technopolis Group based on CORDIS

An example of an electronics project funded under Horizon 2020 supporting the digital transition is described in Box 6.

[87] The methodological report provides detailed insights into the methodology. In particular, the CORDIS data have been merged with the relevant EuroSciVoc data. EuroSciVoc connects projects to relevant scientific concepts, providing a deeper understanding of the research focus.
[88] Cut-off date is March 2024

Box 6: Horizon 2020 and Horizon Europe electronics projects supporting the digital transition

The project LastChance[89] is funded under **Horizon 2020** under the title "New super-energy efficient non-volatile memory manufacturing – bringing Europe back on the semiconductor track". The project started in 2014 and total cost amount EUR 99.39M.

The **LastChance** project aims to assess the global market and technical specifications for **non-volatile static random-access memory (nvSRAM)** products and systems, including a business plan for their introduction. **ANV**, a fabless company, will develop an innovative **SONOS-module** with 90nm technology for **System on Chip (SOC)** solutions, offering ultra-low power **nv memory cells**. This solution combines **Logic**, **DRAM**, and **FLASH** technologies in a cost-effective way, providing fast read/write operations and reducing energy demand by a factor of 10,000 compared to conventional Flash. The core markets include **industrial electronics**, **medical technology**, **sensor technology**, and **data storage**, with the potential for significant growth. The project will conduct a feasibility study to prioritise technical specifications, assess production robustness, and prove the economic viability of the new technology, including a business plan, pilot products, and an IP strategy.

The project Q-ONE is funded under Horizon Europe under the title "Quantum Optical Networks based on Exciton-polaritons". The project started in 2023 and the total cost go up to a total of EUR 67.3 M.

The **Q-ONE project** aims to address a key need in **Quantum Optics and Quantum Information** by developing a novel approach to generating, manipulating, and characterising quantum states of light using **quantum neural networks (QNN)** in integrated photonic devices. The project seeks to solve the problem of recognising quantum states, such as **Fock states** or **entangled pairs**, without requiring correlation measurements. The QNN platform will be **reversible**, allowing users to either characterise an input quantum state or generate an arbitrary quantum state from a classical input. By integrating **neuromorphic computing** and **exciton-polaritons**, Q-ONE targets the development of a new quantum device capable of recognising and generating quantum states without needing extreme interaction strengths. If successful, this project will lead to a fully reconfigurable and reversible **universal quantum platform**, advancing the field of **Quantum Technologies**.

Source: Cordis

3.2.2. Skills underpinning the digital transition

This section analyses the demand and supply side of the labour market in terms of skills relevant for the digital transition. The skills needed for electronics to succeed in the future will be different from those that were required for success in the past. As highlighted by a report of McKinsey, more than 50% of all activities in electronics can be automated with technology. Moreover, customer needs and preferences for how they interact with electronics are continually changing. They are seeking more personalised experiences, seamless interactions across multiple channels, and faster, more convenient services, which in turn requires new types of skills.

Supply of digital skills

In order to analyse trends in the skills of professionals working in the electronics industrial ecosystem, data from LinkedIn has been used. LinkedIn data provides a proxy about self-reported supply of digital skills including moderate and advanced digital skills.

Moderate digital skills have been defined following Cedefop[90] notably including "*five types of skills under the digital skills umbrella such as information processing (e.g. using a search engine and storing information and data); communication (including teleconferencing and application sharing); content creation (such as producing text and*

[89] European Commission (2024) FlAgship demonstration of industrial scale production of nutrient Resources from Mealworms to develop a bioeconomY New Generation. Retrieved from: https://cordis.europa.eu/project/id/837750/reporting

[90] https://www.cedefop.europa.eu/en/data-insights/digital-skills-challenges-and-opportunities

tables, and multimedia content); security (e.g. using a password and encrypting files); and, problem solving (e.g. finding IT assistance and using software tools to solve problems)". (The list of keywords that have been used to track LinkedIn is included in Appendix B).

Advanced digital skills have been defined as a specific group of digital skills in the context of the main digital technologies captured in this project notably in artificial intelligence, cloud computing, connectivity, robotics, Internet of Things, augmented and virtual reality and blockchain (the list of keywords that have been used and are possible to track with the algorithm of LinkedIn is included in Appendix B). LinkedIn data have to be interpreted in the light of its representativeness for electronics and across the EU. An analysis of representativeness is provided in Appendix B and in the related methodological report.

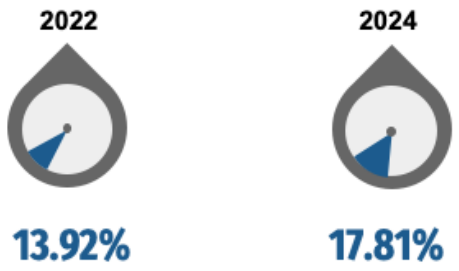
The results indicate that both the share of professionals with moderate and advanced digital skills -

In order to capture the number of professionals working in the sector, occupations related to the industrial ecosystem have been taken into account. A total of 1.32 million professionals were found to be related to the electronics and semiconductor industry on LinkedIn in April 2024. While this means that an estimated 51% of electronics professionals are active on the platform, it is to be noted that some jobs, functions and countries are better represented than others[91], such as in Germany and France. The data shows that about 52% of electronics professionals active on LinkedIn has a master's degree, and 30% a bachelor's degree, hence there is a bias towards professionals with higher educational level.

Digital skills have been identified as skills related to advanced skills such as artificial intelligence, big data, robotics, augmented and virtual reality, blockchain and advanced manufacturing. Notably, within the registered professionals on LinkedIn employed in the electronics industry, **17.81% indicated to have advanced digital skills such as the abovementioned, which is slightly higher than in 2022 where 13.92% were found to have these kind of skills (see Figure 33).** The highest number of professionals with advanced digital skills are concentrated in Germany, Ireland, France, The Netherlands and Romania.

Companies like ASML, STMicroelectronics, Infineon Technologies, Intel Corporation and Apple are among the top employers of professionals with advanced digital skills.

Figure 33: Share of professionals in electronics with skills relevant for the digital transition



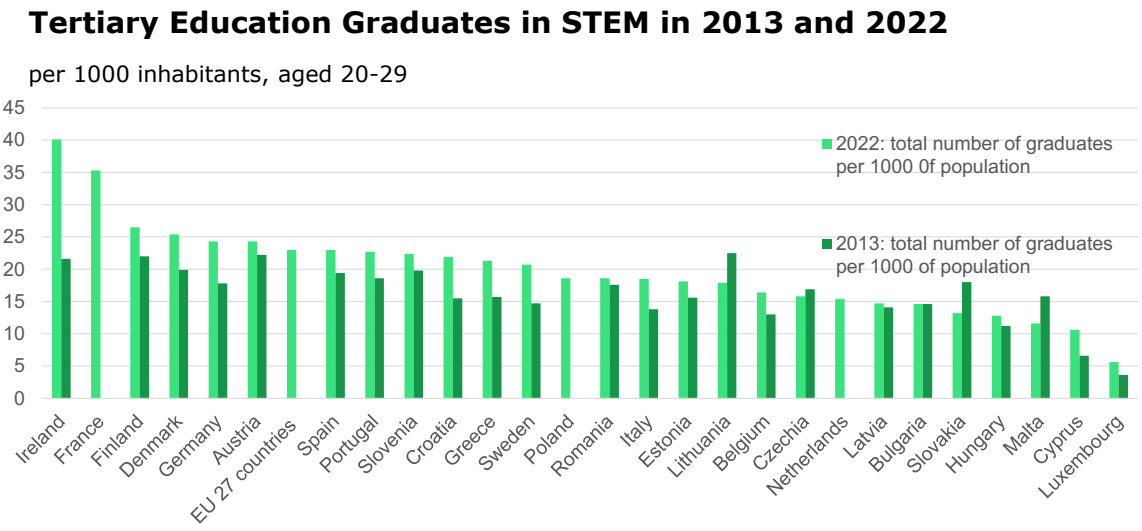
Source: Technopolis Group based on LinkedIn

This skilled workforce is essential for innovation, research, and development within the electronics industry, helping to meet the increasing needs for advanced technologies. The proportion of academic employees in the (micro)electronics sector is exceptionally high and is expected to increase further. This reflects the industry's reliance on highly educated professionals. The availability of a skilled workforce is a significant factor in location

[91] The methodological report outlines how the report corrects this representativeness bias, i.e., by comparing LinkedIn population to the active population and derive a corrective weighting.

decisions for companies within the electronics sector. Regions that can provide a robust talent pool are more likely to attract investment and foster growth.

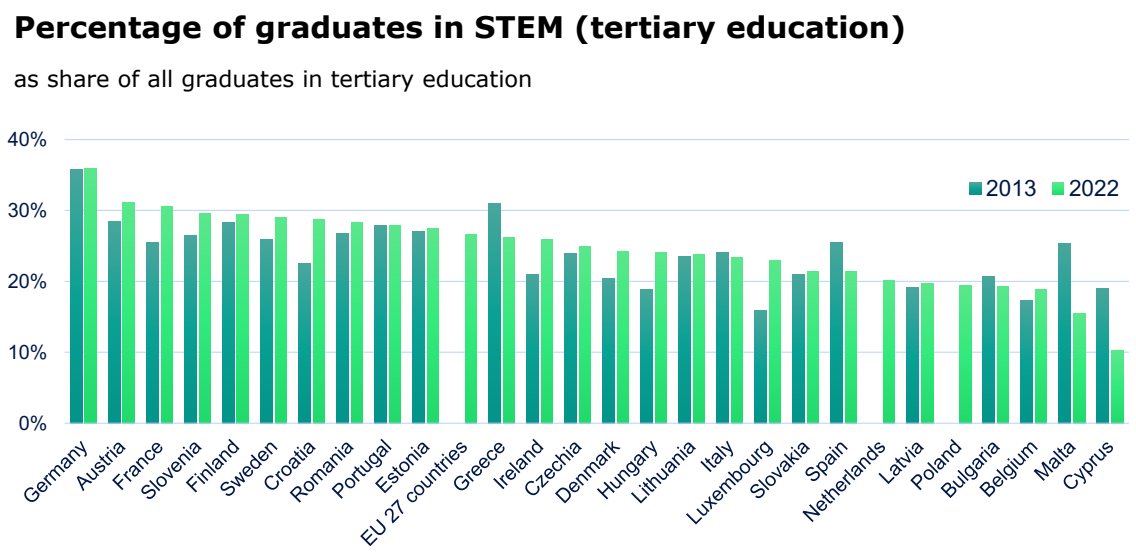
Figure 34: STEM graduates per 1 000 of population



Source: Eurostat. Note: Graduates in tertiary education, in science, math., computing, engineering, manufacturing, construction, Tertiary education (levels 5-8).

The supply of skilled STEM professionals (science, tech, engineering, or math) is critical for the electronics industry's growth and sustainability. Countries with higher intensity of STEM graduates per 1 000 of population, such as Ireland, France, Germany, Austria, Spain, Portugal, Slovenia, Croatia, Greece and Sweden are better positioned to support the evolving demands of the electronics market. The increasing number of tertiary graduates in STEM fields across various EU countries is vital for the electronics industry. However, the supply of STEM professionals does not increase to the extent required to meet the rising needs of industrial ecosystems, including electronics. The majority of EU-countries displays an insignificant increase in the share of STEM graduates. A range of EU countries including Romania, Italy, Estonia, Lithuania Czechia show a declining intensity of STEM graduates in their tertiary education systems while Latvia, Bulgaria, Slovakia, Hungary, Malta, Cyprus and Luxembourg have extremely low concentration of STEM graduates (see Figure below). This trend is concerning as it limits the available talent pool for industries reliant on technical skills.

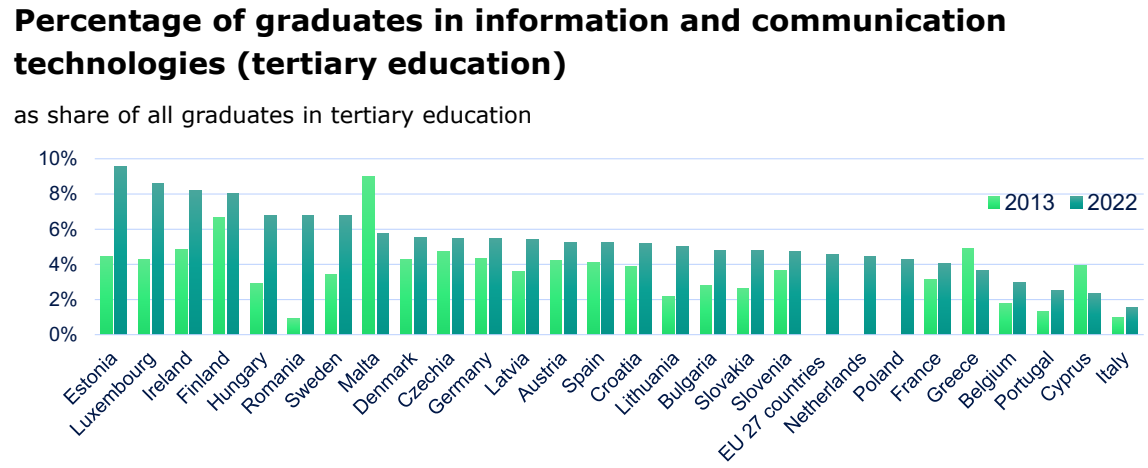
Figure 35: Share of tertiary education STEM graduates as share of all graduates in tertiary education



Source: Eurostat.

Further, the Eurostat data reveals that the majority of EU countries displays a low share – albeit with a somewhat growing trend – of graduates in the area of information and communication technologies (ICT), with significant disparities across the EU (Figure 36). ICT encompasses various disciplines crucial for electronics, including, Computer Science, Information Systems, Telecommunications, Cybersecurity and Data Science and Analytics.

Figure 36: Share of tertiary education graduates in Information and Communication technologies

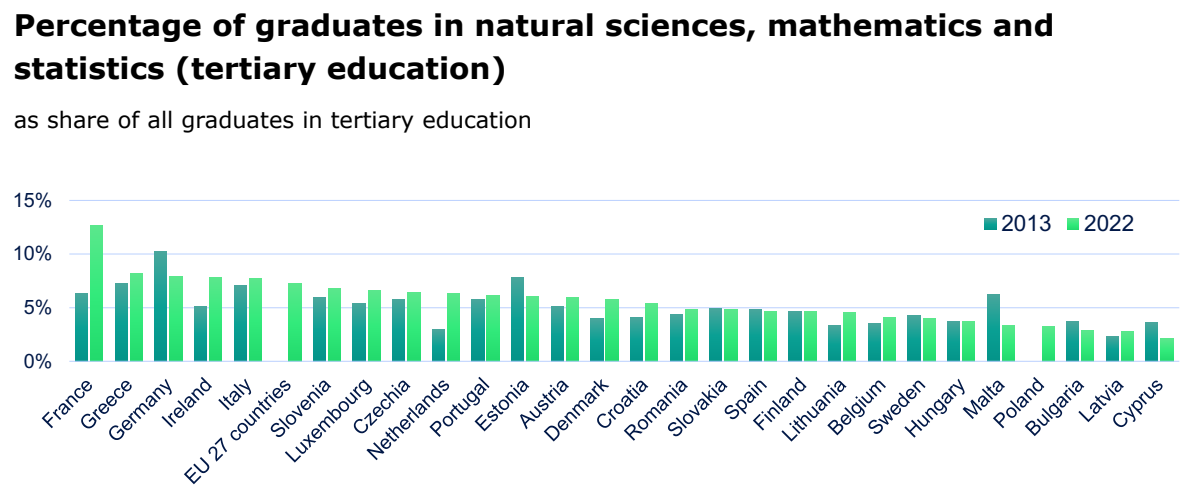


Source: Eurostat.

The share of graduates in Natural Sciences, Mathematics, and Statistics is crucial for fostering innovation and development in modern technologies, including electronics. This field encompasses a range of disciplines that provide foundational knowledge and skills fundamental in various fields, including data analysis, algorithm development, and modelling. Advancements in electronics often require interdisciplinary skills where natural sciences, mathematics and statistics play a pivotal role. While the overall share of graduates in Natural Sciences, Mathematics, and Statistics is relatively low for all EU countries (only France has managed to achieve a significant increase since 2013) (Figure 37), the Eurostat data exhibits disparities across EU countries. Especially Belgium, Sweden, Hungary, Malta, Poland, Bulgaria, Latvia and Cyprus are struggling with extremely low shares, impacting their ability to innovate in key technological sectors, including

electronics. Specialised talent is essential, requiring master's or doctoral degrees in fields like natural language processing, data engineering, or machine learning. Additionally, professionals must possess complex skills such as domain understanding, and the ability to translate specific contexts. Moreover, many top professionals in the EU migrate to leading AI hubs, resulting in talent shortages in AI[92].

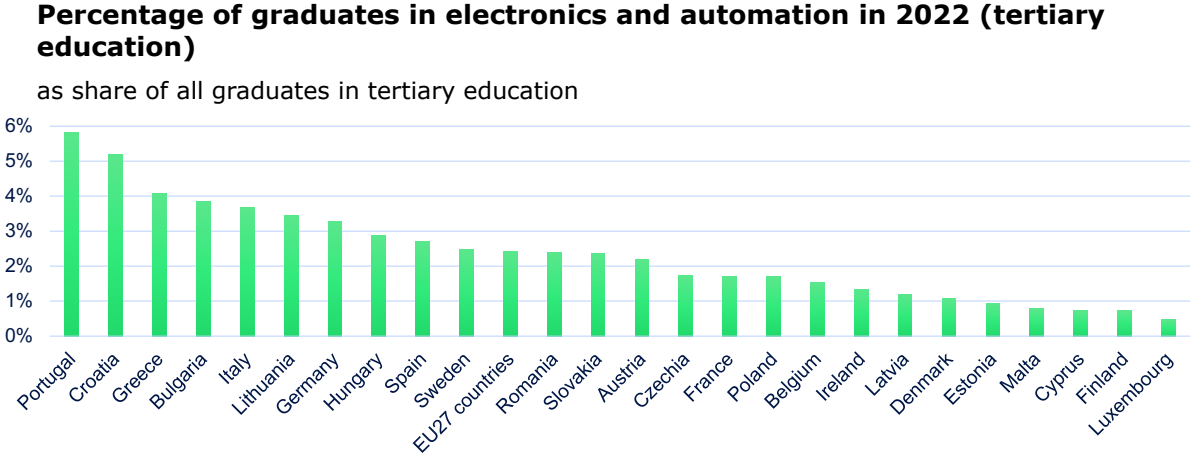
Figure 37: Share of tertiary education graduates in natural sciences, mathematics and statistics



Source: Eurostat.

Of high relevance for the electronics industrial ecosystem are dedicated skills in electronics and automation. Graduates in electronics are crucial for developing, integrating and producing electronics technologies, while ensuring their efficiency and functionality. Automation specialists play a key role in optimising manufacturing processes in the industry. Their skills are key for implementing advanced digital technologies and manufacturing techniques. A strong pipeline of graduates in these areas contributes significantly to the development and growth of electronics industry, enabling countries to enhance their competitiveness in the global market and attract investments. However, most EU countries demonstrate very low shares of graduates in these disciplines ranging between 3% and 1% of all graduates in tertiary education. Only Portugal and Croatia display shares larger than 5% in 2022 (Figure 38).

Figure 38: Share of graduates in tertiary education in electronics and automation as share of all graduates



Source: Eurostat.

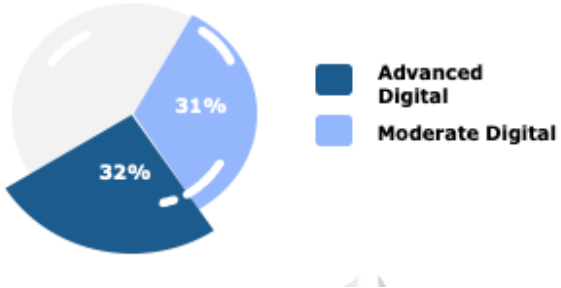
[92] Löser, Alexander; Tresp, Volker (2023): Große Sprachmodelle entwickeln und anwenden. Unter Mitarbeit von Acatech.

Skills demand

Skills demand in the electronics industrial ecosystem has been analysed following the skills intelligence insights of Cedefop, as in the case of the green transition[93]. Two types of skills have been analysed: **moderate skills**, which go beyond basic Microsoft proficiency but do not necessarily involve artificial intelligence, and **advanced digital skills**, such as AI, big data, augmented/virtual reality, or cloud computing.

As it can be observed in the Figure bellow, **32%** of job ads referenced advanced digital skills, while **31%** required moderate digital skills. These results emphasise the growing need for digital expertise in today's job market.

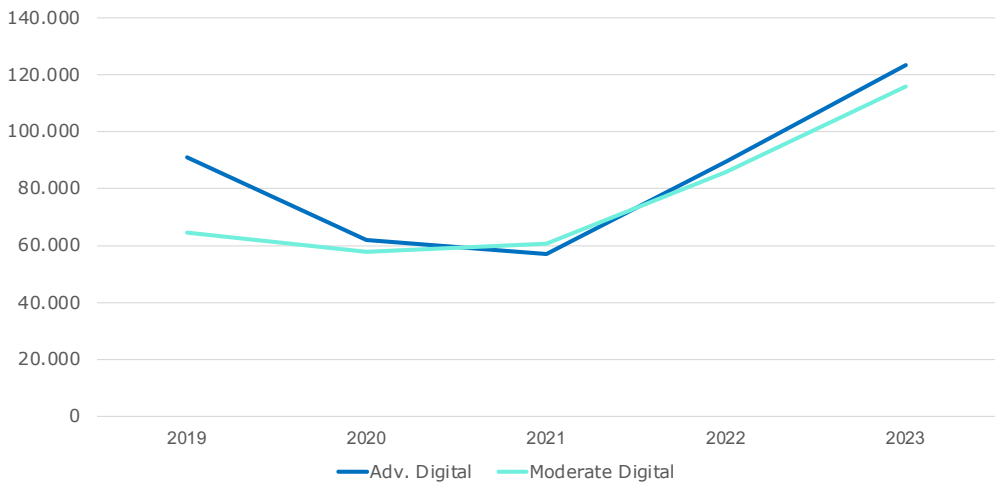
Figure 39: Share of advance and moderate digital skills in the electronic industrial ecosystem for the digital transition



Source: Technopolis Group based on Cedefop.

The results indicate that the requirements for digital skills listed on online job advertisements within the electronics industrial ecosystem has been growing dynamically over the period from 2021-2023. Trends also indicate that an equal number of online job advertisements have been published requiring both moderate and advanced digital skills, underscoring the rising importance of areas such as artificial intelligence, edge computing, and other technological advancements. This trend highlights that companies are increasingly seeking a workforce capable of navigating complex digital landscapes, where proficiency in AI, machine learning, data analytics, and cloud-based technologies is essential for staying competitive. The demand for these skills reflects a broader industry shift toward embracing digital transformation, where employees with expertise in both foundational and advanced digital tools are vital for driving innovation and maintaining operational efficiency.

Figure 40: Share of online job advertisements with a requirement for digital skills in construction in the EU27



Source: Technopolis Group based on analysis of Cedefop data

[93] https://www.cedefop.europa.eu/en/tools/skills-online-vacancies

Increasing disparity between supply and demand

The EU Chips Act aims to double the EU's global market share in semiconductors to 20% by 2030, necessitating additional 350,000 professionals, according to the estimates from Strategy& – the business strategy unit of Price Waterhouse Coopers.[94] The rapid growth of the global semiconductor and electronics markets, coupled with ambitious European targets for the Twin Transition, highlights the critical need for a skilled workforce. However, the current talent shortage is expected to worsen, creating a significant industry-wide talent gap.[95] The effective realisation of the EU Chips Act goals heavily relies on overcoming the talent shortage. If not addressed, this shortage could delay the launch of new fabrication plants and hinder their ability to operate at full capacity, particularly affecting businesses focused on advanced technologies[80].

Experts pointed out that electronics companies in the EU are experiencing a notable shortage of skilled workers in all key areas. Additionally, there is a significant lack of professional competencies in project management and quality management. Key roles required are project managers, reflecting strong demand for leadership in transformation efforts of the industry. AI and Machine Learning specialists follow closely, driven by the industry's focus on advanced technologies[96].

To address the talent gap challenge, various training and recruitment initiatives are underway. Programmes like the Semiconductor Talent Incubation Programme in Saxony and TSMC's collaboration with the Technical University of Dresden aim to prepare students for careers in the industry. Industry organisations in some EU countries and their partners have initiated campaigns to inspire young people to pursue careers in electrical engineering and related fields. Fostering curiosity about science and technology subjects at an early age is vital for increasing interest in STEM fields. Educational initiatives should aim to engage students in hands-on science experiences. Additionally, some EU countries, such as Germany, have simplified access to their labour market for foreign skilled workers to attract talent from non-EU countries.[97] For the majority of electronics companies it is essential to use learning and on-the-job training as a key strategy for achieving their business goals. Additionally, offering higher wages and clearly articulating business purpose are recognised as effective strategies. Other important practices include implementing diversity, equity, and inclusion policies, providing remote and hybrid work opportunities, and delivering effective reskilling and upskilling programmes[96].

Workforce dynamics further complicate the situation, as the semiconductor industry ranks lower in workplace attractiveness. The semiconductor industry faces challenges in branding and attracting talents compared to other tech sectors. The next generation of recruits has different expectations, necessitating innovative management strategies to effectively integrate and train younger employees.[98]

To address the skills gap and align educational outcomes with industry needs, EU countries must invest in educational reforms, promote STEM careers, and strengthen partnerships between academia and industry. Solutions must include expanding the talent pipeline through training programmes supported by companies as well as the EU and EU countries' national government initiatives.

Industry experts emphasised the necessity of an ongoing dialogue between industry and educational institutions to align training programmes with the skills needed in the workplace. This collaboration can help ensure that curricula are relevant and effective in

[94] https://www.strategyand.pwc.com/de/en/industries/telecommunication-media-and-technology/bridging-the-talent-gap.html
[95] https://www.eetimes.eu/semiconductor-capacity-is-up-but-mind-the-talent-gap/
[96] WEF (2023): Future of Jobs Report 2023. World Economic Forum. Online available at https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf.
[97] https://www.eetimes.eu/semiconductor-capacity-is-up-but-mind-the-talent-gap/
[98] https://www.eetimes.eu/semiconductor-capacity-is-up-but-mind-the-talent-gap/

preparing students for careers in the electronics sector. In addition, there is a need for a survey to identify skill and knowledge requirements and their gaps in vocational training.

Effects of digitisation and automation on jobs in electronics

In the electronics industry, the increased adoption of new and frontier technologies is expected to have a substantial effect on jobs. Several technologies are poised to significantly impact job creation, but they also carry risks of displacement. This applies first and foremost to AI, including machine learning and neural networks, necessitating a large number of professionals[96]. The broad adoption of AI can contribute to securing skilled labour through several dimensions. First, AI-driven automation can reduce the future demand for skilled workers by increasing productivity and relieving employees of burdensome tasks. AI can support upskilling by providing personalised training plans for individual workers. However, its adoption may displace some traditional roles.

In the future, the balance of tasks performed by humans and machines in the electronics industry is expected to shift significantly. Currently, humans perform 67% of tasks, while machines handle 33%. By 2027, this is forecasted to change to 56% for humans and 44% for machines, indicating a growing reliance on automation[96]. Increasing automation and process innovations will also contribute – albeit partially - to close the labour gap[80].

As technologies continue to evolve, a solid scientific and technical education will remain crucial. Short, application-oriented training may quickly become outdated, highlighting the need for ongoing education and skill development.

The rise of artificial intelligence (AI) in manufacturing is introducing transformative technologies, processes, and techniques to the electronics industry. Key drivers for the AI adoption across the electronics ecosystem include the industry's reliance on robotics, increasing global demand coupled with labour scarcity, as well as high cost of operational disruptions.[99] Companies are advancing in adoption of AI-based applications, though challenges like high costs and inadequate infrastructure, especially in small and medium-sized enterprises, persist. Among all sectors, electronics companies exhibit the highest rate of AI adoption[96]. Large companies in the electronics sector are generally at the forefront of AI adoption, being better positioned through availability of resources, strong focus on R&D and vast amount of data generated within different internal processes. Smaller firms struggle due to a lack of resources, understanding, infrastructure, and skills, and many businesses face incompatible IT systems. Regulatory challenges, such as the EU's Artificial Intelligence Act, add further complexity.[99] Despite these hurdles, the adoption of AI applications is growing, offering significant potential for task automation and increased productivity. In the EU, companies specialising on activities related to manufacturing of computer, electronic and optical products are experiencing high adoption rates of AI technologies. Companies that belong to manufacturing of electrical equipment and machinery – a further segment of electronics industrial ecosystem – display a somewhat moderate rate of AI adoption. Also, the adoption varies significantly across countries and regions: most EU countries with a strong industrial base and focus on Industry 4.0 display higher adoption rates, while many countries in Eastern Europe still have low to very low adoption rates of AI.

Recently, generative AI has become increasingly integral to electronics manufacturing, providing innovative solutions that enhance design, efficiency, and production processes. It is used in design generation, allowing engineers to explore numerous variations quickly while optimising for specific performance goals, such as weight reduction and thermal efficiency. In component selection, AI can help analyse vast datasets to recommend materials. Generative AI can support electronics manufacturing by generating synthetic data to train visual quality inspection (VQI) systems.[100] Overall, integrating generative AI

[99] https://www.eetimes.eu/the-undeniable-impact-of-ai-on-modern-manufacturing/
[100] This method overcomes the challenge of obtaining large, labelled datasets for detecting defects. VQI systems using this synthetic data can quickly identify defects, boosting production efficiency and throughput.

improves accuracy, speed, and adaptability in manufacturing, resulting in more efficient production.[101]

Further advanced digital technology, such as augmented and virtual reality is likely to be heavily adopted by organisations in electronics in the near future[96]

The interplay between the green and digital transitions represents a significant opportunity for the electronics industry. These two trends complement each other by driving innovation that enhances environmental sustainability while also advancing digital capabilities.

- **Technological Synergy**: Many technologies serve dual purposes, providing benefits for both digital advancement and environmental sustainability. For example, advanced sensors and IoT devices play a crucial role in optimising manufacturing processes, reducing waste, and improving energy efficiency. By integrating these technologies, companies can create smarter systems that monitor and control energy use in real time.
- **Energy Efficiency and Emission Reduction**: Technologies such as AI and data analytics are instrumental in enhancing the efficiency of manufacturing processes. By analysing data from various sensors, companies can gain insights into system performance, enabling them to optimise operations and reduce energy consumption.
- **Waste Reduction**: The use of digital technologies can significantly minimise waste across the production cycle. For instance, predictive maintenance powered by AI can identify potential equipment failures before they occur, reducing downtime and waste associated with defective products. AI can also be deployed to detect defects in real-time, reducing scrap.

3.2.3. Case study: advanced electronics as enabler for AI

Innovative electronics and semiconductors play a vital role in advancing AI capabilities. They provide the necessary hardware for computational power and efficiency needed for complex AI tasks while addressing energy consumption and scalability challenges.[102] Technologies like smaller semiconductor process nodes (7nm, 5nm, and 3nm) increase efficiency by allowing more transistors on a chip. Table 3 outlines pivotal milestones in the development of advanced semiconductors and their essential role in the AI revolution. Application-Specific Integrated Circuits (ASICs) and Field-Programmable Gate Arrays (FPGAs) are or can be tailored for specific AI tasks, optimising performance and flexibility. High-performance components such as graphics processing units (GPUs) and tensor processing units (TPUs) are designed to handle AI workloads effectively, improving processing speed and accuracy[103].

Table 3: Milestones in Semiconductor Technology for AI

Year	Milestone	Technology Node	Key Features and Impacts
1997	Deep Blue	0.6- and 0.35-µm	Early AI implementation, chess computing

[101]https://www.forbes.com/councils/forbestechcouncil/2023/04/26/how-generative-ai-can-be-used-in-electronics-manufacturing/
[102] https://ashamaei.medium.com/role-of-semiconductors-in-ai-analytics-and-edge-computing-f1b81282abfe
[103] CSET (2020): AI Chips: What They Are and Why They Matter. An AI Chips Reference. Center for Security and Emerging Technology. Online available at https://cset.georgetown.edu/wp-content/uploads/AI-Chips%E2%80%94What-They-Are-and-Why-They-Matter-1.pdf.

Year	Milestone	Technology Node	Key Features and Impacts
2012	ImageNet DNN	40 nm	Initiated the current machine learning era
2016	AlphaGo	28 nm	Advanced AI for complex game playing
2020	Initial ChatGPT	5 nm	Enabled large-scale language model training
2023	Current ChatGPT	3 nm	Further enhancements in AI performance

Source: compilation based on Liu and Wong (2024)[104].

Further key technological innovations that drive AI include advanced packaging techniques, such as chip-on-wafer-on-substrate (CoWoS) and system-on-integrated-chips (SoIC), which enable more transistors and improved performance. High-bandwidth memory, achieved by stacking DRAM chips vertically using through-silicon vias (TSVs) or hybrid bonding, is essential for AI workloads. Silicon photonics, which uses optical interconnects for high-speed communication between GPUs and CPUs, is becoming crucial for large-scale AI systems. To sustain the AI revolution, more performant GPUs as well as new computing architectures are needed. Future advancements in semiconductor technology, such as System-Technology Co-Optimisation and 3D integration, are essential for enhancing AI capabilities[104].

Balancing performance and energy efficiency

Energy consumption and efficiency are crucial topics around AI, as the deployment of AI systems often demands substantial computational resources, which lead to significant energy usage. Optimising energy efficiency is essential not only for reducing operational costs but also for minimising environmental impact. Recent advancements in semiconductors and electronics have significantly improved energy-efficient performance. According to Liu and Wong, the semiconductor industry has tripled energy-efficient performance (EEP)[105] of GPUs every two years over the past 15 years, significantly reducing energy consumption and greenhouse gas emissions. This was possible due to various technological innovations, such as:

- **Materials**: Use of advanced materials that improve performance and reduce power consumption;
- **Device integration**: Efficient integration techniques that pack more transistors into a smaller area, reducing power leakage;
- **EUV Lithography**: Extreme Ultraviolet Lithography enables finer patterning of circuits, leading to more energy-efficient chips;
- **Circuit design**: Optimised design techniques that focus on reducing power consumption;
- **System architecture**: Advanced architectures that improve computational efficiency and reduce energy usage.

However, as described in Box 7, the relationship between improved EEP and technology advancement is far more complex. Although EEP gains have led to increased energy

[104] Liu, Mark; Wong, H.-S. Philip (2024): How We'll Reach a 1 Trillion Transistor GPU Advances in semiconductors are feeding the AI boom. In: *IEEE Spectrum*. Online available at https://spectrum.ieee.org/trillion-transistor-gpu.
[105] Energy-Efficient Performance (EEP) is a metric that combines the energy efficiency and computational speed of a system.

efficiency of GPUs, the escalating performance demands of AI applications necessitate increasingly complex technologies with advanced features that consume significantly more energy, often offsetting the benefits of improved EEP. Thus, while EEP improvements signify progress, in practice they do not automatically lead to lower total energy consumption. Therefore, a more balanced approach that addresses both technological advancements and energy efficiency is crucial for sustainable GPU development. However, it is important to note that without advancements in microelectronics achieving high computing power while simultaneously reducing costs and energy expenditure would just not be possible. Thus, these developments go hand in hand. While EEP improvements signify progress, in practice they do not automatically lead to lower total energy consumption due significantly higher performance requirements. Therefore, a more balanced approach that addresses both technological advancements and energy efficiency is crucial for sustainable AI technology development.

Box 7: Demonstrating complex interplay between energy efficiency and increased AI performance requirements[106]

Advanced semiconductor technologies help maximise the performance of GPUs. Modern GPUs designed with cutting-edge semiconductor technologies consume less power while delivering superior performance. This is critical for data centers, where energy consumption accelerates due to the rising application of AI. An average GPU consumes around 200 W per workload. However, modern AI GPUs can consume up to 700 W per workload.[107] To showcase the extent of the impact that advanced electronics has on improved energy efficiency of modern GPUs, we perform calculation based on the assumption of tripling the EEP every two years made by Liu and Wong 2024 and other available data. The estimations are made for average GPUs broadly used in data centers. Potential savings through achieved EEP are compared to the energy consumption and environmental impact of modern GPUs without the EEP effects.

Aspect	Modern GPUs (700 W)	GPUs (200 W)
Power Consumption per GPU per workload	700 watts	200 watts
Annual Utilisation[108]	61%	61%
Annual Energy Consumption per GPU	Approximately 3.74 MWh (3,740,520 Wh)	Approx. 1.07 MWh (1,068,720.00 Wh)
Carbon Intensity of Electricity	0.475 kg CO_2 per kWh	0.475 kg CO_2 per kWh
Annual CO_2 footprint per GPU	Approx. 1,776.7 kg CO_2	Approx. 507.6 kg CO_2
Total GPUs	3,836,000[109]	100,000,000[110]
Total Annual Energy Consumption	Approx. 14,348.63 GWh (14,348,634,720.000 kWh)	Approx. 106,872 GWh (106,872,000,000.00 kWh)
Total annual footprint	6,815,601,49.00 kg CO2 or approx. **6,815,601.49 metric tons CO2**	50.764.200.000,00 kg CO_2 or **50,764,200.00 metric tons CO2**
Energy-Efficient Performance (EEP)		0.05 W per workload (15 years later)
Effective Total Energy Consumption (EEP)		26,718,000.00 kWh (15 years later)
Total Annual Carbon Footprint (EEP)		12,691,050.00 kg CO_2 (15 years later)

[106] For details on assumptions and calculations see Appendix B.
[107] https://www.tomshardware.com/desktops/servers/a-single-modern-ai-gpu-consumes-up-to-37-mwh-of-power-per-year-gpus-sold-last-year-alone-consume-more-power-than-13-million-households
[108] Based on the approx. utilisation rate of advanced GPUs in 2023: https://www.tomshardware.com/desktops/servers/a-single-modern-ai-gpu-consumes-up-to-37-mwh-of-power-per-year-gpus-sold-last-year-alone-consume-more-power-than-13-million-households
[109] This corresponds to the amount of GPUs of this capacity sold in 2023.
[110] Assumed number of GPUs annually in use for the sake of simplicity. The real number can be much higher.

Total Footprint Reduction (EEP)	Over 99%

Conclusion

The tripling of EEP every two years over the past 15 years leads to a dramatic reduction in energy consumption for the same computational workload. This reduction translates into a substantial reduction in CO_2 emissions, contributing to lower environmental impact. When scaled globally, the annual carbon footprint savings could be substantial. For instance, with 100 million GPUs, the total savings could amount to approximately 83.2 million metric tons of CO_2 annually (over 99% CO_2 reduction), when compared to the baseline of 15 years ago.

Overall, performance improvements and EEP gains have enabled several AI applications that would otherwise be prohibitively expensive. However, while improved EEP indicates that GPUs are becoming more efficient, the increased performance requirements of modern AI applications mean that power consumption has also risen significantly. Energy consumption of modern GPUs is driven by their ability to handle complex tasks that their precursors were simply not designed for. Therefore, the claim of a theoretical 99% reduction in energy consumption based on EEP improvements does not hold when considering the complex interplay of technological advancements, performance and workload requirements. Modern GPUs are designed to deliver high performance that often necessitates much higher energy use, thus complicating the relationship between EEP and actual power consumption in practice. The 700 W consumption per workload far outpaces the energy efficiency gains realised through EEP improvements. This escalation in energy consumption largely negates the theoretical benefits of reduced energy use based on EEP improvements.

Source: own estimations and elaborations based on data and assumptions indicated above.

Innovative electronics: key to sustainable AI

Despite efforts to improve efficiency, AI workloads have driven energy consumption of data centres up by 20-40% annually. Major AI tasks like deep learning, machine learning, and natural language processing are highly power-intensive contributing to environmental impacts during both the training and operation of AI systems. To achieve greener AI and more sustainable inference, advances in electronics and related areas are key. In particular, following strategies can be employed to reduce the environmental impact and improve cost-effectiveness of AI[111]:

- **Energy-Efficient Hardware:** GPUs, TPUs, and custom Deep Learning Accelerators (DLAs) designed for better energy efficiency, leading to a smaller carbon footprint.
- **New technologies, such as photonic integrated circuits and silicon photonics** hold huge potential against the background of increasing demands of AI and data centers to reduce power consumption and increase performance. These technologies facilitate high-speed data transmission, which is crucial for handling the massive data rates needed by AI accelerators and modern data infrastructures. In addition, advances in integration and packaging techniques can further enhance the efficiency and scalability of photonics[112].
- **Innovations and optimisations during the design stage:** implementing low power design techniques and use of low-power components, as well as designing for better thermal management.
- **Exploring new computer architecture** to maintain high performance while minimising energy use (e.g. in-memory computing). Traditional architecture separates CPU and memory, causing inefficiencies with data shuttling between them.
- **Optimised AI architectures:** Moving away from traditional CPU-centric architectures to specialised AI-centric architectures, which offer higher performance and lower power consumption.

[111] EE Times (Hg.) (2023b): EE Times Europe. Vol. 6, Number 4. Online available at https://www.eetimes.eu/ee-times-europe-magazine-november-2023/.
[112] EE Times Europe (2024): SPECIAL REPORT: OPTICS & PHOTONICS. Online available at https://www.eetimes.eu/wp-content/uploads/2024/08/24_09_EETE_all-FINALx3.pdf.

- **Heterogeneous integration** of different processors on a single chip further enhances energy efficiency for AI applications.
- **System-level optimisation:** Focusing on hardening data movement, sequencing, efficient networking, and heterogeneous computing to reduce energy per inference operation.
- **Further advances in semiconductor fabrication techniques** to improve manufacturing efficiency, focusing on recycling and reducing waste.

3.3. Impact of digital technologies on industrial competitiveness

The electronics industry is witnessing high demand for advanced digital technologies such as AI, robotics, and the Internet of Things (IoT). Robots, for instance, are now common in manufacturing processes, enhancing efficiency and precision. Anticipated advancements, particularly in fields like AI and sensor technology, are expected to further influence the market demand. The growth of data ecosystems and edge computing represents a significant opportunity for the EU to leverage its strengths in data management and analysis.

There is strong evidence that the adoption of digital technologies like IoT, big data analytics, AI, and robotics can significantly boost the competitiveness of the electronics industry. Automation and robotics can streamline production, reducing costs and cycle times. Big data analytics enables informed decision-making, while AI improves quality control and optimisation of major processes within design, development and manufacturing stages. Additionally, these technologies facilitate mass customisation and predictive maintenance, enhancing operational flexibility and reducing downtime. The broad adoption of advanced digital technologies is no longer an option for the electronics industry, but a necessity to sustain global competition.

The industry undertakes efforts to make supply more efficient and robust through leveraging digital technologies. Traditionally, the focus has been on cost optimisation, but companies are increasingly recognising that minimising costs alone is not enough. A robust and resilient supply chain is essential to counter potential disruptions. Resilience and robustness involve various aspects, including timely availability of raw materials, minimising supply shortages, and identifying and managing risks. By leveraging digital technologies like IoT, data analytics, and AI, companies can improve supply chain monitoring, detect potential risks early, and take appropriate measures to address them.

The EMI survey indicates that the adoption of advanced digital technologies increased productivity between 10-15%. The largest share of respondents witnessed an increase in productivity as a result of AI, followed by robotics (see Figure below).

Figure 41: The impact on productivity of advanced digital technologies as expressed by the share of respondents

	Increased	No change	Decreased
Artificial Intelligence	75%	25%	
Robotics	70%	25%	5%
Cloud computing	66%	34%	
Big Data	65%	35%	
AVR	45%	45%	10%
IoT	45%	53%	
Edge computing	40%	60%	
Blockchain	42%	44%	14%

Source: EMI Enterprise Survey 2024

Appendix A: References

Axenbeck, Janna; Berner, Anne; Kneib, Thomas (2022): What drives the relationship between digitalization and industrial energy demand? Exploring firm-level heterogeneityExploring firm-level heterogeneity. ZEW - Leibniz Centre for European Economic Research. Mannheim (ZEW Discussion Papers, No. 22-059). Online verfügbar unter https://www.econstor.eu/bitstream/10419/266649/1/1826800425.pdf.

Baldé, Cornelis P. et al. (2024): Global E-waste Monitor. International Telecom-munication Union (ITU) and United Nations Institute for Training and Research (UNITAR).

Blaabjerg, Frede; Yang, Yongheng; Kim, Katherine A.; Rodriguez, Jose (2023): Power Electronics Technology for Large-Scale Renewable Energy Generation. In: Proc. IEEE 111 (4), S. 335–355. DOI: 10.1109/JPROC.2023.3253165.

Boakes, L.; Garcia Bardon, M.; Schellekens, V.; Liu, I-Y.; Vanhouche, B.; Mirabelli, G. et al. (2023): Cradle-to-gate Life Cycle Assessment of CMOS Logic Technologies. In: 2023 International Electron Devices Meeting (IEDM). 9-13 Dec. 2023. 2023 International Electron Devices Meeting (IEDM). San Francisco, CA, USA, 12/9/2023 - 12/13/2023. Institute of Electrical and Electronics Engineers. Piscataway, NJ: IEEE, S. 1–4.

Chen, Steve; Gautam, Apoorv; Weig, Florian (2013): Bringing energy efficiency to the fab. McKinsey. Online verfügbar unter https://www.mckinsey.com/~/media/mckinsey/dotcom/client_service/operations/pdfs/bringing_fabenergyefficiency.ashx.

CSET (2020): AI Chips: What They Are and Why They Matter. An AI Chips Reference. Center for Security and Emerging Technology. Online verfügbar unter https://cset.georgetown.edu/wp-content/uploads/AI-Chips%E2%80%94What-They-Are-and-Why-They-Matter-1.pdf.

Dayarathna, Miyuru; Wen, Yonggang; Fan, Rui (2016): Data Center Energy Consumption Modeling: A Survey. In: IEEE Commun. Surv. Tutorials 18 (1), S. 732–794. DOI: 10.1109/COMST.2015.2481183.

EE Times (2023a): 100 Silicon Startups worth watching in 2023. 23 edition. Hg. v. EE Times. Online verfügbar unter https://www.eetimes.com/product/silicon-100-startups-worth-watching-in-2023-final/.

EE Times (Hg.) (2023b): EE Times Europe. Vol. 6, Number 4. Online verfügbar unter https://www.eetimes.eu/ee-times-europe-magazine-november-2023/.

EE Times Europe (2024): SPECIAL REPORT: OPTICS & PHOTONICS. Online verfügbar unter https://www.eetimes.eu/wp-content/uploads/2024/08/24_09_EETE_all-FINALx3.pdf.

GHGRP (2022): GHGRP Overview Report.

Greenpeace (2023): Invisible Emissions. A forecast of tech supply chain emissions and electricity consumption by 2030. Greenpeace. Online verfügbar unter https://www.greenpeace.org/static/planet4-eastasia-stateless/2023/04/620390b7-greenpeace_energy_consumption_report.pdf.

Gupta, Udit; Kim, Young Geun; Lee, Sylvia; Tse, Jordan; Lee, Hsien-Hsin S.; Wei, Gu-Yeon et al. (2020): Chasing Carbon: The Elusive Environmental Footprint of Computing. Online verfügbar unter http://arxiv.org/pdf/2011.02839v1.

Han, Botang; Wang, Dong; Ding, Weina; Han, Lei (2016): Effect of information and communication technology on energy consumption in China. In: Nat Hazards 84 (S1), S. 297–315. DOI: 10.1007/s11069-016-2188-1.

Hess, Julia (2024): Chip Production’s Ecological Footprint: Mapping Climate and Environmental Impact. Stiftung Neue Verantwortung. Online verfügbar unter https://www.interface-eu.org/publications/chip-productions-ecological-footprint.

IDTechEx Research (2024): Sustainable Electronics Manufacturing 2023 - 2033. Online verfügbar unter https://www.idtechex.com/en/research-report/sustainable-electronics-manufacturing-2023-2033/902.

Kamiya, G. and Bertoldi, P. (2024): Energy Consumption in Data Centres and Broadband Communication Networks in the EU. Hg. v. Publications Office of the European Union. uropean Commission, Joint Research Centre. Luxembourg. Online verfügbar unter https://data.europa.eu/doi/10.2760/706491, JRC135926.

Kunkel, S.; Neuhäusler, P.; Matthess, M.; Dachrodt, M. F. (2023): Industry 4.0 and energy in manufacturing sectors in China. In: Renewable and Sustainable Energy Reviews 188, S. 113712. DOI: 10.1016/j.rser.2023.113712.

Lin, Boqiang; Xu, Chongchong (2024): The effects of industrial robots on firm energy intensity: From the perspective of technological innovation and electrification. In: Technological Forecasting and Social Change 203, S. 123373. DOI: 10.1016/j.techfore.2024.123373.

Lin, Weiwei; Yu, Tianhao; Gao, Chongzhi; Liu, Fagui; Li, Tengyue; Fong, Simon; Wang, Yongxiang (2021): A hardware-aware CPU power measurement based on the power-exponent function model for cloud servers. In: Information Sciences 547, S. 1045–1065. DOI: 10.1016/j.ins.2020.09.033.

Liu, Mark; Wong, H.-S. Philip (2024): How We'll Reach a 1 Trillion Transistor GPU Advances in semiconductors are feeding the AI boom. In: IEEE Spectrum. Online verfügbar unter https://spectrum.ieee.org/trillion-transistor-gpu.

Löser, Alexander; Tresp, Volker (2023): Große Sprachmodelle entwickeln und anwenden. Unter Mitarbeit von Acatech.

Matthess, Marcel; Kunkel, Stefanie; Dachrodt, Melissa Fiona; Beier, Grischa (2023): The impact of digitalization on energy intensity in manufacturing sectors – A panel data analysis for Europe. In: Journal of Cleaner Production 397, S. 136598. DOI: 10.1016/j.jclepro.2023.136598.

McKinsey (2022): Keeping the semiconductor industry on the path to net zero. Online verfügbar unter https://www.mckinsey.com/industries/semiconductors/our-insights/keeping-the-semiconductor-industry-on-the-path-to-net-zero.

McKinsey & Company (2023): Reinventing our economy from within. How Europe's start-up ecosystems can learn from each other to ignite and scale up entrepreneurship. Online verfügbar unter https://www.mckinsey.de/~/media/mckinsey/locations/europe%20and%20middle%20east/deutschland/publikationen/2023-09-22%20reinventing%20our%20economy%20from%20within/reinventing%20our%20economy%20from%20within.pdf.

Mulvik, I., et al. (2021): Education for environmental sustainability: policies and approaches in European Union Member States. Hg. v. Luxembourg: Publications Office of the European Union, 2021. European Commission. Online verfügbar unter https://www.oekolog.at/dokumente/97/Education_for_Environmental_Sustainability_-_EC_Report_-_2022.pdf.

Pelé, Anne-Françoise (2024): VC Discernment Is Critical Amid AI Surge, Celesta's Top Exec Says. In: EE Times Europe. Online verfügbar unter https://www.eetimes.eu/wp-content/uploads/2024/08/24_09_EETE_all-FINALx3.pdf.

PWC (2023): Future Proofing the Electronics Industry. The case for circular business models. PWC. Online verfügbar unter https://circulareconomy.europa.eu/platform/sites/default/files/2023-11/Circular%20electronics.pdf.

Schulte, Patrick; Welsch, Heinz; Rexhäuser, Sascha (2016): ICT and the Demand for Energy: Evidence from OECD Countries. In: Environ Resource Econ 63 (1), S. 119–146. DOI: 10.1007/s10640-014-9844-2.

Semi; Semiconductor Climate Consortium; Boston Consulting Group (2023): Transparency, Ambition, and Collaboration: Advancing the Climate Agenda of the Semiconductor Value Chain. Online verfügbar unter https://discover.semi.org/rs/320-QBB-055/images/Transparency-Ambition-and-Collaboration-BCG-SEMI-SCC-20230919.pdf.

Tembey, G., et al. (2023): A Net Zero Plan for the Semiconductor Industry. Boston Consulting Group. Online verfügbar unter https://www.bcg.com/publications/2023/a-plan-to-reduce-semiconductor-emissions#:~:text=Manufacturing%20more%20advanced%20semiconductors%20requires,not%20peak%20until%20about%202045.

Vertiv (2023): ENERGY LOGIC: CALCULATING AND PRIORITIZING YOUR DATA CENTER IT EFFICIENCY ACTIONS. Online verfügbar unter https://www.vertiv.com/4a162d/globalassets/documents/white-papers/energy-logic-2-wp-en-na-sl-24632_51018_2.pdf.

Wang, En-Ze; Lee, Chien-Chiang; Li, Yaya (2022): Assessing the impact of industrial robots on manufacturing energy intensity in 38 countries. In: Energy Economics 105, S. 105748. DOI: 10.1016/j.eneco.2021.105748.

WBCSD and WRI (2004): The Greenhouse Gas Protocol. A Corporate Accounting and Reporting Standard. Revised version. World Business Council for Sustainable Development. Online verfügbar unter https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf.

WEF (2023): Future of Jobs Report 2023. World Economic Forum. Online verfügbar unter https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf.

World Robotics (2022): World Robotics 2022 – Industrial Robots, VDMA Services GmbH, Frankfurt am Main, Germany, 2022. IFR Statistical Department, VDMA Services GmbH. Online verfügbar unter https://ifr.org/img/worldrobotics/Executive_Summary_WR_Industrial_Robots_2022.pdf.

Wu, Hui; Han, Mengyu; Shen, Yang (2023): Technology-driven energy revolution: the impact of digital technology on energy efficiency and its mechanism. In: Front. Energy Res. 11, Artikel 1242580. DOI: 10.3389/fenrg.2023.1242580.

Yole (2024): Power Converters for xEV, Green Energy and Industrial 2024. Yole Intelligence.

Appendix B: Methodological notes

Table 4: Summary of sustainability goals and progress made of selected electronics companies in the EU

Company	Sustainability goals	Strategies for achievement	Progress	Source:
Infineon Technologies (Germany, Austria)	• Carbon neutrality by 2030; • Reduce emissions by 70 % by 2025 compared to 2019; • Water usage reduction targets; • Waste management and waste reduction targets; • Promote recycling initiatives; • Sustainable product design principles; • Circular economy initiatives (e.g. on waste management)	• Implementation of energy efficiency programmes as well as the switch to green electricity in the EU and North America • Implementation and expansion of gas abatement concepts • Waste management strategie • Water consumption reduction strategies	• Reduction of scope 1 and scope 2 emissions by 56.8 % by the end of 2023 compared to 2019; • Reduction of waste generation by ca. 36% between 2019 and 2023 (from 5.6 grams per EUR of revenues in 2019 to 3.6 in 2023); • Reduction of water consumption between 2019 and 2023 by ca. 29%: (from 0.31 litres per EUR of revenues to 0.22); • Reduction of PFC emissions by ca. 57% in 2023 compared to 2019	Infenion Sustainability Report 2023
ASML (the Netherlands)	• Net zero emissions across entire value chain by 2040 • Reduction of 25.2% of scope 1 and 2 emissions in 2025, compared to base year 2019, • Reduction of 35.3% in scope 3	• Close collaboration with value chain stakeholders on sustainability goals • Circular economy focus trough 1) preventing waste; 2) Extending the lifetime and productivity • of systems and assets, 3)	• 100% of the electricity used in the EU and US is from renewable sources • The overall scope 3 emissions are increasing due to the growth of ASML • Reuse rate of parts returned	ASML Annual Report 2023

Company	Sustainability goals	Strategies for achievement	Progress	Source:
	emissions by 2025 compared to 2019 • Reuse rate of parts returned from field and factory of 95% by 2025; • Recycling rates of 90% by 2025 • Zero waste from operations to landfill or incineration by 2030	Reusing resources, parts, tools, packaging and goods across the value chain 4) Recycling materials; • Waste management and waste prevention strategies	from field and factory in 2023 – 88% • Recycling rate of 55% in 2023	
Ericsson (Sweden)	• Net Zero GHG emissions across the value chain for scope 1-3 by 2040 • Emission reductions of total GHG emissions in the value chain by 50% by 2030 • Emission reduction of scope 1 and 2 by 90% by 2030 • Reduction of energy consumption of typical new radio base station sites by 40% by 2025 • Supply chain engagement: Have 350 high-emitting and strategic suppliers set their own 1.5 °C-aligned emissions	• Sustainable product design • Circular economy principles: work on increasing product take-back volumes and sale of refurbished equipment and awareness raising campaigns • Purchase of renewable energy	• Total value chain emissions in 2023 compared to 2020 +24% (The increase is largely explained by an increase in downstream emissions in the use-phase of sold products.[113] • Scope 1 and 2 emissions: -38% in 2023 compared to 2020 • Reduction of energy consumption of typical new radio base station sites by 30% in 2023 compared to 2021 • 237 suppliers with accepted targets in 2023 • Share of renewable	Sustainability & CR report 2023

[113] There has been a geographical shift in sales to markets, such as India, where a relatively larger share of electricity is generated using fossil fuels.

Company	Sustainability goals	Strategies for achievement	Progress	Source:
	reduction targets		energy at facilities increased to 70% in 2023	
STMicroelectronics	• Generate at least 33% of revenues from Sustainable Technologies by 2027 • Reduce emissions by 50% by 2025 compared to 2018; • Be carbon neutral by 2027 in all direct and indirect emissions from scope 1 and 2; • Adopt 100% renewable energy sources by 2027; • Reduce energy consumption by at least 150 GWh per year by 2027. • Improve water efficiency by 20% by 2025 vs 2016. • Recycle at least 50% of the water used each year. • Ensure an annual landfill waste rate below 3%; • Reuse or recycle 95% of waste by 2025.	• Implementing eco-design; create at least 50% of new products recognized as Sustainable Technology; • By increasing renewable energy sourcing through energy procurement and green energy installations. • Purchasing Renewable Energy Certificates; • By minimizing transportation emissions • By offsetting remaining emissions through reforestation and compensation programmes by 2027 • By implementing energy conservation projects, such as improving energy efficiency in production, facilities, and processes • Reducing waste and promoting circular economy	• Reduction of scope 1 and scope 2 emissions in 2022 by 40% compared to 2018 • Total reduction of CO2 emissions by 59% between 2016 and 2022 • Consumption of green energy increased from 21% in 2018 to 57% in 2022 • PFC emissions reduced by 46% in 2022 compared to 2016 • Reduced water consumption by 12% in 2022 compared to 2016 • Water recycled and reused in 2022 - 42% • Increase of hazardous waste by 49% in 2022 compared to 2018 • Increase of waste recycling by 30% in 2022 compared to 2018	STMicroelectronics Sustainability Report 2023; Excel Sheet on Sustainability Report

Company	Sustainability goals	Strategies for achievement	Progress	Source:
ON Semiconductor	• Achieve net zero emissions by 2040 (Net Zero 2040) across Scopes 1, 2 and 3 • Using 50 percent renewable energy by 2030 and 100 percent renewable energy by 2040	Reduction of scope 1 emissions through: • Optimising of manufacturing processes, reduction of fossil fuel combustion • Implementing efficient manufacturing technologies suitable to lower GWP gases and electrification • Substitution of process gases with lower GWP • Point-of use abatement devices Reduction of scope 2 emissions through: • Investment and purchase of renewable energy • Energy Conservation & Efficiency • System Optimization Reduction of scope 3 emissions through: • targeted supplier engagement programme Further strategies: • Design and manufacturing of products that help reduce the environmental footprint	• Reduction of scope 1 emissions by 18% in 2023 compared to 2022 (base year) • Slight increase of scope 2 emissions in 2023 • Reduction of scope 3 emissions by 27% in 2023 • Waste reduction by 19% in 2023 • Increase of End-of-Life Treatment of sold products by 55% in 2023	

Company	Sustainability goals	Strategies for achievement	Progress	Source:
Schneider Electric	Targets by 2025: • Carbon neutrality in operations (scope 1 and scope 2) (including CO2 offsets); • 80% green revenue • 800 million tons of CO2 emission savings since 2018; • 1000 top suppliers to reduce CO2 emissions by 50%; • 50% green material content in products; • 100% of primary and secondary packaging free from single-use plastics and use recycled cardboard; Further goals: • Carbon neutrality across entire value chain (including CO2 offsets) by 2040 • Eco-design of new products	• Implementing "Schneider Sustainability Impact (SSI)" programme. • Implementing the key initiatives of company's Net-Zero roadmap • Implementing biodiversity action plan • Schneider Electric launched a comprehensive support programme in April 2021 and named it "The Zero Carbon Project". Under the programme, it provides tools and resources in order to help suppliers set and achieve their climate targets. • Investing in and developing innovative solutions that deliver immediate and lasting decarbonization	Achieved targets in 2023 • 6.13/10 Schneider Sustainability Impact score, outperformed 2023 target (6.00/10) • 63% sustainable packaging (vs. 45% in 2022) • 553M Tonnes of saved and avoided CO2 emissions for customers since 2018 • Increase of green material content in products by 22% since 2020 • Reduced CO2 emissions from top 1,000 suppliers' operations by 27% since 2020 • Electricity sourced from renewables – 88% • Improved energy efficiency at facilities by 15% since 2020 • Product revenues covered with Green Premium – 81% • Local biodiversity conservation and restoration programmes +66% since 2020	Schneider Electrics 2023 Sustainable Development Report

Company	Sustainability goals	Strategies for achievement	Progress	Source:
			• 'Waste-to-Resource' sites – 137 • Avoid primary resource consumption through 'take-back at end-of-use'– 311,229 metric tons (increase of 98% since 2020 • Water conservation strategy and action plan for sites in water-stressed areas +73% since 2020	
Elmos Semiconductor (Germany)	• Reduce GHG emissions by 40% for scope 1 and 2 by 2026, compared to the 2022 baseline • Become climate neutral for its own activities (Scope 1 & 2) by 2035 • Reduce waste and increase re- and upcycling	• Procurement of renewable electricity • LED building lighting at the Dortmund HQs • Internal water treatment and recycling system • PV-system to be installed at Dortmund HQs • Switching to more environmentally friendly refrigerants for the chillers in Dortmund • Re- & upcycling, focusing on critical and rare materials	• Reduction of scope 1 and 2 emissions by 10% in 2023 compared to 2022 • 72% of Group sales make a substantial contribution to environmental protection and higher efficiency.	Elmos Semiconductor SE ESG/Sustainability strategy; Elmos Semiconductor SE ESG Presentation March 2024
Ams Osram (Austria)	• Carbon neutrality of own operations by 2030 • 100% renewable energy for electricity by 2028	• Reduction of greenhouse gas emissions by purchasing more green energy and certificates • Implemented technology roadmaps for	• No reduction of Scope 1 emissions in 2023 compared to 2020 • GHG Scope 2 emissions (market-based) reduction by	Sustainability Report 2023

Company	Sustainability goals	Strategies for achievement	Progress	Source:
	• Implemented energy efficiency measures for 20% • CO_2 reduction by 2028 • Near- and long-term targets for reductions of Scope 3 emissions • Water and waste reduction • Expand wafer substrate recycling • Reduction of plastic by 200 t by 2026 within plastic-free packaging strategy • Expansion of the circular portfolio of LED modules for automotive lighting • Start LCA approach, first analysis in 2024	resource-efficient processes. • Initiated recycling of germanium (producing 63,000 new 6-inch wafers by end of 2023). • Launched "Eco-Friendly Packaging" project for automotive aftermarket, transitioning to plastic-free cardboard by mid-2024. • Reduce resource usage throughout the product life cycle. • Shift towards eco-friendly packaging and minimize plastic use.	13% in 2023 compared to 2023 • GHG Scope 2 emissions (location-based) – reduction by 12% vs 2020 • Metric tons of CO_2 emissions from own activities (Scope 1 and 2) per EUR 1 million operating output – increase by 16% in 2023 compared to 2021 • Water withdrawal – 6% in 2023 compared to 2020 • Specific water withdrawal per EUR 1 million operating output increase by 21% in 2023 compared to 2020	
Melexis (Belgium)	• Zero carbon emissions. • Contribute to circular packaging. • Responsible sourcing of materials.	• Invest in energy-efficient production technologies. • Implement initiatives for recycling and responsible material use. • Engage with stakeholders to promote sustainability across the supply chain. • Monitor and measure environmental performance with ISO 14001 certification.	• Materials used: increased by 36% in 2021 compared to 2019 • Increased energy consumption within organisation in 2021 by 12% compared to 2019 • Lowered energy intensity per 1 million sold parts by 21%	Information on the corporate website

Table 5: Sustainability Challenges and Needs of Electronics SMEs

Challenges and Needs of SMEs	Description	Potential mitigation strategies
Lack of Awareness about Ecological Sustainability	Many SMEs may not fully understand the importance of sustainability or how to implement it effectively in their operations	Develop informational campaigns; conduct training programmes and workshops to raise awareness about sustainable practices;
Lack of Intrinsic Motivation	Many SMEs may not feel a strong internal drive to pursue sustainability, often responding only to regulatory pressures.	Showcase successful case studies of SMEs that have benefited from sustainability initiatives to inspire others; Introduce competitions or awards for sustainable practices to create incentives and encourage engagement.
Resource Constraints	SMEs often lack the financial and human resources needed to invest in and integrate sustainable technologies and practices	Provide specific grants and incentives for SMEs that implement sustainable technologies; Support public and private partnerships to facilitate access to resources and technologies.
Bureaucratic Hurdles	Complex regulatory requirements can be overwhelming for SMEs, making compliance difficult and time-consuming	Simplify approval and licensing processes to ease compliance for SMEs; Offer guidance and support to help navigate complex regulatory requirements
Access to Data and Information	SMEs may struggle to obtain reliable data needed (e.g. for LCA) to comply with regulations and assess their environmental impact	Establish a centralised EU database to provide SMEs with relevant information on sustainable practices and legal requirements; Promote collaborations between SMEs and larger companies to share data and best practices.
Pressure from Larger Companies	Larger corporations often set high sustainability standards for their suppliers, which can create pressure on SMEs to comply.	Encourage larger firms to actively support their SME suppliers in achieving sustainability goals; Develop networks and platforms to facilitate knowledge-sharing between SMEs and larger companies
Economic Uncertainty and Survival	Many SMEs are often fighting for their economic survival in global markets, which leads to different priorities, making sustainability a lower priority	Develop tailored support programmes specifically designed for SMEs facing economic challenges; Provide incentives for SMEs to invest in sustainable technologies, demonstrating potential cost savings that can aid their survival.
Integration of New Technologies	SMEs may find it difficult to adopt new technologies due to a lack of knowledge or infrastructure.	Collaborate with research institutions and universities to facilitate access to innovations in sustainable electronics; Promote technology transfer programmes that help SMEs adapt new energy-efficient technologies

Challenges and Needs of SMEs	Description	Potential mitigation strategies
Need for Demonstrating Real Value of Sustainable Technologies Integration	SMEs require concrete examples and data to understand the tangible benefits of integrating sustainable technologies.	Conduct pilot projects showcasing the successful integration of sustainable technologies in SMEs, highlighting cost savings and efficiency improvements, Develop metrics and case studies that quantify the benefits of sustainability initiatives, emphasizing long-term economic gains.

Source: Compilation based on insights from industry expert interviews

Box 8: Environmental impact estimations of overall annual carbon footprint savings resulting from advancements in electronics

To showcase the extent of the impact that advanced electronics has on improved energy efficiency of modern GPUs, we make following assumptions based on available data and own calculations:

Energy-Efficient Performance (EEP) of standard GPU

- **Baseline EEP: Assume that a standard GPU has an EEP of 1**
- **15 years later: If EEP has tripled every two years (according to estimations of (Liu und Wong 2024), the current EEP (15 years later) would be: 1 x $3^{(15/2)}$ = 3,787.995**

Impact on Energy Consumption

- **Baseline: Assume a GPU consumed 200 W of power for a given computational workload.**
- **15 years later: Given the EEP improvement, a modern GPU would consume: 200/3,787.995 ≈ 0.05 W for the same workload.**

Carbon Footprint Reduction

- **According to IEA, the average carbon intensity of electricity generated is 0.475 kg CO_2 per kWh.**
- **Baseline: For an 8-hour operation, the energy consumption would be 200 W x 8 h = 1.6 kWh, resulting in 1.6 x 0.475 kg CO_2/kWh ≈ 0.8 kg CO_2**
- **15 years later: For the same 8-hour operation with the improved EEP, the energy consumption would be 0.05 W x 8 h = 0.42 Wh, resulting in 0.42 x 0.475 kg CO_2/kWh} = 0.0002 kg CO_2**

Estimation

To estimate the overall annual carbon footprint savings resulting from advancements in semiconductor technology, we need to consider the following factors

- **Operational Hours per Year: 24 hours / day 365 days/year = 8,760 hours**
- **Number of GPUs in Use Globally: 100 million (as an example)**
- **Energy Savings per Device:**
- **Annual energy consumption without EEP: 200 W x 8,760 h = 1,752 kWh**
- **Annual energy consumption with EEP (15 years later): 0.05 W x 8,760 h = 0.46 kWh**
- **Annual energy savings per device: 1,752 kWh - 0.46 kWh = 1,751.54 kWh**
- **Carbon Footprint Reduction per Device:**
- **Annual carbon footprint reduction per device: 1,751.54 kWh x 0.475 kg CO_2/kWh = 832,199 kg CO_2**
- **Total Carbon Footprint Savings: 832,199 kg CO_2 x 100,000,000 devices = 83,219,978,031 kg CO_2. Converted to metric tons: = 83,219,978 metric tons CO_2**

Energy consumption and environmental impact of modern GPUs[114]

- **Total GPUs sold in 2023: 3,836,000**
- **Power consumption per GPU: up to 700 watts per AI workload**
- **Annual utilization: 61%**
- **Annual energy consumption of 1 GPU: 700 W x 0.61 x 24 x 365 = 3,740,520 Wh = 3.74 MWh**
- **Total annual energy consumption: 14,348.63 GWh or 14,348,634,720 kWh**
- **Total carbon footprint: 6,815,601,492.00 kg CO2 Converted to metric tons: 6,815,601.5 metric tons CO2**

Source: own estimations

Startup data and venture capital data analysis

Crunchbase and Net Zero Insights

within their description) and additional search in the business descriptions.

LinkedIn data analysis

Green skills – keywords used: Cleantech, Sustainability, Sustainable Development, Sustainable Business, Energy Efficiency, Clean Energy Technologies, Renewable Energy, Wind Energy, Biomass, Biomass Conversion, Solar Energy, Solar Power, Urban Forestry, Forest Ecology, Sustainable Communities, Organic Farming, Organic Gardening, Urban Agriculture, Organic Food, Waste Management, Waste Reduction, Recycling, Water Treatment, Water Resource Management, Water Purification, Green Marketing, Green Printing, Environmental Biotechnology, Environmental Science, Environmental Engineering, Environmental Management Systems, Environmental Protection, Wastewater Treatment, Ecology, Circular Economy, Zero Waste, Waste to Energy, Plastics Recycling, E-Waste, Carbon Reduction Strategies, Carbon Footprinting, Carbon Neutral, Energy Retrofits, Biodiversity, Biodiversity Conservation, Nature Conservation, Advanced Materials, Nanomaterials, Biomaterials, Reuse, Separation Process, Sorting, Equipment Repair, Natural Resource Management, Sustainability Reporting, Green Development, Sustainable Cities, Energy Conservation, Energy Management, Environmental Awareness, Environmental Impact Assessment, Environmental Compliance, Leadership in Energy and Environmental Design (LEED), Environmental Policy, Green Technology, Sustainable Design, Sustainable Architecture, Environmental Consulting, Maintenance and Repair, Solar PV, Solar Cells, Wind Turbines, Wind Turbine Design, Carbon Capture, Low Carbon Technologies, Low Carbon, Renewable Fuels, Renewable Energy Systems, Renewable Resources, Integrated Water Resources Management, Natural Resources, Biodiesel, Bioplastics, Waste Treatment, Waste Water Treatment Plants, Electric Vehicles, Hybrid Electric Vehicles, Multi-modal Transportation, Energy Efficiency Consulting, Recycled Water, Adaptive Reuse, Ecodesign, Life Cycle Assessment, Energy Optimisation, Alternative Fuels, Green Building, Green Infrastructure, Green Purchasing, Biodegradable Products, ISO 14001, EMAS, Environmental Standards

Digital skills – keywords used: data analytics, online platforms, digital payment, online ticketing, Cybersecurity, Intrusion Detection, Malware Detection, Cloud Security, Cybercrime Investigation, Cyber Threat Intelligence (CTI), Cryptography, DLP, Malware Analysis, IDP; Vulnerability Assessment, Certified Information Security Manager (CISM), Computer Forensics, Cloud Infrastructure, Cloud Services, Google Cloud Platform (GCP), SAP Cloud Platform, SAP HANA, Everything as a Service (XaaS), Software as a Service (SaaS), Platform as a Service (PAAS), Infrastructure as a Service (IaaS), Private Clouds, Hybrid Cloud, Cloud Computing, Edge Computing, High Performance Computing (HPC), Serverless Computing, Robotics, Robot, Robotic Surgery, Human-robot Interaction,

[114] Based on data provided in: https://www.tomshardware.com/desktops/servers/a-single-modern-ai-gpu-consumes-up-to-37-mwh-of-power-per-year-gpus-sold-last-year-alone-consume-more-power-than-13-million-households

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